8 The Making of the National Movement: 1870s–1947



Fig. 1 *– Police teargas demonstrators during the Quit India movement*

In the previous chapters, we have looked at:

- The British conquest of territories, and takeover of kingdoms
- Introduction of new laws and administrative institutions
- Changes in the lives of peasants and tribals
- Educational changes in the nineteenth century
- Debates regarding the condition of women
- Challenges to the caste system
- Social and religious reform
- The revolt of 1857 and its aftermath
- The decline of crafts and growth of industries

On the basis of what you have read about these issues, do you think Indians were discontented with British rule? If so, how were different groups and classes dissatisfied?

The Emergence of Nationalism

The above-mentioned developments led the people to ask a crucial question: what is this country of India and for whom is it meant? The answer that gradually emerged was: India was the people of India – *all the people* irrespective of class, colour, caste, creed, language, or gender. And the country, its resources and systems, were meant for all of them. With this answer came the awareness that the British were exercising control over the resources of India and the lives of its people, and until this control was ended, India could not be for Indians.

This consciousness began to be clearly stated by the political associations formed after 1850, especially those that came into being in the 1870s and 1880s. Most of these were led by English-educated professionals such as lawyers. The more important ones were the Poona Sarvajanik Sabha, the Indian Association, the Madras Mahajan Sabha, the Bombay Presidency Association, and of course the Indian National Congress.

Note the name, "Poona Sarvajanik Sabha". The literal meaning of "*sarvajanik*" is "of or for all the people" (*sarva* = all + *janik* = of the people). Though many of these associations functioned in specific parts of the country, their goals were stated as the goals of all the people of India, not those of any one region, community or class. They worked with the idea that the people should be **sovereign** – a modern consciousness and a key feature of nationalism. In other words, they believed that the Indian people should be empowered to take decisions regarding their affairs.

Sovereign – The capacity to act independently without outside interference

The dissatisfaction with British rule intensified in the 1870s and 1880s. The Arms Act was passed in 1878, disallowing Indians from possessing arms. In the same year, the Vernacular Press Act was also enacted in an effort to silence those who were critical of the government. The Act allowed the government to confiscate the assets of newspapers including their printing presses if the newspapers published anything that was found "objectionable". In 1883, there was a furore over the attempt by the government to introduce the Ilbert Bill. The bill provided for the trial of British or European persons by Indians, and sought equality between British and Indian judges in the country. But when white opposition forced the government to withdraw the bill, Indians were enraged. The event highlighted the racial attitudes of the British in India.

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The need for an all-India organisation of educated Indians had been felt since 1880, but the Ilbert Bill controversy deepened this desire. The Indian National Congress was established when 72 delegates from all over the country met at Bombay in December 1885. The early leadership – Dadabhai Naoroji, Pherozeshah Mehta, Badruddin Tyabji, W.C. Bonnerji, Surendranath Banerji, Romesh Chandra Dutt, S. Subramania Iyer, among others – was largely from Bombay and Calcutta. Naoroji, a businessman and **publicist** settled in London, and for a time member of the British Parliament, guided the younger nationalists. A retired British official, A.O. Hume, also played a part in bringing Indians from the various regions together.

Publicist – Someone who publicises an idea by circulating information, writing reports, speaking at meetings

Source 1

Who did the Congress seek to speak for?

A newspaper, *The Indian Mirror*, wrote in January 1886:

> The First National Congress at Bombay … is the nucleus of a future Parliament for our country, and will lead to the good of inconceivable magnitude for our countrymen.

Badruddin Tyabji addressed the Congress as President in 1887 thus:

> this Congress is composed of the representatives, not of any one class or community of India, but of all the different communities of India.



Fig. 2 – Dadabhai Naoroji

Naoroji's book *Poverty and Un-British Rule in India* offered a scathing criticism of the economic impact of British rule.

A nation in the making

It has often been said that the Congress in the first twenty years was "moderate" in its objectives and methods. During this period, it demanded a greater voice for Indians in the government and in administration. It wanted the Legislative Councils to be made more representative, given more power, and introduced in provinces where none existed. It demanded that Indians be placed in high positions in the government. For this purpose, it called for civil service examinations to be held in India as well, not just in London.

The demand for Indianisation of the administration was part of a movement against racism, since most important jobs at the time were monopolised by white

Activity

From the beginning the Congress sought to speak for, and in the name of, *all* the Indian *people*. Why did it choose to do so?

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Repeal – To undo law; to officially end the validity of something such as a law

officials, and the British generally assumed that Indians could not be given positions of responsibility. Since British officers were sending a major part of their large salaries home, Indianisation, it was hoped, would also reduce the drain of wealth to England. Other demands included the separation of the judiciary from the executive, the **repeal** of the Arms Act and the freedom of speech and expression.

The early Congress also raised a number of economic issues. It declared that British rule had led to poverty and famines: increase in the land revenue had impoverished peasants and zamindars, and exports of grains to Europe had created food shortages. The Congress demanded reduction of revenue, cut in military expenditure, and more funds for irrigation. It passed many resolutions on the salt tax, treatment of Indian labourers abroad, and the sufferings of forest dwellers – caused by an interfering forest administration. All this shows that despite being a body of the educated elite, the Congress did not talk only on behalf of professional groups, zamindars or industrialists.

The Moderate leaders wanted to develop public awareness about the unjust nature of British rule. They published newspapers, wrote articles, and showed how British rule was leading to the economic ruin of the country. They criticised British rule in their speeches and sent representatives to different parts of the country to mobilise public opinion. They felt that the British had respect for the ideals of freedom and justice, and so they would accept the just demands of Indians. What was necessary, therefore, was to express these demands, and make the government aware of the feelings of Indians.

Source 2

In pursuit of gold

This is what a Moderate leader, Dinshaw Wacha, wrote to Naoroji in 1887:

> Pherozeshah is nowadays too busy with his personal work ... They are already rich enough ... Mr. Telang too remains busy. I wonder how if all remain busy in the pursuit of gold can the progress of the country be advanced?

Activity

What problems regarding the early Congress does this comment highlight?

"Freedom is our birthright"

By the 1890s, many Indians began to raise questions about the political style of the Congress. In Bengal, Maharashtra and Punjab, leaders such as Bepin Chandra Pal, Bal Gangadhar Tilak and Lala Lajpat Rai were beginning to explore more radical objectives and methods. They criticised the Moderates for their "politics of prayers", and emphasised the importance of self-reliance and constructive work. They argued that people must rely on their own strength, not on the "good" intentions of the government; people must fight for *swaraj*. Tilak raised the slogan, "Freedom is my birthright and I shall have it!"

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In 1905, Viceroy Curzon partitioned Bengal. At that time Bengal was the biggest province of British India and included Bihar and parts of Orissa. The British argued for dividing Bengal for reasons of administrative convenience. But what did "administrative convenience" mean? Whose "convenience" did it represent? Clearly, it was closely tied to the interests of British officials and businessmen. Even so, instead of removing the non-Bengali areas from the province, the government separated East Bengal and merged it with Assam. Perhaps the main British motives were to curtail the influence of Bengali politicians and to split the Bengali people.

The partition of Bengal infuriated people all over India. All sections of the Congress – the Moderates and the Radicals, as they may be called – opposed it. Large public meetings and demonstrations were organised and novel methods of mass protest developed. The struggle that unfolded came to be known as the Swadeshi movement, strongest in Bengal but with echoes elsewhere too – in deltaic Andhra for instance, it was known as the Vandemataram Movement.



Fig. 3 – *Balgangadhar Tilak*
Notice the name of the newspaper that lies on the table. *Kesari*, a Marathi newspaper edited by Tilak, became one of the strongest critics of British rule.



Fig. 4 – *Thousands joined the demonstrations during the Swadeshi movement*

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Fig. 5 – *Lala Lajpat Rai*
A nationalist from Punjab, he was one of the leading members of the Radical group which was critical of the politics of petitions. He was also an active member of the Arya Samaj.

The Swadeshi movement sought to oppose British rule and encourage the ideas of self-help, *swadeshi* enterprise, national education, and use of Indian languages. To fight for *swaraj*, the radicals advocated mass mobilisation and boycott of British institutions and goods. Some individuals also began to suggest that **"revolutionary violence"** would be necessary to overthrow British rule.

The opening decades of the twentieth century were marked by other developments as well. A group of Muslim landlords and nawabs formed the All India Muslim League at Dacca in 1906. The League supported the partition of Bengal. It desired separate electorates for Muslims, a demand conceded by the government in 1909. Some seats in the **councils** were now reserved for Muslims who would be elected by Muslim voters. This tempted politicians to gather a following by distributing favours to their own religious groups.

Meanwhile, the Congress split in 1907. The Moderates were opposed to the use of boycott. They felt that it involved the use of force. After the split, the Congress came to be dominated by the Moderates with Tilak's followers functioning from outside. The two groups reunited in December 1915. Next year, the Congress and the Muslim League signed the historic Lucknow Pact and decided to work together for representative government in the country.

Revolutionary violence
The use of violence to make a radical change within society

Council – An appointed or elected body of people with an administrative, advisory or representative function

The Growth of Mass Nationalism

After 1919, the struggle against British rule gradually became a mass movement, involving peasants, tribals, students and women in large numbers and occasionally factory workers as well. Certain business groups too began to actively support the Congress in the 1920s. Why was this so?

The First World War altered the economic and political situation in India. It led to a huge rise in the defence expenditure of the Government of India. The government in turn increased taxes on individual incomes and business profits. Increased military expenditure and the demands for war supplies led to a sharp rise in prices which created great difficulties for the common people. On the other hand, business groups reaped fabulous profits from the war. As you have seen (Chapter 6), the war created a demand for industrial goods (jute bags, cloth, rails) and caused a decline of imports from other countries into India. So

Activity

Find out which countries fought the First World War.

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Indian industries expanded during the war, and Indian business groups began to demand greater opportunities for development.

The war also lead the British to expand their army. Villages were pressurised to supply soldiers for an alien cause. A large number of soldiers were sent to serve abroad. Many returned after the war with an understanding of the ways in which imperialist powers were exploiting the peoples of Asia and Africa and with a desire to oppose colonial rule in India.

Furthermore, in 1917, there was a revolution in Russia. News about peasants' and workers' struggles and ideas of socialism circulated widely, inspiring Indian nationalists.

The advent of Mahatma Gandhi

It is in these circumstances that Mahatma Gandhi emerged as a mass leader. As you may know, Gandhiji, aged 46, arrived in India in 1915 from South Africa. Having led Indians in that country in non-violent marches against racist restrictions, he was already a respected leader, known internationally. His South African campaigns had brought him in contact with various types of Indians: Hindus, Muslims, Parsis and Christians; Gujaratis, Tamils and north Indians; and upper-class merchants, lawyers and workers.

Mahatma Gandhi spent his first year in India travelling throughout the country, understanding the people, their needs and the overall situation. His earliest



Fig. 6 *– Founders of the Natal Congress, Durban, South Africa, 1895*

In 1895, along with other Indians, Mahatma Gandhi established the Natal Congress to fight against racial discrimination. Can you identify Gandhiji? He is standing at the centre in the row at the back, wearing a coat and tie.

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interventions were in local movements in Champaran, Kheda and Ahmedabad where he came into contact with Rajendra Prasad and Vallabhbhai Patel. In Ahmedabad, he led a successful millworkers' strike in 1918.

Let us now focus in some detail on the movements organised between 1919 and 1922.

The Rowlatt Satyagraha

In 1919, Gandhiji gave a call for a *satyagraha* against the Rowlatt Act that the British had just passed. The Act curbed fundamental rights such as the freedom of expression and strengthened police powers. Mahatma Gandhi, Mohammad Ali Jinnah and others felt that the government had no right to restrict people's basic freedoms. They criticised the Act as "devilish" and tyrannical. Gandhiji asked the Indian people to observe 6 April 1919 as a day of non-violent opposition to this Act, as a day of "humiliation and prayer" and *hartal* (strike). *Satyagraha Sabhas* were set up to launch the movement.

Activity

Find out about the Jallianwala Bagh massacre. What is Jallianwala Bagh? What atrocities were committed there? How were they committed?



***Fig. 7** – The walled compound in which General Dyer opened fire on a gathering of people*
The people are pointing to the bullet marks on the wall.

The Rowlatt Satyagraha turned out to be the first all-India struggle against the British government although it was largely restricted to cities. In April 1919, there were a number of demonstrations and *hartals* in the country and the government used brutal measures to suppress them. The Jallianwala Bagh atrocities, inflicted by General Dyer in Amritsar on Baisakhi day (13 April), were a part of this repression. On learning about the massacre, Rabindranath Tagore expressed the pain and anger of the country by renouncing his **knighthood**.

Knighthood – An honour granted by the British Crown for exceptional personal achievement or public service

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During the Rowlatt Satyagraha, the participants tried to ensure that Hindus and Muslims were united in the fight against British rule. This was also the call of Mahatma Gandhi who always saw India as a land of *all* the people who lived in the country – Hindus, Muslims and those of other religions. He was keen that Hindus and Muslims support each other in any just cause.

Khilafat agitation and the Non-Cooperation Movement

The Khilafat issue was one such cause. In 1920, the British imposed a harsh treaty on the Turkish Sultan or Khalifa. People were furious about this as they had been about the Jallianwala massacre. Also, Indian Muslims were keen that the Khalifa be allowed to retain control over Muslim sacred places in the erstwhile Ottoman Empire. The leaders of the Khilafat agitation, Mohammad Ali and Shaukat Ali, now wished to initiate a full-fledged Non-Cooperation Movement. Gandhiji supported their call and urged the Congress to campaign against "Punjab wrongs" (Jallianwala massacre), the Khilafat wrong and demand *swaraj*.

The Non-Cooperation Movement gained momentum through 1921–22. Thousands of students left government-controlled schools and colleges. Many lawyers such as Motilal Nehru, C.R. Das, C. Rajagopalachari and Asaf Ali gave up their practices. British titles were surrendered and legislatures boycotted. People lit public bonfires of foreign cloth. The imports of foreign cloth fell drastically between 1920 and 1922. But all this was merely the tip of the iceberg. Large parts of the country were on the brink of a formidable revolt.

People's initiatives

In many cases, people resisted British rule non-violently. In others, different classes and groups, interpreting Gandhiji's call in their own manner, protested in ways that were not in accordance with his ideas. In either case, people linked their movements to local grievances. Let us look at a few examples.

In Kheda, Gujarat, Patidar peasants organised non-violent campaigns against the high land revenue demand of the British. In coastal Andhra and interior Tamil Nadu, liquor shops were **picketed**. In the Guntur district of Andhra Pradesh, tribals and poor peasants staged a number of "forest *satyagrahas*", sometimes sending their cattle into forests without paying grazing fee. They were protesting because the colonial state

Source 3

The eternal law of suffering

What did Mahatma Gandhi mean by *ahimsa* (non-violence)? How could *ahimsa* become the basis of struggle? This is what Gandhiji said:

> Non-violence comes to us through doing good continually without the slightest expectation of return, ... That is the indispensable lesson in non-violence ... In South Africa ... I succeeded in learning the eternal law of suffering as the only remedy for undoing wrong and injustice. It means positively the law of non-violence. You have to be prepared to suffer cheerfully at the hands of all and sundry and you will wish ill to no one, not even to those who may have wronged you.
>
> *Mahatma Gandhi, 12 March 1938*

Picket – People protesting outside a building or shop to prevent others from entering

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Mahants – Religious functionaries of Sikh gurdwaras

Illegal eviction – Forcible and unlawful throwing out of tenants from the land they rent

had restricted their use of forest resources in various ways. They believed that Gandhiji would get their taxes reduced and have the forest regulations abolished. In many forest villages, peasants proclaimed *swaraj* and believed that "Gandhi Raj" was about to be established.

In Sind (now in Pakistan), Muslim traders and peasants were very enthusiastic about the Khilafat call. In Bengal too, the Khilafat-Non-Cooperation alliance gave enormous communal unity and strength to the national movement.

In Punjab, the Akali agitation of the Sikhs sought to remove corrupt ***mahants*** – supported by the British – from their gurdwaras. This movement got closely identified with the Non-Cooperation Movement. In Assam, tea garden labourers, shouting *"Gandhi Maharaj ki Jai"*, demanded a big increase in their wages. They left the British-owned plantations amidst declarations that they were following Gandhiji's wish. Interestingly, in the Assamese Vaishnava songs of the period, the reference to Krishna was substituted by "Gandhi Raja".

The people's Mahatma

We can see from the above that sometimes people thought of Gandhiji as a kind of messiah, as someone who could help them overcome their misery and poverty. Gandhiji wished to build class unity, not class conflict, yet peasants could imagine that he would help them in their fight against zamindars, and agricultural labourers believed he would provide them land. At times, ordinary people credited Gandhiji with their *own* achievements. For instance, at the end of a powerful movement, peasants of Pratapgarh in the United Provinces (now Uttar Pradesh) managed to stop **illegal eviction** of tenants; but they felt it was Gandhiji who had won this demand for them. At other times, using Gandhiji's name, tribals and peasants undertook actions that did not conform to Gandhian ideals.

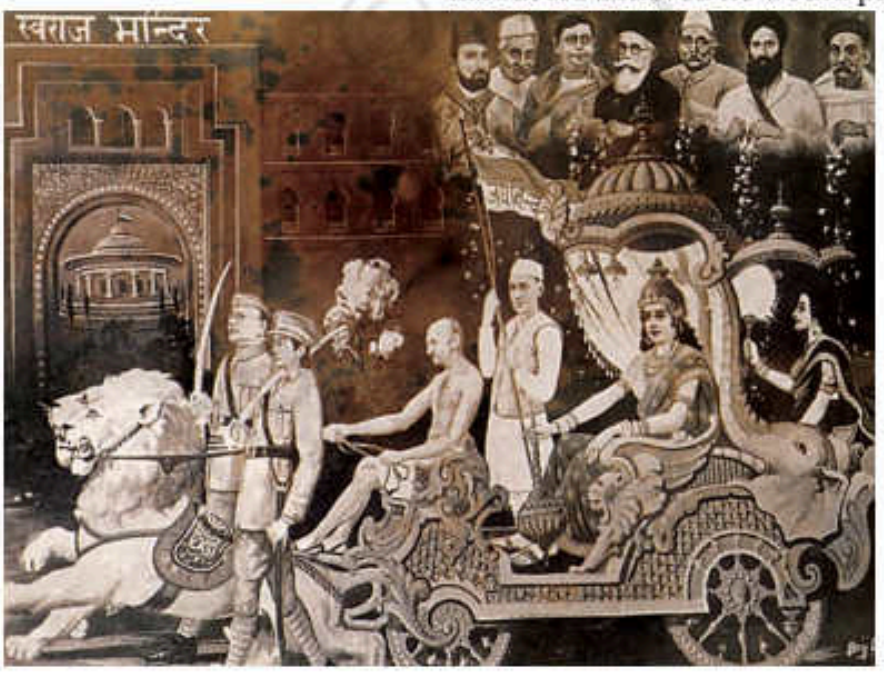


Fig. 8 – *A popular representation of Mahatma Gandhi*

In popular images too Mahatma Gandhi is often shown as a divine being occupying a place within the pantheon of Indian gods. In this image, he is driving Krishna's chariot, guiding other nationalist leaders in the battle against the British.

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Source 4

"It was he who got *bedakhli* stopped in Pratapgarh"

The following is an extract from a CID report on the kisan movement in Allahabad district, January 1921:

> The currency which Mr. Gandhi's name has acquired even in the remotest villages is astonishing. No one seems to know quite who or what he is, but it is an accepted fact that what he says is so, and what he orders must be done. He is a Mahatma or *sadhu*, a Pundit, a Brahmin who lives at Allahabad, even a *deota* … the real power of his name is to be traced back to the idea that it was he who got *bedakhli* [illegal eviction] stopped in Pratapgarh … as a general rule, Gandhi is not thought of as being antagonistic to Government, but only to the *zamindars* … We are for Gandhiji and the Sarkar.

Activity

Read Source 4. According to this report, how did people view Mahatma Gandhi? Why do you think they felt that he was opposed to zamindars but not to the government? Why do you think they were in favour of Gandhiji?

The happenings of 1922–1929

Mahatma Gandhi, as you know, was against violent movements. He abruptly called off the Non-Cooperation Movement when in February 1922, a crowd of peasants set fire to a police station in Chauri Chaura. Twenty- two policemen were killed on that day. The peasants were provoked because the police had fired on their peaceful demonstration.

Once the Non-Cooperation movement was over, Gandhiji's followers stressed that the Congress must undertake constructive work in the rural areas. Other leaders such as Chitta Ranjan Das and Motilal Nehru argued that the party should fight elections to the councils and enter them in order to influence government policies. Through sincere social work in villages in the mid-1920s, the Gandhians were able to extend their support base. This proved to be very useful in launching the Civil Disobedience movement in 1930.

Two important developments of the mid-1920s were the formation of the Rashtriya Swayamsevak Sangh (RSS), a Hindu organisation, and the Communist Party of India. These parties have held very different ideas about the kind of country India should be. Find out about their ideas with the help of your teacher. The revolutionary nationalist Bhagat Singh too was active in this period.



***Fig. 9** – Chitta Ranjan Das*
A major figure in the freedom movement, Das was a lawyer from East Bengal. He was especially active in the Non-Cooperation Movement.

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Fig. 10 – *Demonstrators oppose the Simon Commission*

In 1927, the British government in England decided to send a commission headed by Lord Simon to decide India's political future. The Commission had no Indian representative. The decison created an outrage in India. All political groups decided to boycott the Commission. When the Commission arrived, it was met with demonstrations with banners saying "Simon Go Back".

The decade closed with the Congress resolving to fight for *Purna Swaraj* (complete independence) in 1929 under the presidentship of Jawaharlal Nehru. Consequently, "Independence Day" was observed on 26 January 1930 all over the country.



Fig. 11 – *Bhagat Singh*

"It takes a loud voice to make the deaf hear. *Inquilab Zindabad !*"

Revolutionary nationalists such as Bhagat Singh, Chandra Shekhar Azad, Sukhdev and others wanted to fight against the colonial rule and the rich exploiting classes through a revolution of workers and peasants. For this purpose, they founded the Hindustan Socialist Republican Association (HSRA) in 1928 at Ferozeshah Kotla in Delhi. On 17 December, 1928, Bhagat Singh, Azad and Rajguru assassinated Saunders, a police officer who was involved in the lathi-charge that had caused the death of Lala Lajpat Rai.

On 8 April, 1929, Bhagat Singh and B.K. Dutt threw a bomb in the Central Legislative Assembly. The aim, as their leaflet explained, was not to kill but "to make the deaf hear", and to remind the foreign government of its callous exploitation.

Bhagat Singh, Sukhdev and Rajguru were executed on March 23, 1931. Bhagat Singh's age at that time was only 23.

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The March to Dandi

Purna Swaraj would never come on its own. It had to be fought for. In 1930, Gandhiji declared that he would lead a march to break the salt law. According to this law, the state had a monopoly on the manufacture and sale of salt. Mahatma Gandhi along with other nationalists reasoned that it was sinful to tax salt since it is such an essential item of our food. The Salt March related the general desire of freedom to a specific grievance shared by everybody, and thus did not divide the rich and the poor.

Gandhiji and his followers marched for over 240 miles from Sabarmati to the coastal town of Dandi where they broke the government law by gathering natural salt found on the seashore, and boiling sea water to produce salt.



Fig. 12 – *Mahatma Gandhi breaking the salt law by picking up a lump of natural salt, Dandi, 6 April 1930*

Women in the freedom struggle: Ambabai from Karnataka

Women from diverse backgrounds participated in the national movement. Young and old, single and married, they came from rural and urban areas, from both conservative and liberal homes. Their involvement was significant for the freedom struggle, for the women's movement, and for themselves personally.

Both British officials and Indian nationalists felt that women's participation gave the national struggle an immense force. Participation in the freedom movement brought women out of their homes. It gave them a place in the professions, in the governance of India, and it could pave the way for equality with men.

What such participation meant for women is best recounted by them. Ambabai of Karnataka had been married at age twelve. Widowed at sixteen, she picketed foreign cloth and liquor shops in Udipi. She was arrested, served a sentence and was rearrested. Between prison terms she made speeches, taught spinning, and organised *prabhat pheris*. Ambabai regarded these as the happiest days of her life because they gave it a new purpose and commitment.

Women, however, had to fight for their right to participate in the movement. During the Salt Satyagraha, for instance, even Mahatma Gandhi was initially opposed to women's participation. Sarojini Naidu had to persuade him to allow women to join the movement.

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Fig. 13 – *Sarojini Naidu with Mahatma Gandhi, Paris, 1931*
Active in the national movement since the early 1920s, Naidu was a significant leader of the Dandi March. She was the first Indian woman to become President of the Indian National Congress (1925).

Peasants, tribals and women participated in large numbers. A business federation published a pamphlet on the salt issue. The government tried to crush the movement through brutal action against peaceful *satyagrahis*. Thousands were sent to jail.

The combined struggles of the Indian people bore fruit when the Government of India Act of 1935 prescribed **provincial autonomy** and the government announced elections to the provincial legislatures in 1937. The Congress formed governments in 7 out of 11 provinces.

In September 1939, after two years of Congress rule in the provinces, the Second World War broke out. Critical of Hitler, Congress leaders were ready to support the British war effort. But in return they wanted that India be granted independence after the war. The British refused to concede the demand. The Congress ministries resigned in protest.

Provincial autonomy
Capacity of the provinces to make relatively independent decisions while remaining within a federation

Source 5

Veer Lakhan Nayak was hanged

Baji Mohammed, President of the Nabrangpur Congress, Orissa in the 1930s, reports:

> On August 25, 1942 … nineteen people died on the spot in police firing at Paparandi in Nabarangpur. Many died thereafter from their wounds. Over 300 were injured. More than a thousand were jailed in Koraput district. Several were shot or executed. Veer Lakhan Nayak (a legendary tribal leader who defied the British) was hanged.

Nayak, Baji tells us, was not worried about being executed, only sad that he would not live to see freedom's dawn.

Baji Mohammad mobilised 20,000 people to join the national struggle. He offered *satyagraha* many times over. He participated in protests against the Second World War and in the Quit India movement, and served long jail terms.

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Fig. 14 – *Quit India movement, August 1942*
Demonstrators clashed with the police everywhere. Many thousands were arrested, over a thousand killed, many more were injured.

Bose and the INA



Fig. 15 – *Subhas Chandra Bose*

A radical nationalist, with socialist leanings, Bose did not share Gandhiji's ideal of *ahimsa,* though he respected him as the "Father of the Nation". In January 1941, he secretly left his Calcutta home, went to Singapore, via Germany, and raised the Azad Hind Fauj or the Indian National Army (INA). To free India from British control, in 1944, the INA tried to enter India through Imphal and Kohima but the campaign failed. The INA members were imprisoned and tried. People across the country, from all walks of life, participated in the movement against the INA trials.

Quit India and Later

Mahatma Gandhi decided to initiate a new phase of movement against the British in the middle of the Second World War. The British must quit India immediately, he told them. To the people he said, "do or die" in your effort to fight the British – but you must fight non-violently. Gandhiji and other leaders were jailed at once but the movement spread. It specially attracted peasants and the youth who gave up their studies to join it. Communications and symbols of state authority were attacked all over the country. In many areas the people set up their own governments.

The first response of the British was severe repression. By the end of 1943, over 90,000 people were arrested, and around 1,000 killed in police firing. In many areas, orders were given to machine-gun crowds from airplanes. The rebellion, however, ultimately brought the Raj to its knees.

Towards Independence and Partition

Meanwhile, in 1940 the Muslim League had moved a resolution demanding "Independent States" for Muslims in the north-western and eastern areas of the country. The resolution did not mention partition or Pakistan. Why did the League ask for an autonomous arrangement for the Muslims of the subcontinent?



Fig. 16 – *Jawaharlal Nehru listens to Mahatma Gandhi before the Bombay session of the Congress, July 1946*
Gandhiji's disciple, a Congress Socialist, and an internationalist, Nehru was a leading architect of the national movement and of free India's economy and polity.

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Fig. 17 – *Maulana Azad with other members at the Congress Working Committee, Sevagram, 1942*

Azad was born in Mecca to a Bengali father and an Arab mother. Well-versed in many languages, Azad was a scholar of Islam and an exponent of the notion of *wahadat-i-deen*, the essential oneness of all religions. An active participant in Gandhian movements and a staunch advocate of Hindu-Muslim unity, he was opposed to Jinnah's two-nation theory.

Fig. 18 – *Chakravarti Rajagopalachari speaking to Gandhiji before the Gandhi-Jinnah talks, 1944*

A veteran nationalist and leader of the Salt Satyagraha in the south, C. Rajagopalachari, popularly known as Rajaji, served as member of the Interim Government of 1946 and as free India's first Indian Governor-General.





Fig. 19 – *Sardar Vallabhbhai Patel played an important role in the negotiations for independence during 1945–47*

Patel hailed from an impoverished peasant-proprietor family of Nadiad, Gujarat. A foremost organiser of the freedom movement from 1918 onwards, Patel served as President of the Congress in 1931.



Fig. 20 – *Mohammad Ali Jinnah with Mahatma Gandhi, Bombay, September 1944*

An ambassador of Hindu-Muslim unity until 1920, Jinnah played an important role in the making of the Lucknow Pact. He reorganised the Muslim League after 1934, and became the major spokesperson for the demand for Pakistan.

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From the late 1930s, the League began viewing the Muslims as a separate "nation" from the Hindus. In developing this notion, it may have been influenced by the history of tension between some Hindu and Muslim groups in the 1920s and 1930s. More importantly, the provincial elections of 1937 seemed to have convinced the League that Muslims were a minority, and they would always have to play second fiddle in any democratic structure. It feared that Muslims may even go unrepresented. The Congress's rejection of the League's desire to form a joint Congress-League government in the United Provinces in 1937 also annoyed the League.

The Congress's failure to mobilise the Muslim masses in the 1930s allowed the League to widen its social support. It sought to enlarge its support in the early 1940s when most Congress leaders were in jail. At the end of the war in 1945, the British opened negotiations between the Congress, the League and themselves for the independence of India. The talks failed because the League saw itself as the sole spokesperson of India's Muslims. The Congress could not accept this claim since a large number of Muslims still supported it.

Elections to the provinces were again held in 1946. The Congress did well in the **"General" constituencies** but the League's success in the seats reserved for Muslims was spectacular. It persisted with its demand for "Pakistan". In March 1946, the British cabinet sent a three-member mission to Delhi to examine this demand and to suggest a suitable political framework for a free India. This mission suggested that India should remain united and constitute itself as a loose confederation with some autonomy for Muslim-majority areas. But it could not get the Congress and the Muslim League to agree to specific details of the proposal. Partition now became more or less inevitable.

After the failure of the Cabinet Mission, the Muslim League decided on mass agitation for winning its Pakistan demand. It announced 16 August 1946 as "Direct Action Day". On this day riots broke out in Calcutta, lasting several days and resulting in the death of thousands of people. By March 1947, violence spread to different parts of northern India.

Many hundred thousand people were killed and numerous women had to face untold brutalities during the Partition. Millions of people were forced to flee their homes. Torn asunder from their homelands, they were reduced to being refugees in alien

"General" constituencies
Election districts with no reservations for any religious or other community



Fig. 21 *– Khan Abdul Ghaffar Khan, the Pashtun leader from the North West Frontier Province, with his colleagues at a peace march through Bihar, March 1947*

Also known as Badshah Khan, he was the founder of the Khudai Khidmatgars, a powerful non-violent movement among the Pathans of his province. Badshah Khan was strongly opposed to the Partition of India. He criticised his Congress colleagues for agreeing to the 1947 division.



Fig. 22 *– Refugees from riot-torn Punjab gather in New Delhi, in search of shelter and food*

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lands. Partition also meant that India changed, many of its cities changed, and a new country – Pakistan – was born. So, the joy of our country's independence from British rule came mixed with the pain and violence of Partition.

Let's imagine

Imagine that you are involved in the Indian national movement. Based on your reading of this chapter, briefly discuss your preferred methods of struggle and your vision of a free India.

Let's recall

1. Why were people dissatisfied with British rule in the 1870s and 1880s?
2. Who did the Indian National Congress wish to speak for?
3. What economic impact did the First World War have on India?
4. What did the Muslim League resolution of 1940 ask for?

Let's discuss

5. Who were the Moderates? How did they propose to struggle against British rule?
6. How was the politics of the Radicals within the Congress different from that of the Moderates?
7. Discuss the various forms that the Non-Cooperation Movement took in different parts of India. How did the people understand Gandhiji?
8. Why did Gandhiji choose to break the salt law?
9. Discuss those developments of the 1937–47 period that led to the creation of Pakistan.

Let's do

10. Find out how the national movement was organised in your city, district, area or state. Who participated in it and who led it? What did the movement in your area achieve?
11. Find out more about the life and work of any two participants or leaders of the national movement and write a short essay about them. You may choose a person not mentioned in this chapter.

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Notes

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Notes

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Distribution of Minerals

Do you know?
You can always tell if a rock contains copper because then the rock looks blue in colour.

Minerals occur in different types of rocks. Some are found in igneous rocks, some in metamorphic rocks while others occur in sedimentary rocks. Generally, metallic minerals are found in igneous and metamorphic rock formations that form large plateaus. Iron-ore in north Sweden, copper and nickel deposits in Ontario, Canada, iron, nickel, chromites and platinum in South Africa are examples of minerals found in igneous and metamorphic rocks. Sedimentary rock formations of plains and young fold mountains contain non-metallic minerals like limestone. Limestone deposits of Caucasus region of France, manganese deposits of Georgia and Ukraine and phosphate beds of Algeria are some examples. Mineral fuels such as coal and petroleum are also found in the sedimentary strata.

Asia

China and India have large iron ore deposits. The continent produces more than half of the world's tin.

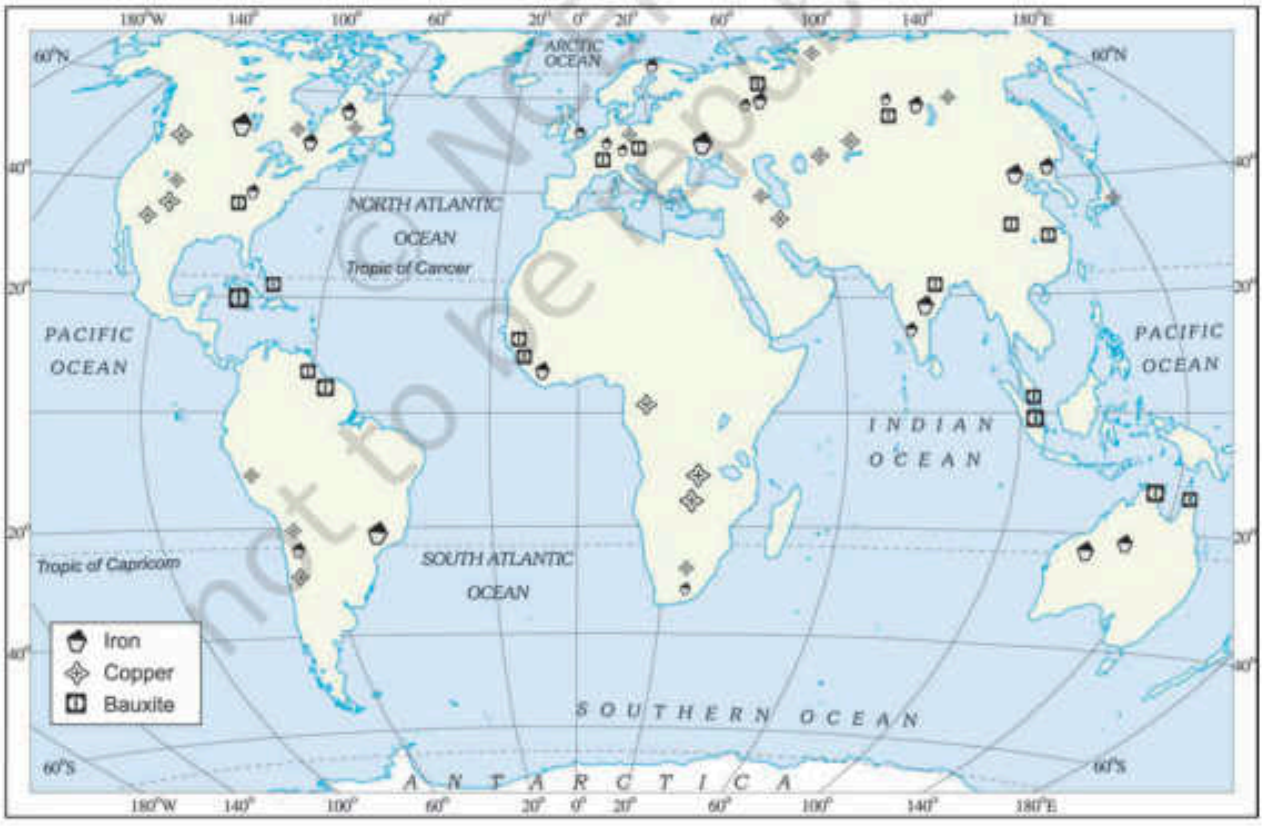


Fig. 3.5: World: Distribution of Iron, Copper and Bauxite

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China, Malaysia and Indonesia are among the world's leading tin producers. China also leads in production of lead, antimony and tungsten. Asia also has deposits of manganese, bauxite, nickel, zinc and copper.

Europe

Europe is the leading producer of iron-ore in the world. The countries with large deposits of iron ore are Russia, Ukraine, Sweden and France. Minerals deposits of copper, lead, zinc, manganese and nickel are found in eastern Europe and European Russia.

Do you know?
Switzerland has no known mineral deposit in it.

North America

The mineral deposits in North America are located in three zones: the Canadian region north of the Great Lakes, the Appalachian region and the mountain ranges of the west. Iron ore, nickel, gold, uranium and copper are mined in the Canadian Shield Region, coal in the Appalachians region. Western Cordilleras have vast deposits of copper, lead, zinc, gold and silver.

Let's do
Identify the Canadian Shield, the Appalachians, Western Cordilleras and Lake Superior with the help of an atlas.

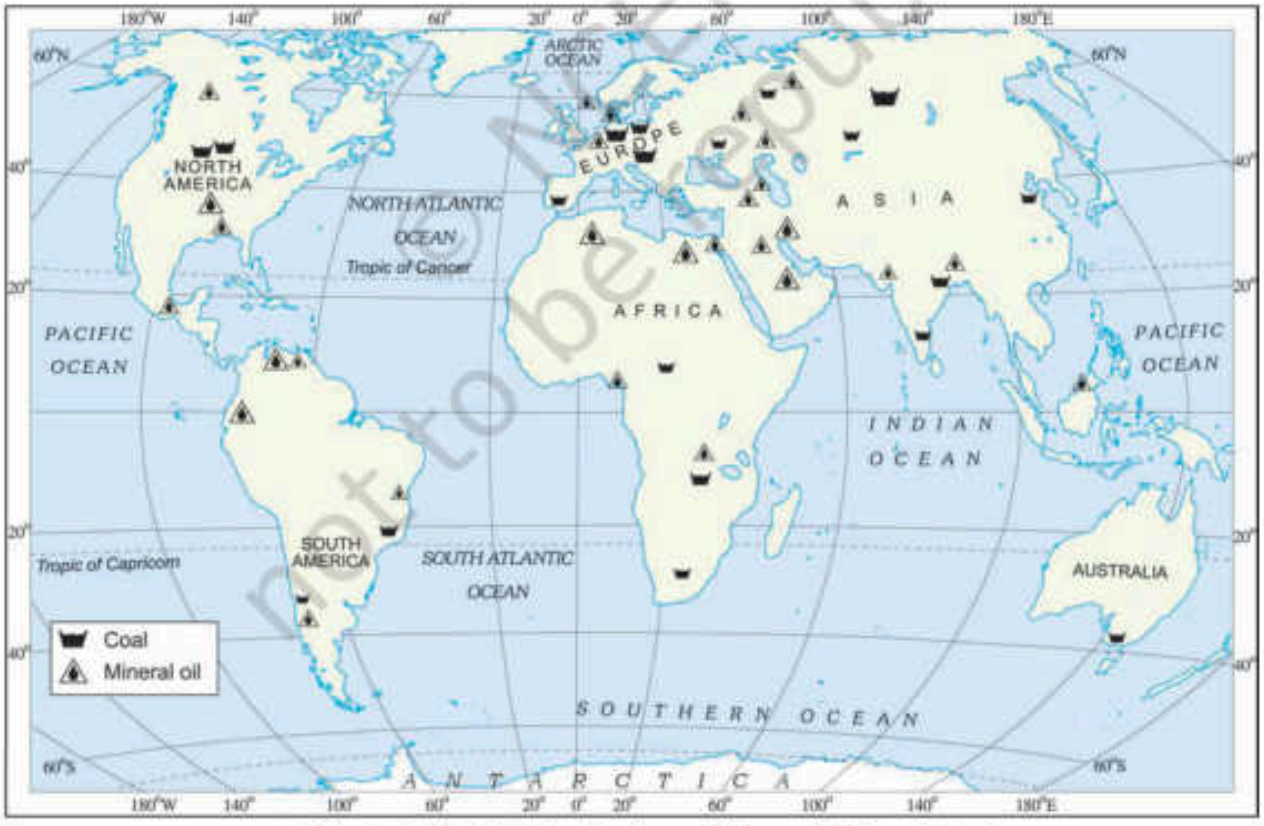


Fig 3.6: World: Distribution of Mineral Oil and Coal

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South America

Brazil is the largest producer of high grade iron-ore in the world. Chile and Peru are leading producers of copper. Brazil and Bolivia are among the world's largest producers of tin. South America also has large deposits of gold, silver, zinc, chromium, manganese, bauxite, mica, platinum, asbestos and diamond. Mineral oil is found in Venezuela, Argentina, Chile, Peru and Columbia.

Do you know?

- A green diamond is the rarest diamond.
- The oldest rocks in the world are in Western Australia. They date from 4,300 million years ago, only 300 million years after the earth was formed.

Africa

Africa is rich in mineral resources. It is the world's largest producer of diamonds, gold and platinum. South Africa, Zimbabwe and Zaire produce a large portion of the world's gold. The other minerals found in Africa are copper, iron ore, chromium, uranium, cobalt and bauxite. Oil is found in Nigeria, Libya and Angola.

Australia

Australia is the largest producer of bauxite in the world. It is a leading producer of gold, diamond, iron ore, tin and nickel. It is also rich in copper, lead, zinc and manganese. Kalgoorlie and Coolgardie areas of western Australia have the largest deposits of gold.

Antarctica

The geology of Antarctica is sufficiently well known to predict the existence of a variety of mineral deposits, some probably large. Significant size of deposits of coal in the Transantarctic Mountains and iron near the Prince Charles Mountains of East Antarctica is forecasted. Iron ore, gold, silver and oil are also present in commercial quantities.

Uses of Minerals

Minerals are used in many industries. Minerals which are used for gems are usually hard. These are then set in various styles for jewellery. Copper is another metal used in everything from coins to pipes. Silicon, used in the computer industry is obtained from quartz. Aluminum obtained from its ore bauxite is used in automobiles and airplanes, bottling industry, buildings and even in kitchen cookware.

Let's do

List uses of any five minerals.



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CONVENTIONAL SOURCES OF ENERGY

Oil

Advantage	Disadvantage
• Easier to transport (tankers) • Basis of petro-chemical industry	• Depletion of oxygen due to oil spillage and gas leakage • Pollutants released caused acid rain • Exploration of new fuel is not easy

Natural Gas

Advantage
• Easier to transport (Pipelines) • Cleaner than oil and coal • Cheaper than oil

Fire Wood

Advantage	Disadvantage
• Easy access • Provides energy to a large number of people	• Collection is time consuming • Polluting • Promoting green house effect • Deforestation

Coal

Advantage	Disadvantage
• Extensively available • Efficient conversion to electricity	• Polluting source • Bulky to transport

Hydel Power

Advantage	Disadvantage
• Non-polluting • Promotes irrigation and fishing • Cheap	• Displacement of local community • Inundates low lying regions • Expensive to setup

Fig 3.8: Conventional Sources of Energy



Fig 3.9: A woman carrying firewood in North East India

conventional energy. The reserves of these minerals are limited. The rate at which the growing world population is consuming them is far greater than the rate of their formation. So, these are likely to be exhausted soon.

Coal

This is the most abundantly found fossil fuel. It is used as a domestic fuel, in industries such as iron and steel, steam engines and to generate electricity. Electricity from coal is called **thermal power**. The coal which we are using today was formed millions of years ago when giant ferns and swamps got buried under the layers of earth. Coal is therefore referred to as **Buried Sunshine**.



Fig 3.10: A view of a Thermal Power Station

The leading coal producers of the world are China, USA, Germany, Russia, South Africa and France. The coal producing areas of India are Raniganj, Jharia, Dhanbad and Bokaro in Jharkhand.

Petroleum

The petrol that keeps your car running as well as the oil that keeps your cycle from squeaking, both began as a thick black liquid called Petroleum. It is found between the layers of rocks and is drilled from oil fields located in off-shore and coastal areas. This is then sent to refineries which process the crude oil and produce a variety of products like diesel, petrol, kerosene, wax, plastics and lubricants. Petroleum and its derivatives are called **Black Gold** as they are very valuable. The chief petroleum producing countries are Iran, Iraq, Saudi Arabia and Qatar. The other major producers are USA, Russia, Venezuela, and Algeria. The leading producers in India are Digboi in Assam, Bombay High in Mumbai and the deltas of Krishna and Godavari rivers.



Fig 3.11: Crude Oil

Word Origin

The word petroleum is derived from Latin words –*Petra* meaning rock, *oleum* meaning oil. So, petroleum means rock oil.

Natural Gas

Natural gas is found with petroleum deposits and is released when crude oil is brought to the surface. It can be used as a domestic and industrial fuel. Russia, Norway, UK and the Netherlands are the major producers of natural gas.

In India Jaisalmer, Krishna Godavari delta, Tripura and some areas off shore in Mumbai have natural gas resources. Very few countries in the world have sufficient natural gas reserves of their own.

The sharp increase in our consumption of fossil fuels has led to their depletion at an alarming rate. The toxic pollutants released from burning these fuels are also a cause for concern. Unchecked burning of fossil fuel is like an unchecked dripping tap which will eventually run dry. This has led to the tapping of various non-conventional sources of energy that are cleaner alternatives to fossil fuels.

Do you know?

Compressed natural gas (CNG) is a popular eco-friendly automobile fuel as it causes less pollution than petroleum and diesel.

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Do you know?
Norway was the first country in the world to devlop hydroelectricity.

Hydel Power

Rain water or river water stored in dams is made to fall from heights. The falling water flows through pipes inside the dam over turbine blades placed at the bottom of the dam. The moving blades then turn the generator to produce electricity. This is called hydro electricity. The water discharged after the generation of electricity is used for irrigation. One fourth of the world's electricity is produced by hydel power. The leading producers of hydel power in the world are Paraguay, Norway, Brazil, and China. Some important hydel power stations in India are Bhakra Nangal, Gandhi Sagar, Nagarjunsagar and Damodar valley projects.

Do you know?
The site of the world's first solar and wind powered bus shelter is in Scotland.

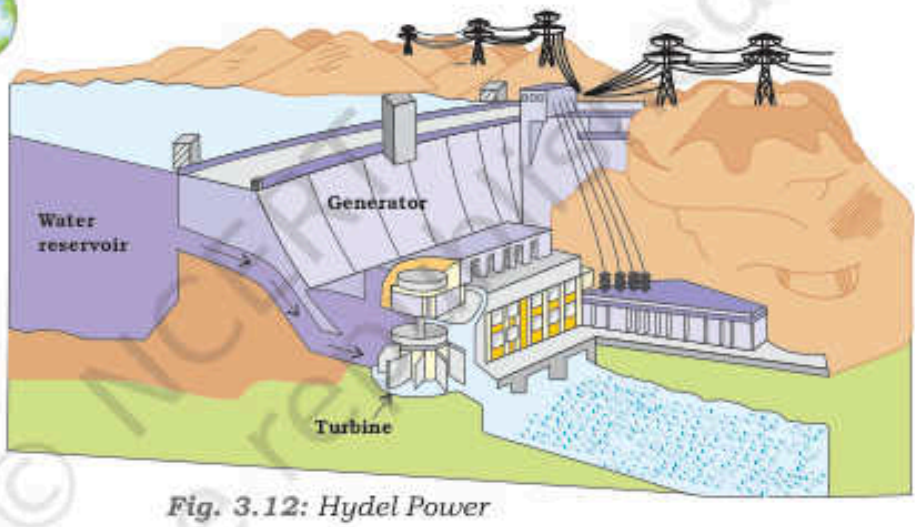


Fig. 3.12: Hydel Power

Non-conventional Sources of Energy

The increasing use of fossil fuels is leading to its shortage. It is estimated that if the present rate of consumption continues, the reserves of these fuel will get exhausted. Moreover, their use also causes environmental pollution. Therefore, there is need for using non-conventional sources such as solar energy, wind energy, tidal energy which are renewable.



Fig. 3.13: Salal Hydroelectric Project Jammu and Kashmir

Solar energy

Sun's heat and light energy can be felt by us every day. Solar energy trapped from the sun can be used in solar cells to produce electricity. Many of these cells are joined into solar panels to generate

Cultural: Places with religion or cultural significance attract people. Varanasi, Jerusalem and Vatican city are some examples.

Economic: Industrial areas provide employment opportunities. Large number of people are attracted to these areas. Osaka in Japan and Mumbai in India are two densely populated areas.

Glossary

Life expectancy
It is the number of years that an average person can expect to live.

Population Change

The population change refers to change in the number of people during a specific time. The world population has not been stable. It has increased manifold as seen in the Fig 5.3. Why? This is actually due to changes in the number of births and deaths. For an extremely long period of human history, until the 1800s, the world's population grew steadily but slowly. Large numbers of babies were born, but they died early too. This was as there were no proper health facilities. Sufficient food was not available for all the people. Farmers were not able to produce enough to meet the food requirements of all the people. As a result the total increase in population was very low.

In 1804, the world's population reached one billion. A hundred and fifty five years later, in 1959, the world's population reached 3 billion. This is often called population explosion. In 1999, 40 years later, the population doubled to 6 billion. The main reason for this growth was that with better food supplies and medicine, deaths were reducing, while the number of births still remained fairly high.

Births are usually measured using the **birth rate** i.e. the number of live births per 1,000 people. Deaths are usually measured using the **death rate** i.e. the number of deaths per 1,000 people. **Migrations** is the movement of people in and out of an area.

Births and deaths are the natural causes of population change. The difference between the birth rate and the death rate of a country is called the **natural growth rate**.

The population increase in the world is mainly due to rapid increase in natural growth rate.

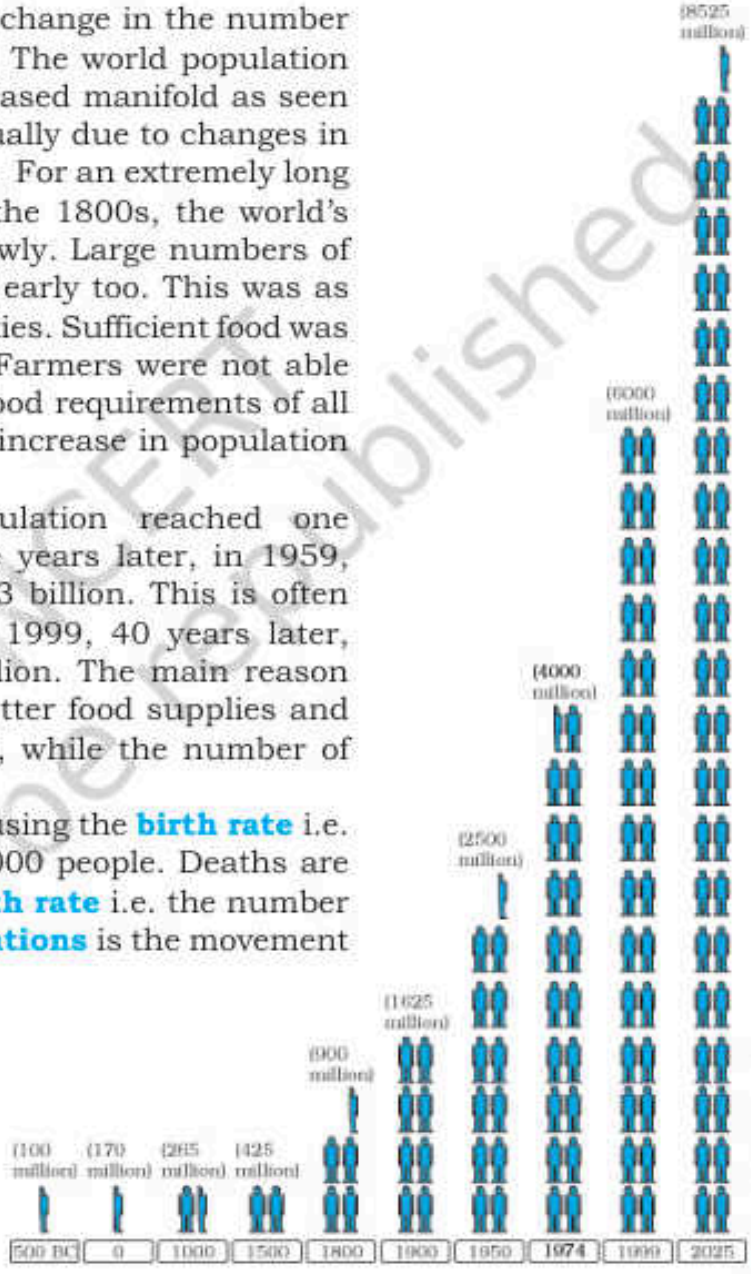


Fig 5.3: World Population Growth

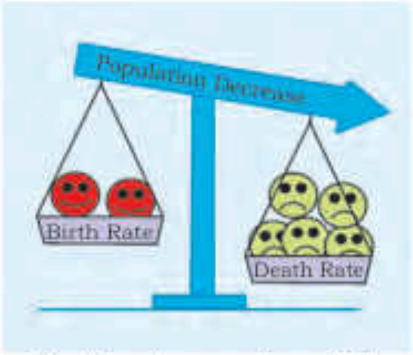
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Birth rate more than death rate: population increase



Birth rate and death rate same: population stays the same



Death rate more than birth rate:population decreases

***Fig 5.4:** Balance of Population*

Glossary

Immigration When a person enters a new country.

Emigration When a person leaves a country.

Migration is another way by which population size changes. People may move within a country or between countries. **Emigrants** are people who leave a country; **Immigrants** are those who arrive in a country.

Countries like the United States of America and Australia have gained in-numbers by **in-migration** or **immigration**. Sudan is an example of a country that has experienced a loss in population numbers due to **out-migration** or **emigration**.

The general trend of international migrations is from the less developed nations to the more developed nations in search of better employment opportunities. Within countries large number of people may move from the rural to urban areas in search of employment, education and health facilities.

Patterns of Population Change

Rates of population growth vary across the world (Fig 5.5). Although, the world's total population is rising rapidly, not all countries are experiencing this growth. Some countries like Kenya have high population growth rates. They had both high birth rates and death rates. Now, with improving health care, death rates have fallen, but birth rates still remain high leading to high growth rates.

In other countries like United Kingdom, population growth is slowing because of both low death and low birth rates.

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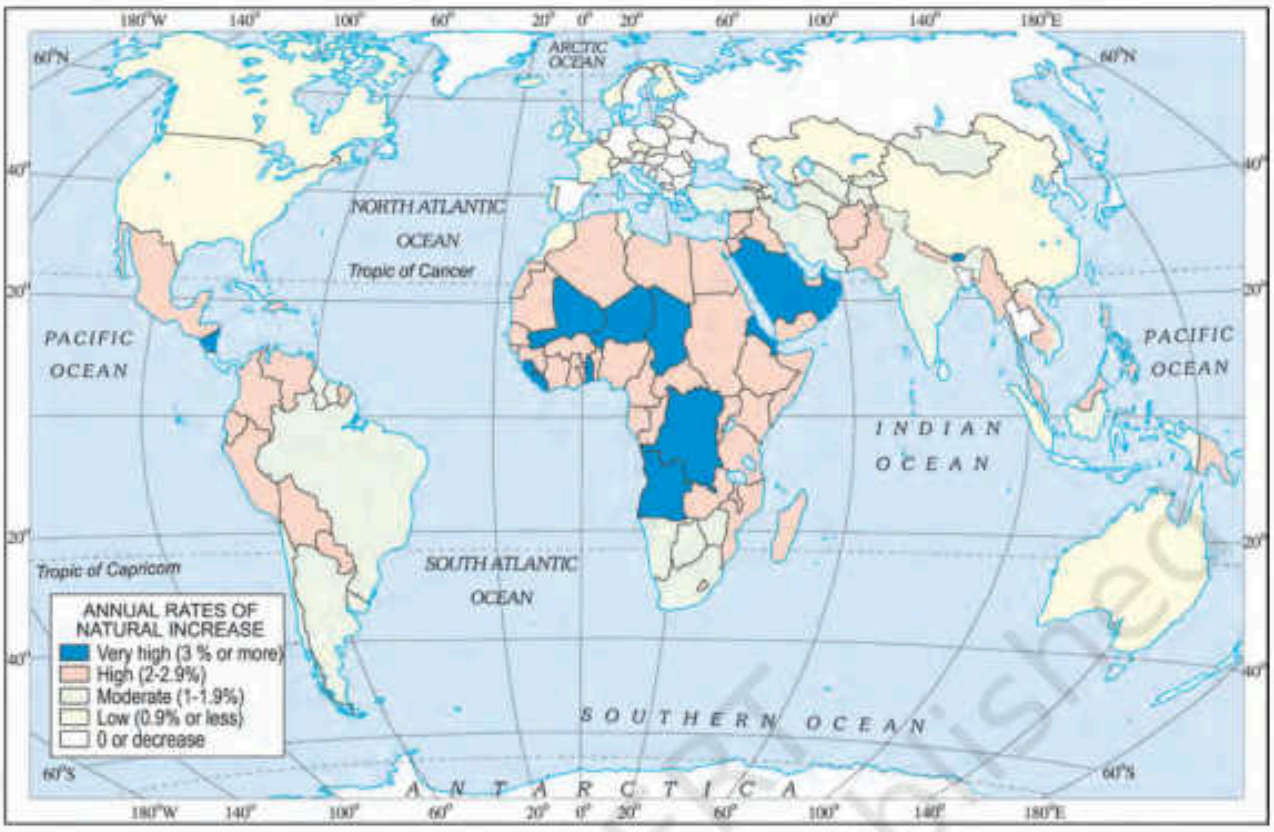


Fig. 5.5: World: Differing rates of population growth

Population Composition

How crowded a country is, has little to do with its level of economic development. For example, both Bangladesh and Japan are very densely populated but Japan is far more economically developed than Bangladesh.

To understand the role of people as a resource, we need to know more about their qualities. People vary greatly in their age, sex, literacy level, health condition, occupation and income level. It is essential to understand these characteristics of the people. Population composition refers to the structure of the population.

The composition of population helps us to know



***Think:** Every human being is potential resource for the society. What will be your contribution as a human resource?*

how many are males or females, which age group they belong to, how educated they are and what type of occupations they are employed in, what their income levels and health conditions are.

An interesting way of studying the population composition of a country is by looking at the population pyramid, also called an age-sex pyramid.

A population pyramid shows

- The total population divided into various age groups, e.g., 5 to 9 years, 10 to 14 years.
- The percentage of the total population, subdivided into males and females, in each of those groups.

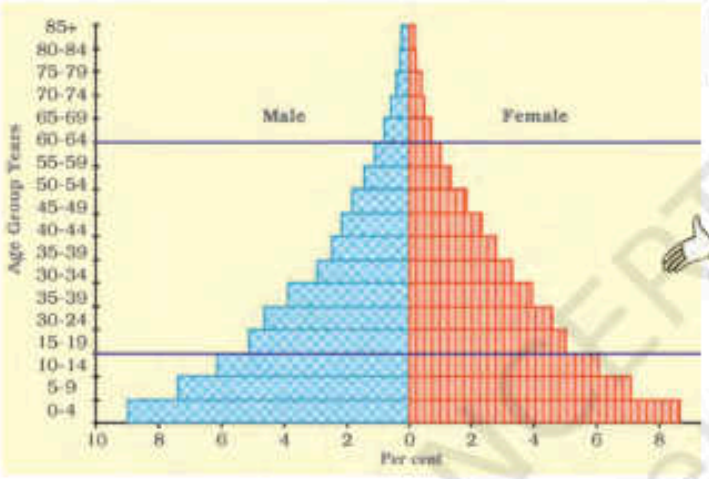


Fig. 5.6: Population Pyramid









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The shape of the population pyramid tells the story of the people living in that particular country. The numbers of children (below 15 years) are shown at the bottom and reflect the level of births. The size of the top shows the numbers of aged people (above 65 years) and reflects the number of deaths.

The population pyramid also tells us how many dependents there are in a country. There are two groups of dependents — young dependents (aged below 15 years) and elderly dependents (aged over 65 years). Those of the working age are the economically active.

The population pyramid of a country in which birth and death rates bothe are high is broad at the base and rapidly narrows towards the top. This is because although, many children are born, a large percentage of them die in their infancy, relatively few become adults and there are very few old people. This situation is typified by the pyramid shown for Kenya (Fig 5.7).

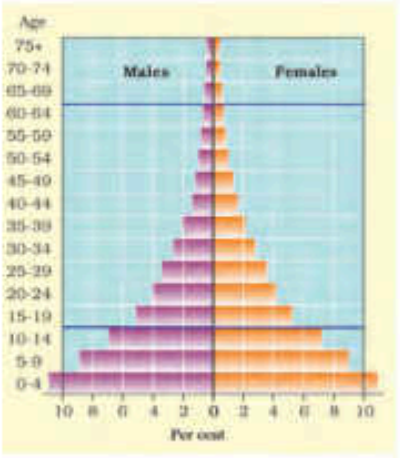


Fig. 5.7: Population Pyramid of Kenya

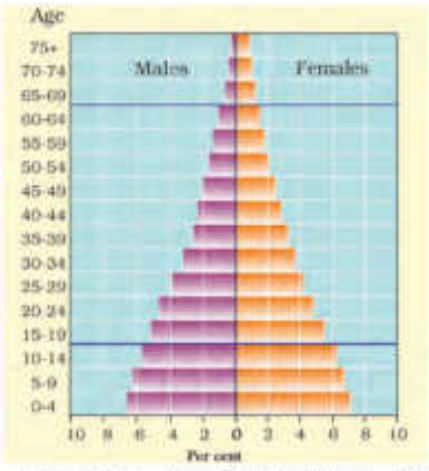


Fig. 5.8: Population Pyramid of India

In countries where death rates (especially amongst the very young) are decreasing, the pyramid is broad in the younger age groups, because more infants survive to adulthood. This can be seen in the pyramid for India (Fig 5.8). Such populations contain a relatively large number of young people and which means a strong and expanding labour force.

In countries like Japan, low birth rates make the pyramid narrow at the base (Fig 5.9). Decreased death rates allow numbers of people to reach old age.

Skilled, spirited and hopeful young people endowed with a positive outlook are the future of any nation. We in India are fortunate to have such a resource. They must be educated and provided skills and opportunities to become able and productive.

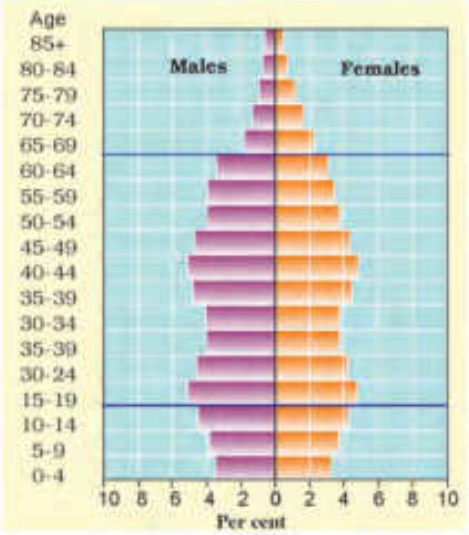


Fig. 5.9: Population Pyramid of Japan

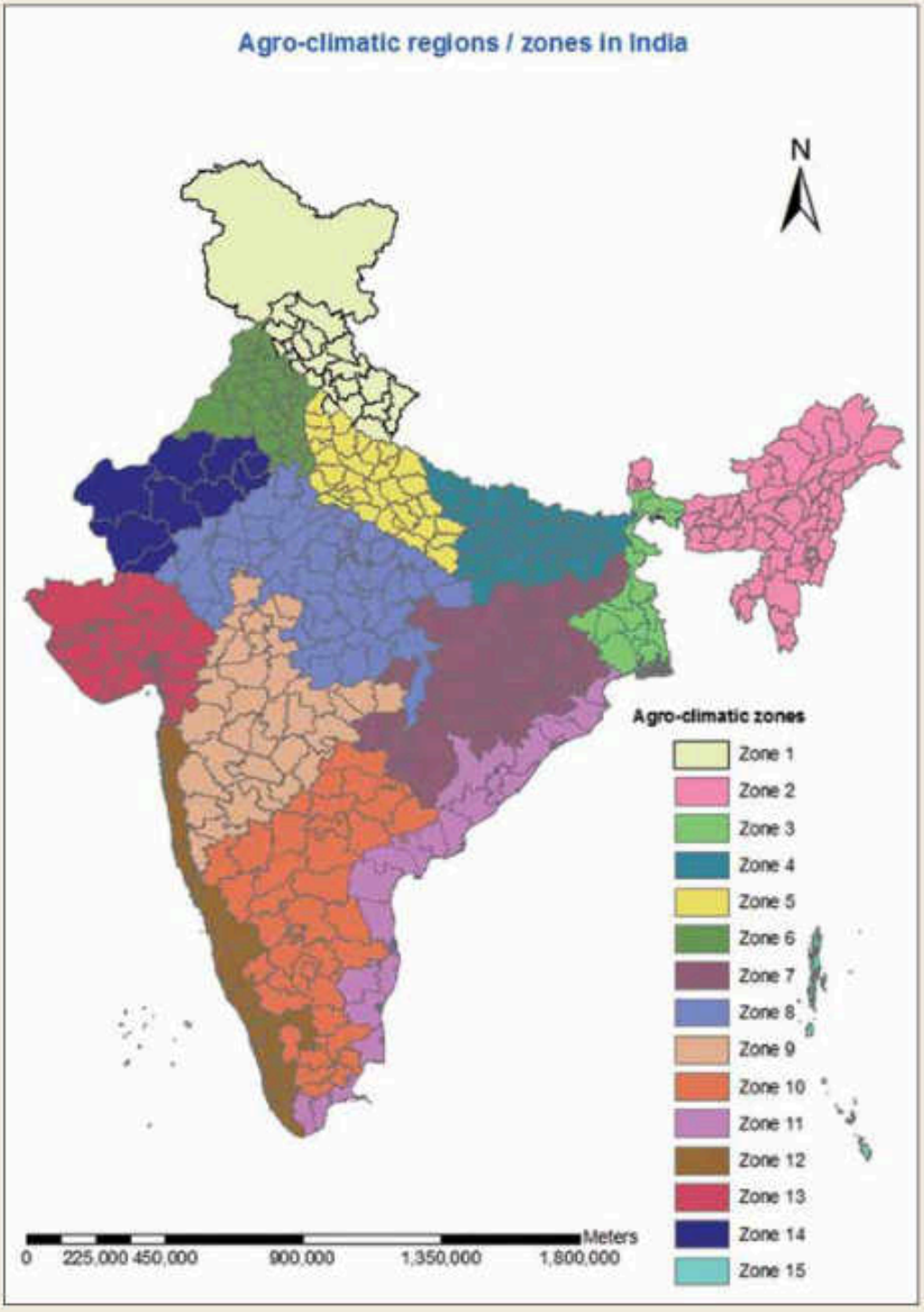


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Table 1.2: Agro-climatic regions/zones in India

S.No.	Agro-climatic regions/zones	States represented
I	**Western Himalayan region**	Himachal Pradesh, Jammu & Kashmir, Uttarakhand
II	**Eastern Himalayan region**	Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tripura, West Bengal
III	**Lower Gangetic plain region**	West Bengal
IV	**Middle Gangetic plain region**	Uttar Pradesh, Bihar
V	**Upper Gangetic plain region**	Uttar Pradesh
VI	**Trans Gangetic plain region**	Chandigarh, Delhi, Haryana, Punjab, Rajasthan
VII	**Eastern plateau and hills region**	Chhattisgarh, Jharkhand, Madhya Pradesh, Maharashtra, Odisha, West Bengal
VIII	**Central plateau and hills region**	Madhya Pradesh, Rajasthan, Uttar Pradesh
IX	**Western plateau and hills region**	Madhya Pradesh, Maharashtra
X	**Southern plateau and hills region**	Andhra Pradesh, Karnataka, Tamil Nadu
XI	**East coast plains and hills region**	Andhra Pradesh, Odisha, Puducherry, Tamil Nadu
XII	**West coast plains and ghat region**	Goa, Karnataka, Kerala, Maharashtra, Tamil Nadu
XIII	**Gujarat plains and hills region**	Gujarat, Dadra & Nagar Haveli, Daman & Diu
XIV	**Western dry region**	Rajasthan
XV	**Island region**	Andaman & Nicobar Islands, Lakshadweep

Source : Planning Commission (Khanna, 1989) has identified 15 resource development regions in the country, 14 in the main land and remaining one in the islands of Bay of Bengal and Arabian Sea.


Agro-climatic regions / zones in India
N
Agro-climatic zones
Zone 1
Zone 2
Zone 3
Zone 4
Zone 5
Zone 6
Zone 7
Zone 8
Zone 9
Zone 10
Zone 11
Zone 12
Zone 13
Zone 14
Zone 15
Meters
0 225,000 450,000 900,000 1,350,000 1,800,000

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FOREWORD

Forests occupy a unique position in so far as climate change is concerned. They are closely linked to socio-economic systems, particularly those of the forest dependent communities in the developing countries. India is one of the few countries where forest and tree cover have increased in recent years transforming country's forests into a net sink of Carbon dioxide owing to national policies aimed at conservation and sustainable management of forests. As per the latest assessment, forests and tree cover has increased from 23.4% in 2005 to 24.4% of the geographical area in 2017. Government of India's long term goal is to bring 33% of its geographical area under forest and tree cover eventually.

India has always been in the forefront of REDD+ negotiations at the United Nations Framework Convention on Climate Change (UNFCCC) and now preparing for implementing REDD+ at national level in order to attract result based finance for REDD+. India has already submitted its National Forest Reference Level for REDD+ to UNFCCC and Government of India has also released India's National REDD+ Strategy in 2018. The National REDD+ Strategy when implemented in completely will be a step forward in achieving India's NDC target of capturing additional 2.5 to 3 billion tones of CO_2 through additional forest and tree cover by 2030. The National REDD+ Strategy indicates Government of India's commitment to implement REDD+ by optimally exploring the mitigation potential of forestry sector in the country. It also recognizes the active participation of local communities and tribal people in conservation and sustainable management of forests.

Scientific management of forests based on principles of sound siliviculture has been in practice for more than 150 years in India. Various principles of forest management in India are of direct relevance to REDD+ activities. This report on 'Mainstreaming of REDD+ Activities in Forest Management Plan' is a sincere contribution by ICFRE towards India's preparedness for early and effective implementation of REDD+ down to the level of forest administrative units. I am hopeful that this knowledge product under the trans-boundary 'REDD+ Himalaya Project' in collaboration with ICIMOD will serve to accelerate the REDD+ activities in India and help in mobilizing result based finance. I congratulate the team of Biodiversity and Climate Change Divisions, Directorate of Research, ICFRE to bring this apt publication.

Date: 18/11/2018

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Acknowledgement

Scientific management of forests in India started in 1864 and is more than 150 years old now. With the change in time and keeping pace with emerging national issues, societal and environmental needs, various aspects of forest management has also been adapted to these changing scenarios. This report on 'Mainstreaming of REDD+ Activities in Forest Management Plan' was conceived and realized during the implementation of trans-boundary REDD+ Himalaya Project in collaboration with ICIMOD. We are grateful to Dr. Suresh Gairola, Director General, ICFRE for providing invaluable guidance and consistent support during the entire course of implementation of REDD+ Himalaya Project. Our special gratitude to Shri S. D. Sharma, Dy. Director General (Research), ICFRE for his valuable inputs and continuous support in implementation of the project. This report has been possible with the strenuous efforts of the entire REDD+ Himalaya Project team at ICFRE, Dehradun. Authors would like to acknowledge suggestions and valuable inputs from the transboundary REDD+ Himalaya Project partners at ICIMOD especially Dr. Bhaskar Singh Karky, REDD+ Programme Coordinator and Mr. Nabin Bhattarai.

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(Authors)



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Abbreviation Used

CO_2	Carbon dioxide
FRCBR	Forest Research Centre for Bamboo and Rattan
GIS	Geographical Information System
GIZ	Deutsche Gesellschaftfür Internationale Zusammenarbeit
GPS	Global Positioning System
HoFF	Head of Forest Force
ICFRE	Indian Council of Forestry Research and Education
ICIMOD	International Centre for Integrated Mountain Development
JFM	Joint Forest Management
JFMC	Joint Forest Management Committee
MoEF	Ministry of Environment and Forests
MoEF&CC	Ministry of Environment, Forest and Climate Change
MRV	Measurement, Reporting and Verification
NDCs	Nationally Determined Contributions
NDE	National Designated Entity
NGC	National Governing Council
NFP	National Forest Policy
NGO	Non-Governmental Organization
NTFPs	Non-Timber Forest Products
PCCF	Principal Chief Conservator of Forests
REDD+	Reducing emissions from deforestation and forest degradation, and role of conservation, sustainable management of forests and enhancement of forest carbon stocks in developing countries
SMF	Sustainable Management of Forest
UNFCCC	United Nations Framework Convention on Climate Change
UT	Union Territory

Executive Summary

India is a mega-biodiversity country where forests and tree cover account for about 24.39% (80.21 million ha) of its geographical area. The progressive conservation oriented forest policies, intensive sustainable management of forests and afforestation programmes in India are contributing towards reduction in CO_2 emissions and improvement of carbon stocks in forests. Scientific management of forests in India began in 1864 with creation of a organised forest service, and the promulgation of 1865 Forest Act, later on revised in 1878. In 1927, the new Indian Forest Act was passed which is still in force.

The National Forest Policy 1988 was a paradigm shift in the forestry sector. It differed from the previous policies of independent India where forests were being looked upon mainly as a source of revenue. The objective of the 1988 policy was to ensure that the rights of the forest dependent people are protected. India recognizes the conservation, expansion and improvement in the quality of our forests as a major national priority. India stands to gain a lot from a global REDD+ mechanism. It has specifically opened the possibilities for the country to expect compensation for its pro-conservation oriented approach and sustainable management of forests resulting in even further increase of forest cover and thereby its forest carbon stocks.

About 30% of the forest area of the country are being managed through more than 1,18,213 Joint Forest Management Committees (JFMCs). There is ample scope and opportunities for integrating REDD+ initiative with the community managed forest and Joint Forest Management (JFM). Promoting and integrating REDD+ actions in JFM activities to increase carbon sequestration will meet both national objectives of climate change mitigation and international obligations as a responsive member of international community. India's National REDD+ Strategy has been aligned with the precepts of the National Forest Policy 1988. The strategy focuses on creation of trained human resource capable of carrying out forest related measurements at all levels of REDD+ implementation. The strategy will support empowerment of youth cadres as community foresters and also ensure adherence to the gender centred sensitivity and transparency in forest governance. The strategy will also support green skill development programme for developing forestry related specialised skill.

Forest management of the country is already attuned to implement REDD+, however, certain components and aspects of forest management will need to be strengthened by appropriate infrastructure and capacity building to support good performance of REDD+. Development and implementation of REDD+ at national level will require close coordination and strong linkages between all stakeholders of forest sector. Adequate technical and institutional capabilities are needed for effective implementation of REDD+ across the country. Strengthening forest department and other partner agencies, strengthening and capacity building of local institutions and building a cadre of community youth for REDD+ are proposed to enhance technical and institutional capabilities for implementation of REDD+. A REDD+ curriculum at forestry universities and forestry training institutions, and creating a network of universities to share experiences on integrating climate change and REDD+ knowledge will contribute to fill the gap of trained human resource for REDD+ implementation in India.

Forest Working Plan has been the main instrument of forest planning for scientific management of forests. Forest Working Plan includes the area specific scientific prescriptions for proper management of forests of a particular forest division, while working schemes are prepared for smaller areas for a specific purpose or for forest areas under the control/ ownership of such bodies as private, village, municipal, cantonment, autonomous district council (especially in north eastern states), etc. All forests are to be sustainably managed under the prescriptions of a Forest Working Plan.

The new Forest Working Plan Code-2014 is more or less tuned with REDD+ activities. According to the revised code, the forest management planning must provide for sustainable management of forests and its biodiversity. Principles of National Forest Working Plan Code-2014 addressed various essential components of REDD+ in forest management. Implementation of REDD+ is one of the objectives of this Working Plan code. It considers REDD+ as a basic mechanism at the division level in forest departments for the sustainable management of forests and enhancement of forest carbon stocks.

The current forest policies, regulations and forestry practices in India are well tuned for mainstreaming of REDD+ in forest management plans. The new National Forest Working Plan Code 2014, National REDD+ Strategy for India 2018 and the draft National Forest Policy 2018 give an ample opportunity for facilitating REDD+ while developing forest management plan. Reshaping conventional forest management practices and aligning then with the concept of REDD+ specially within the framework of community forest management also poses a great challenge to forest managers.

There is significant potential in achieving REDD+ objectives in community controlled/ managed forested landscapes. Implementing REDD+ Himalaya Project in the state of Mizoram shows that by analysing the drivers and deforestation and forest degradation, suitable strategies were be developed for addressing these drivers and also contribute to income generation activities of the forest dependent communities. REDD+ has the potential to further promote national forest governance structure, provide safeguards, both environmental and social by way of full and effective participation of relevant stakeholders. REDD+ activities needs to be expanded to the area of forests managed by communities. REDD+ pilot projects and readiness activities for mainstreaming REDD+ in Forest Management Plan in these areas needed to test the benefit-sharing mechanisms, and mechanisms for monitoring, reporting, and verification (MRV) of carbon benefits.



the speed of light (unlike EM waves, sound waves, which you studied in Chapter 10, Sound Waves: Characteristics and Applications, are mechanical waves and require a medium to travel). The speed of light in vacuum is 3×10^8 ms^{-1}. EM waves cover a wide range of frequencies, from high frequency or short wavelength radiation (**gamma rays** and **X-rays**) to low frequency or long wavelength radiation (**infrared** and **radio waves**). The high frequency EM waves, such as gamma rays and X-rays have very high energy and can be harmful for life on Earth.

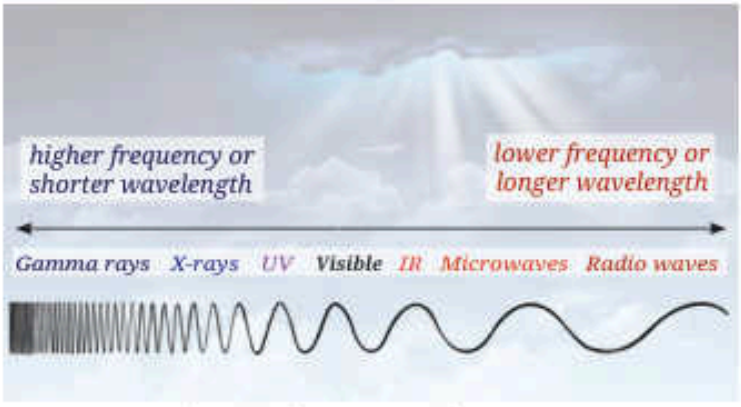


Fig. 13.2: Electromagnetic spectrum

The entire range of electromagnetic radiation called **electromagnetic spectrum** is shown in Fig. 13.2. However, the solar radiation reaching Earth is concentrated mainly in the ultraviolet (UV), visible and infrared (IR) range—about 99 per cent of the Sun's energy falls within these wavelengths. These three regions of the spectrum shape the Earth's climate and support life. Gamma rays and X-rays are mostly filtered by the Earth's upper atmosphere, while microwaves and radio waves carry very little energy to significantly warm the Earth.

Short wavelength UV radiation is mostly absorbed by the ozone layer in the upper atmosphere, protecting life and contributing to some atmospheric heating. Visible light from the Sun reaches the Earth's surface and provides energy for photosynthesis, which is the primary source of food for most organisms. It also partly warms the land and water. Infrared radiation warms the Earth's surface, which then re-radiates this heat back into the atmosphere. A portion of this outgoing heat is trapped by the greenhouse gases, such as carbon dioxide (CO_2), methane (CH_4) and water vapour, keeping the Earth warm enough to support life.

Ready to Go Beyond

The UV rays lie in the wavelength range of 100 nm to 400 nm [1 nanometre (nm) = 10^{-9} m] and have a much higher energy than the visible light. Prolonged exposure to these rays can damage the eyes and skin, and increase the risk of cancer. Therefore, people should use UV protective glasses and sunscreen if they are likely to be exposed to UV rays. UV rays are also useful in killing germs in water purifiers and help power fluorescent lights.

The amount of the Sun's radiation that reaches the Earth's surface is called **insolation**. It is responsible for warming the Earth's surface and its atmosphere. The average amount of solar energy received per unit time per unit area that is perpendicular to the Sun's rays at the top of the Earth's atmosphere is called the **solar constant**. Its value is approximately 1.4 kilowatts per square metre (1.4 kWm^{-2}), or about 1400 joules per second per square metre (1400 J s^{-1}m^{-2}). It represents the Sun's energy available on the Earth before any absorption, scattering or reflection occurs in the atmosphere.

The solar constant is important because it helps scientists understand the Earth's energy balance, climate and weather patterns. However, some of this energy is absorbed and scattered by gases, clouds and dust particles in the atmosphere before reaching the surface of the Earth. As a result, the maximum insolation reaching the Earth's surface is lower than the solar constant and is about 1 kWm^{-2} under clear sky conditions. India, due to its geographical location in the tropical and sub-tropical regions, receives abundant sunlight throughout the year. This makes solar insolation highly significant as it drives the southwest monsoon, which influences India's climate and agriculture. It also offers immense potential for harnessing solar energy as a renewable and sustainable source of energy.

Bridging Science and Society

Anna Mani, India's pioneering atmospheric scientist, mapped solar insolation across India in the 1950s. Along with S. Rangarajan, she later published ***Solar Radiation Over India*** in 1982, creating the country's first **insolation atlas**. Her pioneering measurements showed India's vast solar energy potential. Today, this is being realised through the large-scale deployment of solar power across the country. India is laying the foundation for a resilient, sustainable and solar-powered energy future that sets global benchmarks.

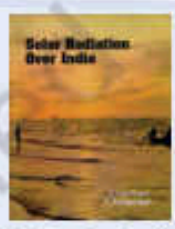


Fig. 13.3: Solar Radiation Over India by Anna Mani and S. Rangarajan

Example 13.1: How much solar energy will be received by a 1 m^2 area in one hour, if the insolation on the surface of the Earth were 1 kWm^{-2}?

Answer: E = Intensity × area × time

$E = 1 \times 1000\ J\,s^{-1}\,m^{-2} \times 1\ m^2 \times 3600\ s$

$E = 3600000\ J = 3.6 \times 10^6\ J$

(This amount of energy is approximately equal to the energy needed to melt 5 kg of ice and heat the water obtained to 100 °C. It is also equal to one unit of electricity used in a household, as shown on an electricity bill).

Think as a Scientist

An interesting estimation problem that helps us to appreciate the enormous amount of energy that we get from the Sun is to estimate how much of the Earth's surface would be needed to be covered with solar panels to supply all the electric power that our country uses today (Fig. 13.4). To make this estimate, you can find these numbers on the internet, assume some insolation on the Earth's surface and consider that some fraction of this energy is converted into electricity. You will probably find that even a fraction of the area of the Thar desert, if covered with solar panels, could supply India's electricity needs.



Fig. 13.4: Solar panel in the desert

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How does the Sun's radiation interact with the Earth's surface and atmosphere, and warm the Earth? We will discuss more about it in the following sections.

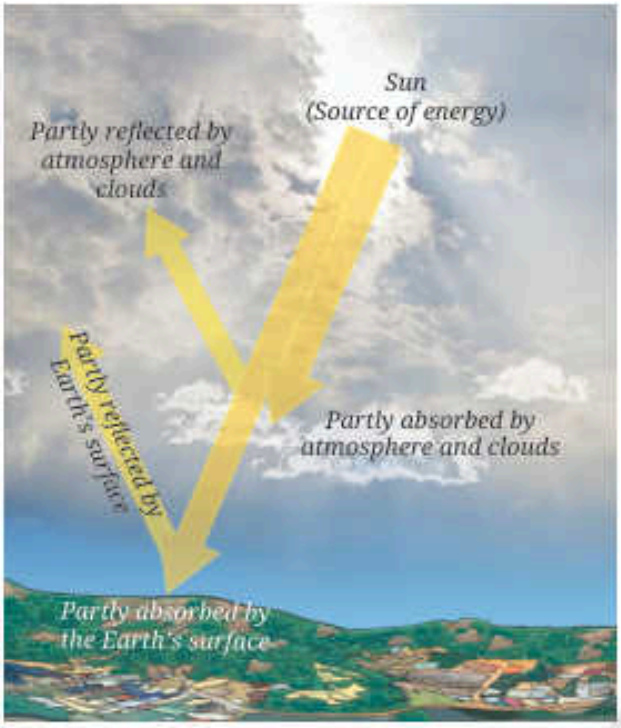


Fig. 13.5: Solar radiation and warming of the Earth

13.1.1 Interaction of solar radiation on the Earth's surface

Different materials absorb and heat up differently under sunlight. You have already learnt how land heats up faster than water. There may be some variation due to the material forming the land and the colour of the soil. Dark surfaces absorb more sunlight, while light coloured surfaces reflect more and remain comparatively cooler. For example, dark coloured roads heat up more quickly, while light coloured surfaces remain comparatively cooler. You have also learnt why dark coloured clothes feel hotter than white ones in summer.

Grade 7 Curiosity Chapter 7

The fraction of solar radiation reflected by a surface is called its **albedo** (the word 'albedo' comes from Latin which means whiteness). High albedo surfaces stay cool because they reflect more light, while low albedo ones heat up more quickly as they reflect less and absorb more light (Fig. 13.5). Let us try to find out the albedo of some of the common objects we come across in our daily lives by conducting Activity 13.2.

Activity 13.2: Let us find out

Complete Table 13.1 using information from authentic sources like websites and books.

Note

Ensure that you do not share any personal information while using the internet.

Table 13.1: Reflection of solar radiation by surfaces of materials

S. No.	Materials	Albedo
1.	Snow	0.80 – 0.90
2.	Ice	0.50 – 0.70
3.	Crushed rock	0.25 – 0.30
4.	Light coloured soil	
5.	Black soil	
6.	Ocean water	

From Table 13.1, you must have seen that snow and ice have high albedo, meaning they reflect a large proportion of the incoming solar radiation. This makes polar regions very cold. On the other hand, surfaces like black soil and ocean water have lower albedo as they absorb more solar radiation, and hence, they are relatively warmer.

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You have also learnt in the earlier grades that all objects radiate heat. We feel hot inside a concrete house at night especially during hot summer, this is due to the heat re-radiated by the concrete during the night. However, traditional houses built with thick mud and wooden walls offer cool conditions even in hot summer due to less re-radiation.

Grade 7
Curiosity
Chapter 7

Threads of Curiosity

Urban Heat Island Effect

Cities are warmer than their surrounding rural areas, especially during summer and at night. Compared to rural areas, cities have more built-up areas, including buildings of steel, concrete, brick, and roads made up of concrete and asphalt. All these materials absorb solar radiation and retain heat. The re-radiated heat from these materials warms cities more than the surrounding rural areas. This increases energy demand for air conditioning further stressing the urban ecosystems. Rural areas and forests have more vegetation, and stay cool through shade and plant transpiration. The urban heat island effect shows how human land use can alter the local climate (Fig. 13.6).



Fig. 13.6: Urban heat island effect in a city

The amount of radiation received by a particular region on the Earth's surface also depends on its latitude. The uneven heating due to the latitude of a place is important as it determines the atmospheric conditions of that place.

13.1.2 Latitude and Earth's shape

You have already learnt in earlier grades that the Earth is spherical. As a result, the Sun's rays strike different latitudes at different angles on the Earth. The Sun's radiation falling on the equatorial region is concentrated over a smaller area, where it is spread over a larger area in the polar region. This is why, equatorial regions remain relatively warm throughout the year, whereas polar regions experience much colder conditions. This uneven heating creates temperature differences between the equator and the poles.

Grade 7
Exploring Society:
India and Beyond
Part 1
Chapter 3

You have also learnt how the Earth's spherical shape and the tilt of the Earth's axis of rotation give rise to seasons, and the changing length of daytime during one revolution of the Earth around the Sun. Thus, solar radiation is not evenly distributed across the globe. This uneven heating of the Earth's surface across the globe drives global winds and ocean currents.

Grade 7
Curiosity
Chapter 12

13.1.3 Role of the atmosphere

The air surrounding the Earth is called atmosphere. It is held in place by the force of Earth's gravity. It consists mainly of nitrogen (78%) and oxygen (21%) along with small amounts of argon, carbon dioxide, water vapour, and other gases.

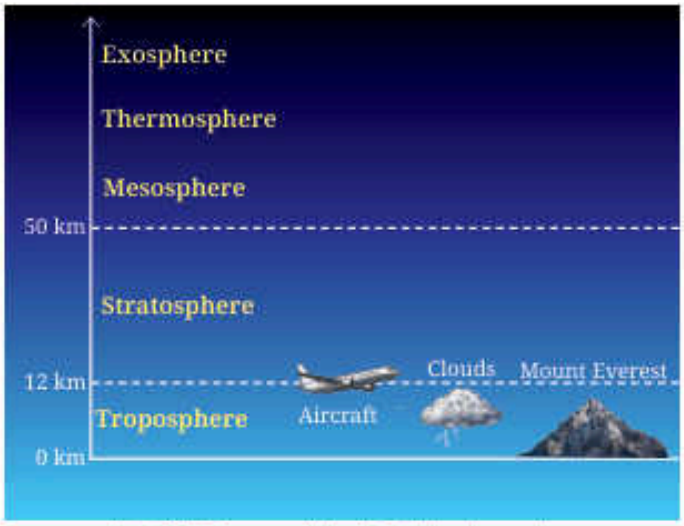


Fig. 13.7: Layers of the Earth's atmosphere

The Earth's atmosphere has a layered structure. This structure of the atmosphere is important as it helps explain key weather patterns (Fig. 13.7). Nearly, all weather phenomena take place in the troposphere (with an average height of about 12 km), which is heated from the Earth's surface. In this layer, temperature decreases with height (~6.5 °C/km). As the warm air rises, it drives winds and storms. The height of the troposphere is maximum above the equator and lowest above the polar regions.

In the stratosphere (12–50 km), where the ozone layer is located, ozone absorbs UV rays and heats up the atmosphere, so temperature rises with height. This increase in temperature with height in the stratosphere calms the layer due to the lack of vertical mixing of air, keeping weather confined to the troposphere. Above the stratosphere are the mesosphere, thermosphere and exosphere, but they play only a minor role in regulating climate on the surface of the Earth. At about 100 km above the Earth is the start of the region, we call the outer space. These atmospheric layers show how the atmosphere regulates energy flow, shaping weather, climate and life.

Table 13.2: Key atmospheric layers

Layer	Approximate altitudes	Features
Troposphere	0–12 km	Weather formation; temperature decreases with height
Stratosphere	12–50 km	Ozone layer absorbs UV; temperature increases with height

The atmosphere plays two crucial roles in protecting the life on the Earth. Firstly, it partly absorbs the incoming solar radiation. The ozone layer blocks the harmful UV rays. Clouds and other gases also absorb some sunlight before it reaches the surface of the Earth (Fig. 13.5). Secondly, it traps outgoing heat. The Earth's surface absorbs sunlight and re-radiates it in the infrared region. Greenhouse gases like CO_2, CH_4 and water vapour absorb this re-radiated heat, preventing it from escaping into space. Without the atmosphere, the Earth would be too cold for life to survive. However, excess CO_2 from human activities enhances the greenhouse effect, causing global warming, which if left unchecked could make the Earth uninhabitable.

You have also learnt that Venus is hotter than Mercury, even though Mercury is closer to the Sun. This is because of the presence of an atmosphere on Venus, which has led to an uncontrolled greenhouse effect.

Grade 8 Curiosity Chapter 13

Meet a Scientist



K.R. Ramanathan, India's atmospheric scientist, climbed to an altitude of 18,000 feet in the Himalayas in 1934 to measure the ozone levels, and discovered that they were lower than expected. His work laid the foundation for understanding how UV absorption varies with altitude and pollution. Later on, he led early monsoon forecasting efforts.

Pause and Ponder

1. Visit the website given below and study the effect of the concentration of greenhouse gas on surface temperature, https://phet.colorado.edu/en/simulations/greenhouse-effect

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The atmosphere is continuously modified by natural processes, influencing weather, climate and energy balance. When solar radiation and cosmic rays interact with the atmosphere, they influence weather and climate, and affect communication systems. Thus, atmosphere makes the Earth a unique planet.

Threads of Curiosity

Why is the ozone layer so important?

The ozone layer is vital to life on the Earth because it acts as a protective shield, absorbing harmful UV radiation from the Sun. When ozone molecules are destroyed faster than they are naturally formed, the layer becomes thinner and less effective. In the late 20th century, human-made chemicals called chlorofluorocarbons (CFCs), used in refrigerators and aerosols, caused severe ozone loss over Antarctica, which came to be known as the ozone hole. Increased UV radiation on the Earth can harm the living organisms and the ecosystems. A global agreement known as the Montreal Protocol reduced the use of CFCs and the ozone layer is now slowly recovering, showing the power of international scientific cooperation.

13.2 Uneven Heating Causes Wind and Ocean Currents

Grade 8 Curiosity Chapter 6

You have learnt that wind is the movement of air from a region of high pressure to a region of low pressure. These pressure differences are mainly caused by the uneven heating of the Earth's surface by the Sun. Uneven heating of land and water is also responsible for phenomena, such as land and sea breezes, which you have studied earlier. The same basic idea helps us understand the formation of winds and ocean currents on vastly different scales.

13.2.1 Local winds

Uneven heating of the Earth's surface also produces **local winds**, such as valley and mountain breezes. In mountainous regions, the slopes and the valley floor do not heat up and cool down at the same rate. During the day, the mountain slopes facing the Sun are heated more rapidly than the valley floor. As a result, the air over the slopes becomes warm and rises, creating a low pressure region. Cooler air from the valley moves up the slopes to replace the rising warm air. This flow of air is called a **valley breeze** (Fig. 13.8a).



Fig. 13.8 (a): Valley breeze

After sunset, the situation reverses. The mountain slopes lose heat faster and become cooler, while the valley floor remains relatively warmer. The air over the slopes becomes cooler and denser, and flows down into the valley. This is known as a **mountain breeze** (Fig. 13.8b). Such daily changes in the direction of the wind are commonly experienced in hilly regions like Shimla, Dehradun, and other Himalayan valleys. These local winds influence the weather conditions, agriculture and daily life in mountainous regions. They help regulate temperature, moisture conditions and support soil and crop health. Thus, uneven heating of the Earth's surface sets up winds and forms an important part of the Earth's atmospheric circulation.



Fig. 13.8 (b): Mountain breeze

13.3 Biogeochemical Cycles

Living organisms constantly exchange matter and energy with the air, water, soil and rocks around them. This continuous interaction between the non-living (abiotic) and living (biotic) components of the Earth results in the transfer of matter and energy across various spheres of the Earth, as discussed earlier. This process ensures that essential nutrients, such as carbon, nitrogen and oxygen are recycled, and remain available to support life on the Earth. This cyclic movement of matter and energy between the abiotic and biotic components is called the **biogeochemical cycle**. There is a dynamic relationship between different ecosystems. This interconnectedness helps ecosystems recover from disturbances or degradation and maintain environmental balance.

We will examine the water, carbon, nitrogen and oxygen cycles. We will try to understand why their balance matters, and explore how they interconnect across spheres and respond to human impacts.

13.3.1 Water cycle

Grade 7 Curiosity Chapter 7

You have studied the water cycle and learnt terms like evaporation, transpiration, condensation, precipitation, infiltration and groundwater. The water cycle is summarised in Fig. 13.12. In this cycle, water evaporates from water bodies, such as rivers, lakes, oceans, etc., and condenses to form clouds. It returns as precipitation in the form of rain, hail or snow to the surface and finally flows back to the ocean. Some of the water seeps through soil and rocks beneath the Earth's surface. Water dissolves minerals from soil and rocks. It also supports all terrestrial organisms and transports these nutrients to oceans, and support marine life.

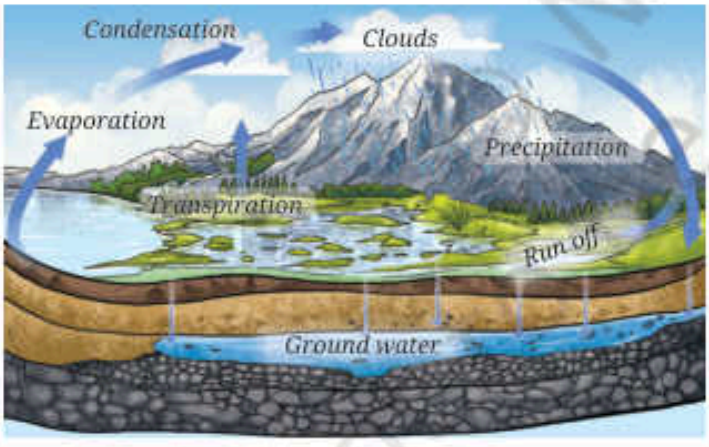


Fig. 13.12: Water cycle

Climate change is now affecting the water cycle. For example, a warmer atmosphere holds more moisture, causing heavier rains in some areas (like intensified monsoons) and droughts elsewhere. Melting glaciers add more water to rivers, raising sea levels in the long run and threatening coastal cities, such as Mumbai and Chennai. Sudden bursts of intense rainfall result in more run off that erodes soil and less infiltration reduces the recharge of groundwater, which in turn, makes sustaining agriculture difficult, especially during dry months. In this way, the water cycle links the cryosphere (glaciers), hydrosphere (rivers and oceans), atmosphere (moisture), geosphere (soil erosion and decreased infiltration), and biosphere (crops and fisheries), all of which are affected by global warming.

13.3.2 Carbon cycle

Carbon forms the backbone of life. Every protein, carbohydrate, fat and DNA molecule contains carbon. It circulates continuously between the atmosphere (CO_2 gas), biosphere (plants and animals), geosphere (carbonate

rocks and fossil fuels, such as coal and oil), and the hydrosphere (dissolved CO_2 and marine shells) as shown schematically in Fig. 13.13. Different parts of the carbon cycle operate at very different time scales. In the fast cycle, which happens over days to years, plants convert atmospheric CO_2 into glucose using sunlight through photosynthesis. CO_2 is released back into the atmosphere through respiration. Animals eat plants and/or other animals. When they die, CO_2 returns to air because of decomposition.

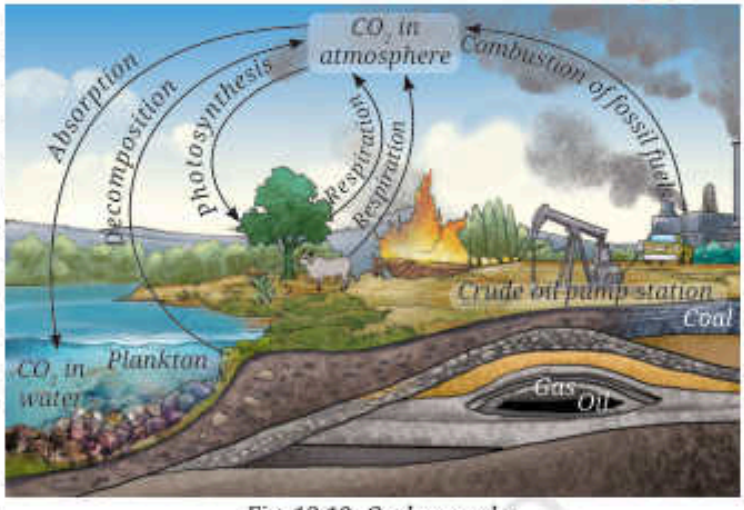


Fig. 13.13: Carbon cycle

In the slow cycle, which happens over millions of years, dead plants and animals get buried, and are converted to fossil fuels, such as coal, oil and gas. These fuels are burnt to provide energy for various needs like heating, cooking, transportation and industrial purposes. When fossil fuels are burnt, carbon is released back as CO_2 on a very short time scale. The atmosphere and ocean water continuously exchange CO_2. The ocean water absorbs atmospheric CO_2 to form carbonate and bicarbonate ions. Phytoplankton use them for photosynthesis. Some marine organisms also use them to form shells. When organisms die, they sink to the ocean floor and their organic matter is stored as carbon for a long period.

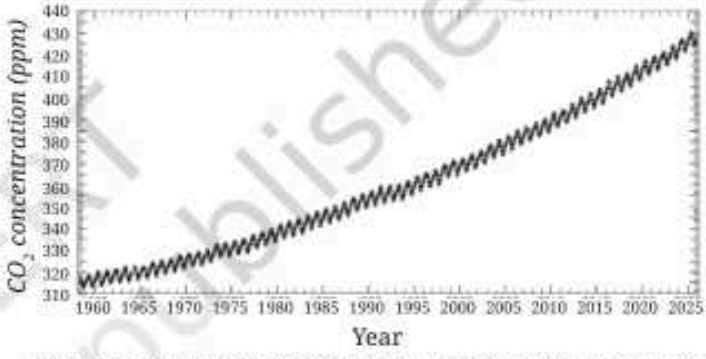


Fig. 13.14: Atmospheric CO_2 concentration (1960 – 2025), the Keeling curve. The sawtooth pattern shows seasonal dips from yearly plant growth in the Northern Hemisphere absorbing CO_2

Human activities like burning fossil fuels and deforestation have raised atmospheric CO_2 by about 35% since 1960 as shown in Fig. 13.14 (315 ppm to 420 ppm; ppm denotes parts per million), an unprecedented rise in the history of human civilisation. While some amounts of carbon dioxide are necessary to keep the Earth warm enough to sustain life, the balance is critical. Excessive amounts of CO_2 intensify the greenhouse effect leading to global warming, melting of glaciers and Arctic sea ice, rising of sea level and more extreme weather conditions. In India, this may lead to more intense monsoons (warmer air holds more moisture) and threats to agriculture from its changing rainfall patterns. Though fossil fuels still power much of our electricity generation, India is rapidly increasing renewable energy sources, which will help minimise the carbon released into the atmosphere.

Pause and Ponder

4. The CO_2 dissolved in the ocean is disturbed when the global temperature increases. What will happen to marine life?

Threads of Curiosity

Carbon constitutes ~ 49% of dry weight of living organisms. Out of the total quantity of global carbon, 71% carbon is found in oceans. This oceanic reservoir regulates the amount of carbon dioxide in the atmosphere.

Do you know that the atmosphere holds only a very small fraction (about 1%) of the total global carbon?

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13.3.3 Nitrogen cycle

Nitrogen is an essential element for the synthesis of proteins and nucleic acids in all living organisms. The largest reservoir of nitrogen is in our atmosphere. However, nitrogen gas (N_2) is rather non-reactive, and cannot be directly used by plants and animals. It must first be converted to soluble compounds that living beings can absorb. The overall movement of nitrogen between air, soil, water and organisms is called the **nitrogen cycle** (Fig. 13.15). The nitrogen cycle contains several steps — nitrogen fixation, assimilation, **ammonification**, **nitrification** and **denitrification**.

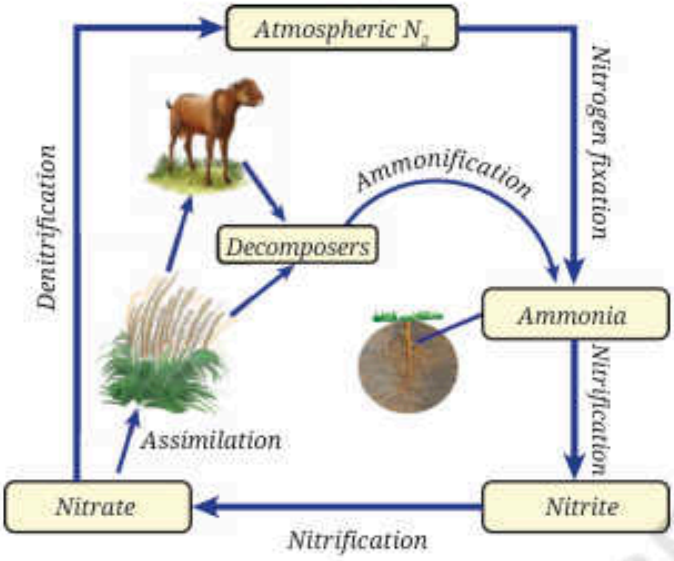


Fig. 13.15: Nitrogen cycle

Nitrogen-fixing bacteria, such as *Rhizobium* in the root nodules of legumes and *Azotobacter* in the soil, convert atmospheric N_2 into ammonia (NH_3). Nitrifying bacteria like *Nitrosomonas* convert ammonia into nitrite (NO_2^-), while *Nitrobacter* convert nitrite into nitrate (NO_3^-). This process is known as **nitrification**. Plants assimilate these nitrogen compounds from the soil, whereas animals obtain nitrogen by consuming plants or other animals. When plants and animals die or produce waste, decomposers like bacteria and fungi break the organic matter, returning nitrogen compounds like ammonia to the soil. This process is known as **ammonification**. Denitrifying bacteria, such as *Pseudomonas* convert some nitrates back into nitrogen gas. This process is known as **denitrification**. This completes the cycle and maintains a balance of nitrogen in ecosystems. Lightning also contributes to the fixation of nitrogen oxides.

Ready to Go Beyond

During lightning strikes, a tiny amount of atmospheric nitrogen is fixed to produce nitrogen oxides. Today most of the nitrogen is fixed artificially via the Haber-Bosch process (early 1900s) of making ammonia from atmospheric nitrogen, which produces most of the fertilisers used today. This 'Bread from Air' revolutionised agriculture, enabling India's Green Revolution and feeding billions. More than half the nitrogen atoms in the human body come from the Haber-Bosch process. However, this reaction is energy intensive (uses ~ 1–2% of global energy), and the overuse of fertilisers has degraded soil and water.

13.3.4 Oxygen cycle

Oxygen is one of the Earth's most abundant elements. About 21% of the atmosphere consist of free oxygen gas (O_2). It is an essential component of most biological molecules like carbohydrates, proteins, nucleic acids and fats. Oxygen also exists in combined forms — in the Earth's crust as

metal oxides and minerals, and in the air as carbon dioxide. While discussing the oxygen cycle, we focus on processes that regulate the oxygen level in the atmosphere (Fig. 13.16). Organisms like plants and animals use oxygen for respiration, and releases CO_2. Combustion of fuels uses oxygen and releases CO_2. Plants restore oxygen through photosynthesis using sunlight, water and CO_2 to form glucose and release O_2. This balance between consumption (respiration and combustion) and production (photosynthesis) circulates oxygen between the atmosphere, land, oceans and living organisms, sustaining life across all spheres of the Earth.

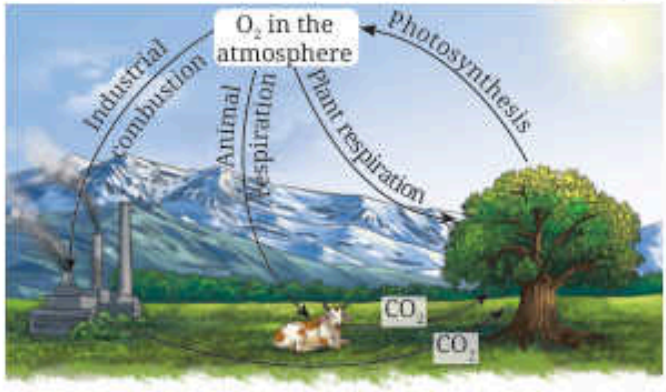


Fig. 13.16: Oxygen cycle

Pause and Ponder

5. What would happen to plants and animals on Earth if the biogeochemical cycles were disrupted and stopped? Explain by giving a few examples.

13.4 Human Impact on Earth's Processes

Grade 8 Curiosity Chapter 13

You have learnt about the impact of climate change earlier, in which we examined the crises our planet Earth is facing. Rising CO_2 levels from the use of fossil fuels leads to extreme weather and biodiversity loss. Here, we will examine how human actions disrupt the biogeochemical cycles across the Earth's spheres. Excess atmospheric CO_2 increases ocean absorption, making sea water more acidic. This could threaten tiny plankton and coral reefs, disrupting the marine ecosystems. However, warmer ocean water reduces the ocean's capacity to absorb CO_2 as an effective carbon sink.

What if ...

photosynthesis stopped, what would happen on the Earth?

Burning of fossil fuels and deforestation saturate natural carbon sinks like forests and oceans. Over the past 50 years, there has been a significant change in the Earth's natural systems. In India, fossil fuels still generate a significant part of our electricity, contributing harmful emissions, although the rapid growth of solar energy offers hope. The excess CO_2 intensifies greenhouse warming, disrupting the carbon cycle.

The overuse of fertilisers in agriculture adds excessive nitrogen via nitrates to rivers and lakes, causing widespread growth of algae (algal blooms) that deplete oxygen and kill fish. This process is called **eutrophication** (Fig. 13.17), which threatens water bodies and coastal fisheries. Deforestation also creates multiple effects across air, water, land and living organisms. Clearing forests results in decreased photosynthesis and reduced transpiration, which can lead to decline in the local rainfall. It also alters surface albedo. Without tree roots to hold the soil together, soil erosion could increase. Over time, habitats could be destroyed, leading to a decline in biodiversity as many species lose their natural homes.



Fig. 13.17: Eutrophication (algal bloom)

Nazism and the Rise of Hitler

In the spring of 1945, a little eleven-year-old German boy called Helmuth was lying in bed when he overheard his parents discussing something in serious tones. His father, a prominent physician, deliberated with his wife whether the time had come to kill the entire family, or if he should commit suicide alone. His father spoke about his fear of revenge, saying, 'Now the **Allies** will do to us what we did to the crippled and Jews.' The next day, he took Helmuth to the woods, where they spent their last happy time together, singing old children's songs. Later, Helmuth's father shot himself in his office. Helmuth remembers that he saw his father's bloody uniform being burnt in the family fireplace. So traumatised was he by what he had overheard and what had happened, that he reacted by refusing to eat at home for the following nine years! He was afraid that his mother might poison him.

Although Helmuth may not have realised all that it meant, his father had been a Nazi and a supporter of Adolf Hitler. Many of you will know something about the Nazis and Hitler. You probably know of Hitler's determination to make Germany into a mighty power and his ambition of conquering all of Europe. You may have heard that he killed Jews. But Nazism was not one or two isolated acts. It was a system, a structure of ideas about the world and politics. Let us try and understand what Nazism was all about. Let us see why Helmuth's father killed himself and what the basis of his fear was.

In May 1945, Germany surrendered to the Allies. Anticipating what was coming, Hitler, his propaganda minister Goebbels and his entire family committed suicide collectively in his Berlin bunker in April. At the end of the war, an International Military Tribunal at Nuremberg was set up to prosecute Nazi war criminals for Crimes against Peace, for War Crimes and Crimes Against Humanity. Germany's conduct during the war, especially those actions which

New words

Allies – The Allied Powers were initially led by the UK and France. In 1941 they were joined by the USSR and USA. They fought against the Axis Powers, namely Germany, Italy and Japan.



Fig. 1 – Hitler (centre) and Goebbels (left) leaving after an official meeting, 1932.

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came to be called Crimes Against Humanity, raised serious moral and ethical questions and invited worldwide condemnation. What were these acts?

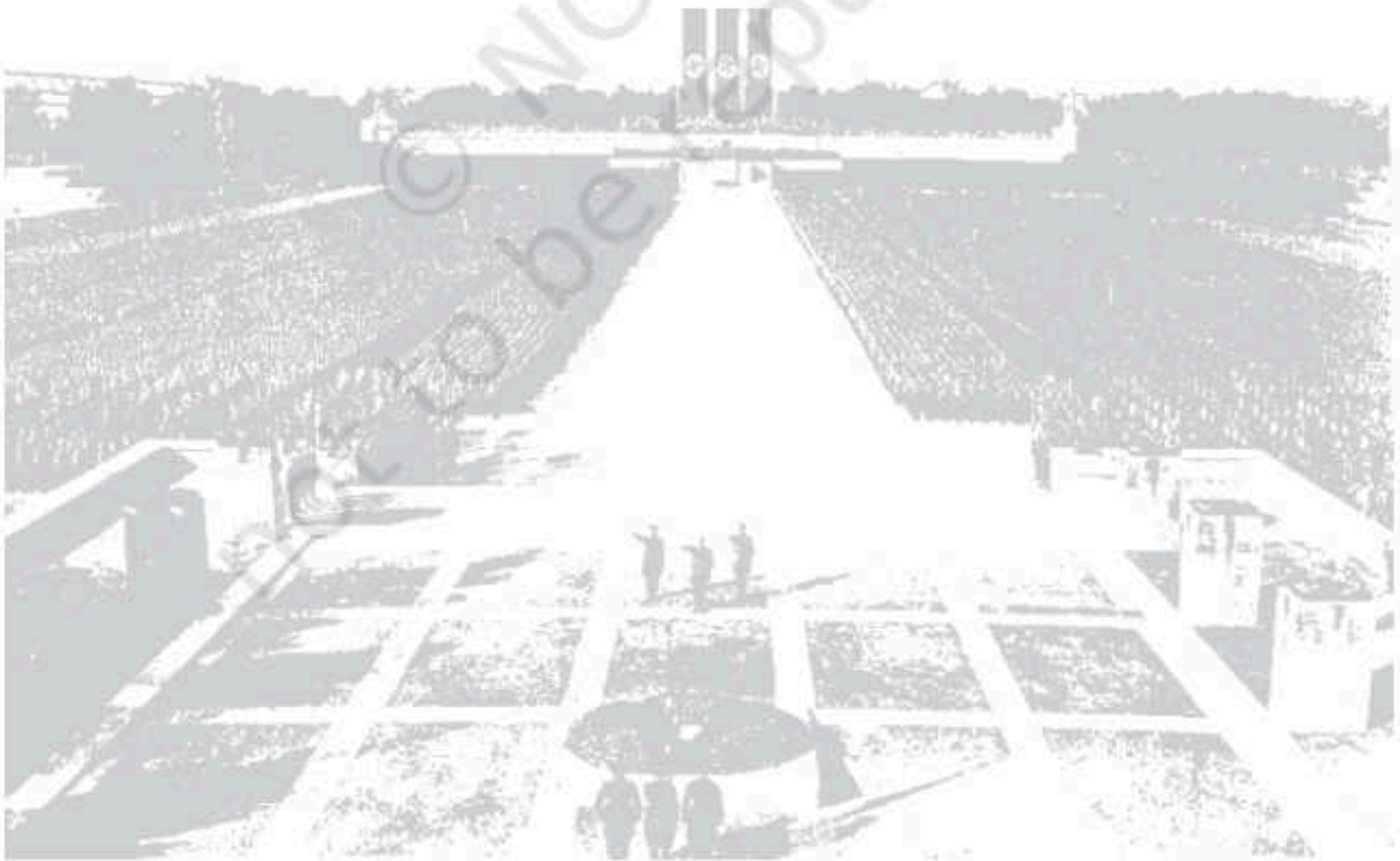
Under the shadow of the Second World War, Germany had waged a **genocidal** war, which resulted in the mass murder of selected groups of innocent civilians of Europe. The number of people killed included 6 million Jews, 200,000 Gypsies, 1 million Polish civilians, 70,000 Germans who were considered mentally and physically disabled, besides innumerable political opponents. Nazis devised an unprecedented means of killing people, that is, by gassing them in various killing centres like Auschwitz. The Nuremberg Tribunal sentenced only eleven leading Nazis to death. Many others were imprisoned for life. The retribution did come, yet the punishment of the Nazis was far short of the brutality and extent of their crimes. The Allies did not want to be as harsh on defeated Germany as they had been after the First World War.

Everyone came to feel that the rise of Nazi Germany could be partly traced back to the German experience at the end of the First World War.

What was this experinece?

New words

Genocidal – Killing on large scale leading to destruction of large sections of people



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1 Birth of the Weimar Republic

Germany, a powerful empire in the early years of the twentieth century, fought the First World War (1914-1918) alongside the Austrian empire and against the Allies (England, France and Russia.) All joined the war enthusiastically hoping to gain from a quick victory. Little did they realise that the war would stretch on, eventually draining Europe of all its resources. Germany made initial gains by occupying France and Belgium. However the Allies, strengthened by the US entry in 1917, won , defeating Germany and the Central Powers in November 1918.

The defeat of Imperial Germany and the abdication of the emperor gave an opportunity to parliamentary parties to recast German polity. A National Assembly met at Weimar and established a democratic constitution with a federal structure. Deputies were now elected to the German Parliament or Reichstag, on the basis of equal and universal votes cast by all adults including women.

This republic, however, was not received well by its own people largely because of the terms it was forced to accept after Germany's defeat at the end of the First World War. The peace treaty at

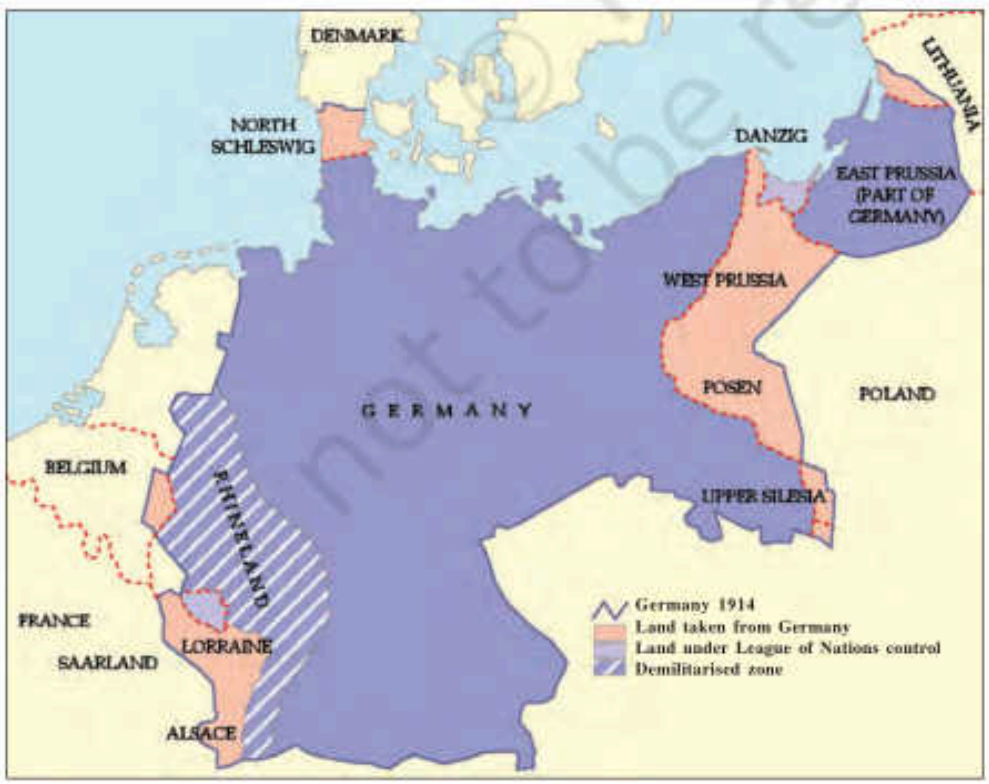


Fig. 2 – Germany after the Versailles Treaty. You can see in this map the parts of the territory that Germany lost after the treaty.

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Versailles with the Allies was a harsh and humiliating peace. Germany lost its overseas colonies, a tenth of its population, 13 per cent of its territories, 75 per cent of its iron and 26 per cent of its coal to France, Poland, Denmark and Lithuania. The Allied Powers demilitarised Germany to weaken its power. The War Guilt Clause held Germany responsible for the war and damages the Allied countries suffered. Germany was forced to pay compensation amounting to £6 billion. The Allied armies also occupied the resource-rich Rhineland for much of the 1920s. Many Germans held the new Weimar Republic responsible for not only the defeat in the war but the disgrace at Versailles.

1.1 The Effects of the War

The war had a devastating impact on the entire continent both psychologically and financially. From a continent of creditors, Europe turned into one of debtors. Unfortunately, the infant Weimar Republic was being made to pay for the sins of the old empire. The republic carried the burden of war guilt and national humiliation and was financially crippled by being forced to pay compensation. Those who supported the Weimar Republic, mainly Socialists, Catholics and Democrats, became easy targets of attack in the conservative nationalist circles. They were mockingly called the 'November criminals'. This mindset had a major impact on the political developments of the early 1930s, as we will soon see.

The First World War left a deep imprint on European society and polity. Soldiers came to be placed above civilians. Politicians and publicists laid great stress on the need for men to be aggressive, strong and masculine. The media glorified trench life. The truth, however, was that soldiers lived miserable lives in these trenches, trapped with rats feeding on corpses. They faced poisonous gas and enemy shelling, and witnessed their ranks reduce rapidly. Aggressive war propaganda and national honour occupied centre stage in the public sphere, while popular support grew for conservative dictatorships that had recently come into being. Democracy was indeed a young and fragile idea, which could not survive the instabilities of interwar Europe.

1.2 Political Radicalism and Economic Crises

The birth of the Weimar Republic coincided with the revolutionary uprising of the Spartacist League on the pattern of the Bolshevik Revolution in Russia. Soviets of workers and sailors were established

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Fig.3 – This is a rally organised by the radical group known as the Spartacist League.
In the winter of 1918-1919 the streets of Berlin were taken over by the people. Political demonstrations became common.

in many cities. The political atmosphere in Berlin was charged with demands for Soviet-style governance. Those opposed to this – such as the socialists, Democrats and Catholics – met in Weimar to give shape to the democratic republic. The Weimar Republic crushed the uprising with the help of a war veterans organisation called Free Corps. The anguished Spartacists later founded the Communist Party of Germany. Communists and Socialists henceforth became irreconcilable enemies and could not make common cause against Hitler. Both revolutionaries and militant nationalists craved for radical solutions.

Political radicalisation was only heightened by the economic crisis of 1923. Germany had fought the war largely on loans and had to pay war **reparations** in gold. This **depleted** gold reserves at a time resources were scarce. In 1923 Germany refused to pay, and the French occupied its leading industrial area, Ruhr, to claim their coal. Germany retaliated with passive resistance and printed paper currency recklessly. With too much printed money in circulation, the value of the German mark fell. In April the US dollar was equal to 24,000 marks, in July 353,000 marks, in August 4,621,000 marks and at

New words

Deplete – Reduce, empty out
Reparation – Make up for a wrong done



***Fig.4** – Baskets and carts being loaded at a bank in Berlin with paper currency for wage payment, 1923. The German mark had so little value that vast amounts had to be used even for small payments.*

98,860,000 marks by December, the figure had run into trillions. As the value of the mark collapsed, prices of goods soared. The image of Germans carrying cartloads of currency notes to buy a loaf of bread was widely publicised evoking worldwide sympathy. This crisis came to be known as hyperinflation, a situation when prices rise phenomenally high.

Eventually, the Americans intervened and bailed Germany out of the crisis by introducing the Dawes Plan, which reworked the terms of reparation to ease the financial burden on Germans.



Fig.5 – Homeless men queuing up for a night's shelter, 1923.

1.3 The Years of Depression

The years between 1924 and 1928 saw some stability. Yet this was built on sand. German investments and industrial recovery were totally dependent on short-term loans, largely from the USA. This support was withdrawn when the **Wall Street Exchange** crashed in 1929. Fearing a fall in prices, people made frantic efforts to sell their shares. On one single day, 24 October, 13 million shares were sold. This was the start of the Great Economic Depression. Over the next three years, between 1929 and 1932, the national income of the USA fell by half. Factories shut down, exports fell, farmers were badly hit and speculators withdrew their money from the market. The effects of this recession in the US economy were felt worldwide.

The German economy was the worst hit by the economic crisis. By 1932, industrial production was reduced to 40 per cent of the 1929 level. Workers lost their jobs or were paid reduced wages. The number of unemployed touched an unprecedented 6 million. On the streets of Germany you could see men with placards around their necks saying, 'Willing to do any work'. Unemployed youths played cards or simply sat at street corners, or desperately queued up at the local employment exchange. As jobs disappeared, the youth took to criminal activities and total despair became commonplace.

The economic crisis created deep anxieties and fears in people. The middle classes, especially salaried employees and pensioners, saw their savings diminish when the currency lost its value. Small businessmen, the self-employed and retailers suffered as their



Fig.6 – Sleeping on the line. During the Great Depression the unemployed could not hope for either wage or shelter. On winter nights when they wanted a shelter over their head, they had to pay to sleep like this.

New words

Wall Street Exchange – The name of the world's biggest stock exchange located in the USA.

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businesses got ruined. These sections of society were filled with the fear of **proletarianisation**, an anxiety of being reduced to the ranks of the working class, or worse still, the unemployed. Only organised workers could manage to keep their heads above water, but unemployment weakened their bargaining power. Big business was in crisis. The large mass of peasantry was affected by a sharp fall in agricultural prices and women, unable to fill their children's stomachs, were filled with a sense of deep despair.

Politically too the Weimar Republic was fragile. The Weimar constitution had some inherent defects, which made it unstable and vulnerable to dictatorship. One was proportional representation. This made achieving a majority by any one party a near impossible task, leading to a rule by coalitions. Another defect was Article 48, which gave the President the powers to impose emergency, suspend civil rights and rule by decree. Within its short life, the Weimar Republic saw twenty different cabinets lasting on an average 239 days, and a liberal use of Article 48. Yet the crisis could not be managed. People lost confidence in the democratic parliamentary system, which seemed to offer no solutions.

New words

Proletarianisation – To become impoverished to the level of working classes.



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2 Hitler's Rise to Power

This crisis in the economy, polity and society formed the background to Hitler's rise to power. Born in 1889 in Austria, Hitler spent his youth in poverty. When the First World War broke out, he enrolled for the army, acted as a messenger in the front, became a corporal, and earned medals for bravery. The German defeat horrified him and the Versailles Treaty made him furious. In 1919, he joined a small group called the German Workers' Party. He subsequently took over the organisation and renamed it the National Socialist German Workers' Party. This party came to be known as the Nazi Party.

In 1923, Hitler planned to seize control of Bavaria, march to Berlin and capture power. He failed, was arrested, tried for treason, and later released. The Nazis could not effectively mobilise popular support till the early 1930s. It was during the Great Depression that Nazism became a mass movement. As we have seen, after 1929, banks collapsed and businesses shut down, workers lost their jobs and the middle classes were threatened with destitution. In such a situation Nazi **propaganda** stirred hopes of a better future. In 1928, the Nazi Party got no more than 2. 6 per cent votes in the Reichstag – the German parliament. By 1932, it had become the largest party with 37 per cent votes.



Fig. 7 – Hitler being greeted at the Party Congress in Nuremberg in 1938.

New words

Propaganda – Specific type of message directly aimed at influencing the opinion of people (through the use of posters, films, speeches, etc.)

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Fig.8 – Nuremberg Rally, 1936.
Rallies like this were held every year. An important aspect of these was the demonstration of Nazi power as various organisations paraded past Hitler, swore loyalty and listened to his speeches.

Hitler was a powerful speaker. His passion and his words moved people. He promised to build a strong nation, undo the injustice of the Versailles Treaty and restore the dignity of the German people. He promised employment for those looking for work, and a secure future for the youth. He promised to weed out all foreign influences and resist all foreign 'conspiracies' against Germany.

Hitler devised a new style of politics. He understood the significance of rituals and spectacle in mass mobilisation. Nazis held massive rallies



Fig.9 — Hitler addressing SA and SS columns.
Notice the sweeping and straight columns of people. Such photographs were intended to show the grandeur and power of the Nazi movement

and public meetings to demonstrate the support for Hitler and instil a sense of unity among the people. The Red banners with the Swastika, the Nazi salute, and the ritualised rounds of applause after the speeches were all part of this spectacle of power.

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Nazi propaganda skilfully projected Hitler as a messiah, a saviour, as someone who had arrived to deliver people from their distress. It is an image that captured the imagination of a people whose sense of dignity and pride had been shattered, and who were living in a time of acute economic and political crises.

2.1 The Destruction of Democracy

On 30 January 1933, President Hindenburg offered the Chancellorship, the highest position in the cabinet of ministers, to Hitler. By now the Nazis had managed to rally the conservatives to their cause. Having acquired power, Hitler set out to dismantle the structures of democratic rule. A mysterious fire that broke out in the German Parliament building in February facilitated his move. The Fire Decree of 28 February 1933 indefinitely suspended civic rights like freedom of speech, press and assembly that had been guaranteed by the Weimar constitution. Then he turned on his arch-enemies, the Communists, most of whom were hurriedly packed off to the newly established **concentration camps**. The repression of the Communists was severe. Out of the surviving 6,808 arrest files of Duesseldorf, a small city of half a million population, 1,440 were those of Communists alone. They were, however, only one among the 52 types of victims persecuted by the Nazis across the country.

On 3 March 1933, the famous Enabling Act was passed. This Act established dictatorship in Germany. It gave Hitler all powers to sideline Parliament and rule by decree. All political parties and trade unions were banned except for the Nazi Party and its affiliates. The state established complete control over the economy, media, army and judiciary.

Special surveillance and security forces were created to control and order society in ways that the Nazis wanted. Apart from the already existing regular police in green uniform and the SA or the Storm Troopers, these included the Gestapo (secret state police), the SS (the protection squads), criminal police and the Security Service (SD). It was the extra-constitutional powers of these newly organised forces that gave the Nazi state its reputation as the most dreaded criminal state. People could now be detained in Gestapo torture chambers, rounded up and sent to concentration camps, deported at will or arrested without any legal procedures. The police forces acquired powers to rule with impunity.

New words

Concentration camp – A camp where people were isolated and detained without due process of law. Typically, it was surrounded by electrified barbed wire fences.

2.2 Reconstruction

Hitler assigned the responsibility of economic recovery to the economist Hjalmar Schacht who aimed at full production and full employment through a state-funded work-creation programme. This project produced the famous German superhighways and the people's car, the Volkswagen.

In foreign policy also Hitler acquired quick successes. He pulled out of the League of Nations in 1933, reoccupied the Rhineland in 1936, and integrated Austria and Germany in 1938 under the slogan, *One people, One empire, and One leader*. He then went on to wrest German-speaking Sudentenland from Czechoslovakia, and gobbled up the entire country. In all of this he had the unspoken support of England, which had considered the Versailles verdict too harsh. These quick successes at home and abroad seemed to reverse the destiny of the country.

Hitler did not stop here. Schacht had advised Hitler against investing hugely in rearmament as the state still ran on deficit financing. Cautious people, however, had no place in Nazi Germany. Schacht had to leave. Hitler chose war as the way out of the approaching

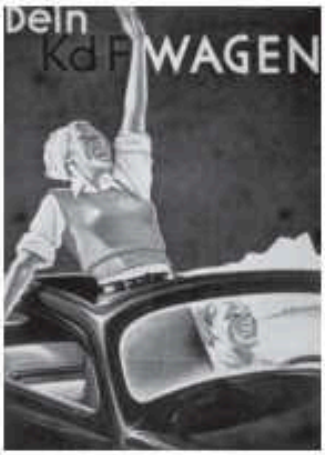


***Fig. 10** – The poster announces: 'Your volkswagen'.*
Such posters suggested that owning a car was no longer just a dream for an ordinary worker.



***Fig. 11** – Expansion of Nazi power: Europe 1942.*

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economic crisis. Resources were to be accumulated through expansion of territory. In September 1939, Germany invaded Poland. This started a war with France and England. In September 1940, a Tripartite Pact was signed between Germany, Italy and Japan, strengthening Hitler's claim to international power. Puppet regimes, supportive of Nazi Germany, were installed in a large part of Europe. By the end of 1940, Hitler was at the pinnacle of his power.

Hitler now moved to achieve his long-term aim of conquering Eastern Europe. He wanted to ensure food supplies and living space for Germans. He attacked the Soviet Union in June 1941. In this historic blunder Hitler exposed the German western front to British aerial bombing and the eastern front to the powerful Soviet armies. The Soviet Red Army inflicted a crushing and humiliating defeat on Germany at Stalingrad. After this the Soviet Red Army hounded out the retreating German soldiers until they reached the heart of Berlin, establishing Soviet hegemony over the entire Eastern Europe for half a century thereafter.

Meanwhile, the USA had resisted involvement in the war. It was unwilling to once again face all the economic problems that the First World War had caused. But it could not stay out of the war for long. Japan was expanding its power in the east. It had occupied French Indo-China and was planning attacks on US naval bases in the Pacific. When Japan extended its support to Hitler and bombed the US base at Pearl Harbor, the US entered the Second World War. The war ended in May 1945 with Hitler's defeat and the US dropping of the atom bomb on Hiroshima in Japan.

From this brief account of what happened in the Second World War, we now return to Helmuth and his father's story, a story of Nazi criminality during the war.



Fig. 12 – Newspapers in India track the developments in Germany.

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3 The Nazi Worldview

The crimes that Nazis committed were linked to a system of belief and a set of practices.

Nazi ideology was synonymous with Hitler's worldview. According to this there was no equality between people, but only a racial hierarchy. In this view blond, blue-eyed, Nordic German Aryans were at the top, while Jews were located at the lowest rung. They came to be regarded as an anti-race, the arch-enemies of the Aryans. All other coloured people were placed in between depending upon their external features. Hitler's racism borrowed from thinkers like Charles Darwin and Herbert Spencer. Darwin was a natural scientist who tried to explain the creation of plants and animals through the concept of evolution and natural selection. Herbert Spencer later added the idea of survival of the fittest. According to this idea, only those species survived on earth that could adapt themselves to changing climatic conditions. We should bear in mind that Darwin never advocated human intervention in what he thought was a purely natural process of selection. However, his ideas were used by racist thinkers and politicians to justify imperial rule over conquered peoples. The Nazi argument was simple: the strongest race would survive and the weak ones would perish. The Aryan race was the finest. It had to retain its purity, become stronger and dominate the world.

The other aspect of Hitler's ideology related to the geopolitical concept of *Lebensraum*, or living space. He believed that new territories had to be acquired for settlement. This would enhance the area of the mother country, while enabling the settlers on new lands to retain an intimate link with the place of their origin. It would also enhance the material resources and power of the German nation.

Hitler intended to extend German boundaries by moving eastwards, to concentrate all Germans geographically in one place. Poland became the laboratory for this experimentation.

3.1 Establishment of the Racial State

Once in power, the Nazis quickly began to implement their dream of creating an exclusive racial community of pure Germans by physically eliminating all those who were seen as 'undesirable' in the

Source A

'For this earth is not allotted to anyone nor is it presented to anyone as a gift. It is awarded by providence to people who in their hearts have the courage to conquer it, the strength to preserve it, and the industry to put it to the plough... The primary right of this world is the right to life, so far as one possesses the strength for this. Hence on the basis of this right a vigorous nation will always find ways of adapting its territory to its population size.'

Hitler, Secret Book, ed. Telford Taylor.

Source B

'In an era when the earth is gradually being divided up among states, some of which embrace almost entire continents, we cannot speak of a world power in connection with a formation whose political mother country is limited to the absurd area of five hundred kilometers.'

Hitler, Mein Kampf, *p. 644.*

Activity

Read Sources A and B

- What do they tell you about Hitler's imperial ambition?
- What do you think Mahatma Gandhi would have said to Hitler about these ideas?

New words

Nordic German Aryans – One branch of those classified as Aryans. They lived in north European countries and had German or related origin.

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extended empire. Nazis wanted only a society of 'pure and healthy Nordic Aryans'. They alone were considered 'desirable'. Only they were seen as worthy of prospering and multiplying against all others who were classed as 'undesirable'. This meant that even those Germans who were seen as impure or abnormal had no right to exist. Under the Euthanasia Programme, Helmuth's father along with other Nazi officials had condemned to death many Germans who were considered mentally or physically unfit.

Jews were not the only community classified as 'undesirable'. There were others. Many **Gypsies** and blacks living in Nazi Germany were considered as racial 'inferiors' who threatened the biological purity of the 'superior Aryan' race. They were widely **persecuted**. Even Russians and Poles were considered subhuman, and hence undeserving of any humanity. When Germany occupied Poland and parts of Russia, captured civilians were forced to work as slave labour. Many of them died simply through hard work and starvation.

Jews remained the worst sufferers in Nazi Germany. Nazi hatred of Jews had a precursor in the traditional Christian hostility towards Jews. They had been stereotyped as killers of Christ and **usurers**.Until medieval times Jews were barred from owning land. They survived mainly through trade and moneylending. They lived in separately marked areas called ghettos. They were often persecuted through periodic organised violence, and expulsion from the land. However, Hitler's hatred of Jews was based on pseudoscientific theories of race, which held that conversion was no solution to 'the Jewish problem'. It could be solved only through their total elimination.

From 1933 to 1938 the Nazis terrorised, **pauperised** and segregated the Jews, compelling them to leave the country. The next phase, 1939-1945, aimed at concentrating them in certain areas and eventually killing them in gas chambers in Poland.

3.2 The Racial Utopia

Under the shadow of war, the Nazis proceeded to realise their murderous, racial ideal. Genocide and war became two sides of the same coin. Occupied Poland was divided up. Much of north-western Poland was annexed to Germany. Poles were forced to leave their homes and properties behind to be occupied by ethnic Germans brought in from occupied Europe. Poles were then herded like



Fig. 13 – Police escorting gypsies who are being deported to Auschwitz, 1943-1944.

New words

Gypsy – The groups that were classified as 'gypsy' had their own community identity. Sinti and Roma were two such communities. Many of them traced their origin to India.
Pauperised – Reduce to absolute poverty
Persecution – Systematic, organised punishment of those belonging to a group or religion
Usurers – Moneylenders charging excessive interest; often used as a term of abuse

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cattle in the other part called the General Government, the destination of all 'undesirables' of the empire. Members of the Polish intelligentsia were murdered in large numbers in order to keep the entire people intellectually and spiritually servile. Polish children who looked like Aryans were forcibly snatched from their mothers and examined by 'race experts'. If they passed the race tests they were raised in German families and if not, they were deposited in orphanages where most perished. With some of the largest ghettos and gas chambers, the General Government also served as the killing fields for the Jews.

Activity

See the next two pages and write briefly:

- What does citizenship mean to you? Look at Chapters I and 3 and write 200 words on how the French Revolution and Nazism defined citizenship.
- What did the Nuremberg Laws mean to the 'undesirables' in Nazi Germany? What other legal measures were taken against them to make them feel unwanted?



***Fig. 14** – This is one of the freight cars used to deport Jews to the death chambers.*

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STEPS TO DEATH

Stage 1: Exclusion 1933–1939

YOU HAVE NO RIGHT TO LIVE AMONG US AS CITIZENS

The Nuremberg Laws of citizenship of September 1935:

1. Only Persons of German or related blood would henceforth be German citizens enjoying the protection of the German empire.
2. Marriages between Jews and Germans were forbidden.
3. Extramarital relations between Jews and Germans became a crime.
4. Jews were forbidden to fly the national flag.

Other legal measures included:

- Boycott of Jewish businesses
- Expulsion from government services
- Forced selling and confiscation of their properties

Besides, Jewish properties were vandalised and looted, houses attacked, synagogues burnt and men arrested in a pogrom in November, 1938, remembered as 'the night of broken glass'

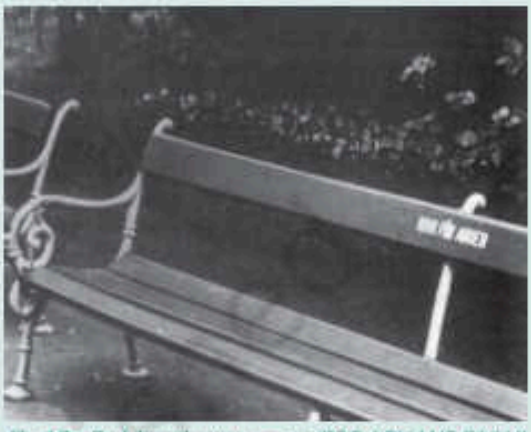


Fig.16 – Park bench announces: 'FOR ARYANS ONLY'

New words

Synagogues – Place of worship for people of Jewish faith



Fig.15 – The sign declares that this North Sea bathing resort is free of Jews.



Fig.17 – 'This is all I have to sell'. Men and women were left with nothing to survive in the ghettos.

Stage 2: Ghettoisation 1940 - 1944

YOU HAVE NO RIGHT TO LIVE AMONG US

From September 1941, all Jews had to wear a yellow Star of David on their breasts. This identity mark was stamped on their passport, all legal documents and houses. They were kept in Jewish houses in Germany, and in ghettos like Lodz and Warsaw in the east. These became sites of extreme misery and poverty. Jews had to surrender all their wealth before they entered a ghetto. Soon the ghettos were brimming with hunger, starvation and disease due to deprivation and poor hygiene.

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Stage 3: Annihilation 1941 onwards:

YOU HAVE NO RIGHT TO LIVE



Fig.18 – Killed while trying to escape. The concentration camps were enclosed with live wires.



Fig.19 – Piles of clothes outside the gas chamber.

Jews from Jewish houses, concentration camps and ghettos from different parts of Europe were brought to death factories by goods trains. In Poland and elsewhere in the east, most notably Belzek, Auschwitz, Sobibor, Treblinka, Chelmno and Majdanek, they were charred in gas chambers. Mass killings took place within minutes with scientific precision.



Fig.20 – A Concentration Camp.



Fig.21 – A concentration camp. A camera can make a death camp look beautiful.



Fig.22 – Shoes taken away from prisoners before the 'Final Solution'.

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4 Youth in Nazi Germany

Hitler was fanatically interested in the youth of the country. He felt that a strong Nazi society could be established only by teaching children Nazi ideology. This required a control over the child both inside and outside school.

What happened in schools under Nazism? All schools were 'cleansed' and 'purified'. This meant that teachers who were Jews or seen as 'politically unreliable' were dismissed. Children were first segregated: Germans and Jews could not sit together or play together. Subsequently, 'undesirable children' – Jews, the physically handicapped, Gypsies – were thrown out of schools. And finally in the 1940s, they were taken to the gas chambers.

'Good German' children were subjected to a process of Nazi schooling, a prolonged period of ideological training. School textbooks were rewritten. Racial science was introduced to justify Nazi ideas of race. Stereotypes about Jews were popularised even through maths classes. Children were taught to be loyal and submissive, hate Jews, and worship Hitler. Even the function of sports was to nurture a spirit of violence and aggression among children. Hitler believed that boxing could make children iron hearted, strong and masculine.

Youth organisations were made responsible for educating German youth in the 'the spirit of National Socialism'. Ten-year-olds had to enter **Jungvolk**. At 14, all boys had to join the Nazi youth organisation – Hitler Youth – where they learnt to worship war, glorify aggression and violence, condemn democracy, and hate Jews, communists, Gypsies and all those categorised as 'undesirable'. After a period of rigorous ideological and physical training they joined the Labour Service, usually at the age of 18. Then they had to serve in the armed forces and enter one of the Nazi organisations.

The Youth League of the Nazis was founded in 1922. Four years later it was renamed Hitler Youth. To unify the youth movement under Nazi control, all other youth organisations were systematically dissolved and finally banned.

New words

Jungvolk – Nazi youth groups for children below 14 years of age.



Fig. 23 – Classroom scene depicting a lesson on racial anti-Semitism.
From Der Giftpilz (The Poison Mushroom) *by Ernst Hiemer (Nuremberg: der Sturmer, 1938), p. 7. Caption reads: 'The Jewish nose is bent at its point. It looks like the number six.'*



Fig. 24 – Jewish teacher and Jewish pupils expelled from school under the jeers of classmates.
From Trau keinem jud auf gruner Heid: Ein Bilderbuch fur Gross und Keorn (Trust No Jew on the Green Heath: a Picture Book for Big and Little), *By Elvira Bauer (Nuremberg: Der Sturmer, 1936).*

Activity

If you were a student sitting in one of these classes, how would you have felt towards Jews?

Have you ever thought of the stereotypes of other communities that people around you believe in? How have they acquired them?

Source: C

All boys between the ages of six and ten went through a preliminary training in Nazi ideology. At the end of the training they had to take the following oath of loyalty to Hitler:

'In the presence of this blood banner which represents our Fuhrer I swear to devote all my energies and my strength to the saviour of our country, Adolf Hitler. I am willing and ready to give up my life for him, so help me God.'

From W. Shirer, The Rise and Fall of the Third Reich

Source: D

Robert Lay, head of the German Labour Front, said:

'We start when the child is three years old. As soon as he even starts to think, he is given a little flag to wave. Then comes school, the Hitler Youth, military service. But when all this is over, we don't let go of anyone. The labour front takes hold of them, and keeps hold until they go to the grave, whether they like it or not.'



***Fig.25** – 'Desirable' children that Hitler wanted to see multiplied.*



***Fig.26** – A German-blooded infant with his mother being brought from occupied Europe to Annexed Poland for settlement.*



***Fig.27** – Jewish children arriving at a death factory to be gassed*

Activity

Look at Figs. 23, 24, and 27. Imagine yourself to be a Jew or a Pole in Nazi Germany. It is September 1941, and the law forcing Jews to wear the Star of David has just been declared. Write an account of one day in your life.

4.1 The Nazi Cult of Motherhood

Children in Nazi Germany were repeatedly told that women were radically different from men. The fight for equal rights for men and women that had become part of democratic struggles everywhere was wrong and it would destroy society. While boys were taught to be aggressive, masculine and steel hearted, girls were told that they had to become good mothers and rear pure-blooded Aryan children. Girls had to maintain the purity of the race, distance

themselves from Jews, look after the home, and teach their children Nazi values. They had to be the bearers of the Aryan culture and race.

In 1933 Hitler said: 'In my state the mother is the most important citizen.' But in Nazi Germany all mothers were not treated equally. Women who bore racially undesirable children were punished and those who produced racially desirable children were awarded. They were given favoured treatment in hospitals and were also entitled to concessions in shops and on theatre tickets and railway fares. To encourage women to produce many children, Honour Crosses were awarded. A bronze cross was given for four children, silver for six and gold for eight or more.

All 'Aryan' women who deviated from the prescribed code of conduct were publicly condemned, and severely punished. Those who maintained contact with Jews, Poles and Russians were paraded through the town with shaved heads, blackened faces and placards hanging around their necks announcing 'I have sullied the honour of the nation'. Many received jail sentences and lost civic honour as well as their husbands and families for this 'criminal offence'.

4.2. The Art of Propaganda

The Nazi regime used language and media with care, and often to great effect. The terms they coined to describe their various practices are not only deceptive. They are chilling. Nazis never used the words 'kill' or 'murder' in their official communications. Mass killings were termed *special treatment, final solution* (for the Jews), *euthanasia* (for the disabled), *selection* and *disinfections*. 'Evacuation' meant deporting people to gas chambers. Do you know what the gas chambers were called? They were labelled 'disinfection-areas', and looked like bathrooms equipped with fake showerheads.

Media was carefully used to win support for the regime and popularise its worldview. Nazi ideas were spread through visual images, films, radio, posters, catchy slogans and leaflets. In posters, groups identified as the 'enemies' of Germans were stereotyped, mocked, abused and described as evil. Socialists and liberals were represented as weak and degenerate. They were attacked as malicious foreign agents. Propaganda films were made to create hatred for Jews. The most infamous film was *The Eternal Jew*. Orthodox Jews were stereotyped and marked. They were shown

Source E

In an address to women at the Nuremberg Party Rally, 8 September 1934, Hitler said:

We do not consider it correct for the woman to interfere in the world of the man, in his main sphere. We consider it natural that these two worlds remain distinct...What the man gives in courage on the battlefield, the woman gives in eternal self-sacrifice, in eternal pain and suffering. Every child that women bring to the world is a battle, a battle waged for the existence of her people.

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Joint Implementation (JI): A mechanism defined in Article 6 of the *Kyoto Protocol*, through which investors (governments or companies) from developed *(Annex B) countries* may implement projects jointly that limit or reduce emissions or enhance *sinks*, and to share the *Emissions Reduction Units (ERU)*. See also *Kyoto Mechanisms*.

Kaya identity: In this identity global emissions are equal to the population size, multiplied by per capita output (*gross world product*), multiplied by the *energy intensity* of production, multiplied by the *carbon intensity* of *energy*.

Kyoto Mechanisms (also referred to as Flexibility Mechanisms): Market-based mechanisms that Parties to the *Kyoto Protocol* can use in an attempt to lessen the potential economic impacts of their commitment to limit or reduce *greenhouse gas (GHG)* emissions. They include *Joint Implementation (JI)* (Article 6), *Clean Development Mechanism (CDM)* (Article 12), and *Emissions trading* (Article 17).

Kyoto Protocol: The Kyoto Protocol to the *United Nations Framework Convention on Climate Change (UNFCCC)* was adopted in 1997 in Kyoto, Japan, at the Third Session of the *Conference of the Parties (COP)* to the UNFCCC. It contains legally binding commitments, in addition to those included in the UNFCCC. Countries included in *Annex B* of the Protocol (most Organisation for Economic Cooperation and Development countries and countries with economies in transition) agreed to reduce their anthropogenic *greenhouse gas (GHG)* emissions (*carbon dioxide (CO_2)*, *methane (CH_4)*, *nitrous oxide (N_2O)*, *hydrofluorocarbons (HFCs)*, *perfluorocarbons (PFCs)*, and *sulphur hexafluoride (SF_6)*) by at least 5% below 1990 levels in the commitment period 2008–2012. The Kyoto Protocol entered into force on 16 February 2005.

Land use (change, direct and indirect): Land use refers to the total of arrangements, activities and inputs undertaken in a certain land cover type (a set of human actions). The term land use is also used in the sense of the social and economic purposes for which land is managed (e.g., grazing, timber extraction and conservation). In urban settlements it is related to land uses within cities and their hinterlands. Urban land use has implications on city management, structure, and form and thus on energy demand, *greenhouse gas (GHG)* emissions, and mobility, among other aspects.

Land use change (LUC): Land use change refers to a change in the use or management of land by humans, which may lead to a change in land cover. Land cover and LUC may have an impact on the surface *albedo*, evapotranspiration, *sources* and *sinks* of GHGs, or other properties of the *climate system* and may thus give rise to *radiative forcing* and/or other impacts on *climate*, locally or globally. See also the IPCC Report on Land Use, Land-Use Change, and Forestry (IPCC, 2000).

Indirect land use change (iLUC): Indirect land use change refers to shifts in land use induced by a change in the production level of an agricultural product elsewhere, often mediated by markets or driven by *policies*. For example, if agricultural land is diverted to fuel production, *forest* clearance may occur elsewhere to replace the former agricultural production. See also *Afforestation*, *Deforestation* and *Reforestation*.

Land use, land use change and forestry (LULUCF): A *greenhouse gas (GHG)* inventory sector that covers *emissions* and removals of GHGs resulting from direct human-induced *land use*, *land use change* and forestry activities excluding *agricultural emissions*. See also *Agriculture, Forestry and Other Land Use (AFOLU)*.

Land value capture: A financing mechanism usually based around transit systems, or other infrastructure and services, that captures the increased value of land due to improved accessibility.

Leakage: Phenomena whereby the reduction in emissions (relative to a *baseline*) in a jurisdiction/sector associated with the implementation of *mitigation policy* is offset to some degree by an increase outside the jurisdiction/sector through induced changes in consumption, production, prices, land use and/or trade across the jurisdictions/sectors. Leakage can occur at a number of levels, be it a project, state, province, nation, or world region. See also *Rebound effect*.

In the context of *Carbon Dioxide Capture and Storage (CCS)*, 'CO_2 leakage' refers to the escape of injected *carbon dioxide (CO_2)* from the storage location and eventual release to the atmosphere. In the context of other substances, the term is used more generically, such as for '*methane (CH_4)* leakage' (e.g., from *fossil fuel* extraction activities), and '*hydrofluorocarbon (HFC)* leakage' (e.g., from refrigeration and air-conditioning systems).

Learning curve/rate: Decreasing cost-prices of technologies shown as a function of increasing (total or yearly) supplies. The learning rate is the percent decrease of the cost-price for every doubling of the cumulative supplies (also called progress ratio).

Least Developed Countries (LDCs): A list of countries designated by the Economic and Social Council of the United Nations (ECOSOC) as meeting three criteria: (1) a low income criterion below a certain threshold of gross national income per capita of between USD 750 and USD 900, (2) a human resource weakness based on indicators of health, education, adult literacy, and (3) an economic vulnerability weakness based on indicators on instability of agricultural production, instability of export of goods and services, economic importance of non-traditional activities, merchandise export concentration, and the handicap of economic smallness. Countries in this category are eligible for a number of programmes focused on assisting countries most in need. These privileges include certain benefits under the articles of the *United Nations Framework Convention on Climate Change (UNFCCC)*. See also *Industrialized/developing countries*.

Levelized cost of conserved carbon (LCCC): See Annex II.3.1.3 for concepts and definition.

Pakistan, Thailand, Laos, Cambodia, Vietnam and the Philippines. The SAsiaM starts in late May/early June and progresses towards the northeast, ending in late September/early October. During the core monsoon season, maxima of SAsiaM precipitation are located over the west coast, north-east and central north India, Myanmar and Bangladesh, whereas minima are located over north-west and south-eastern India, western Pakistan, and south-eastern and northern Sri Lanka. Further details on how SAsiaM is defined and used throughout the Report are provided in Annex V.

West African monsoon (WAfriM)
The West African monsoon (WAfriM) is a seasonal reversal in wind and precipitation whose domain includes Benin, Burkina-Faso, northern Cameroon, Cape Verde, northern Central African Republic, Chad, Gambia, Ghana, Guinea, Guinea Bissau, Ivory Coast, Liberia, Mali, Mauritania, Niger, Nigeria, Senegal, Sierra Leone and Togo. The WAfriM is characterized by the northward progression from May to September of moist low-level south-westerlies from the Gulf of Guinea. In May and June, rainfall essentially remains along the Guinean coast with a maximum occurring near 5°N, followed by a sudden decrease of rainfall, marking the 'short dry season' in the Guinean coast and the monsoon onset in the Sahel. Then rainfall continues to progress northward up to about 18–20°N, with a maximum near 12°N in late August/September, until it retreats starting from October towards the Guinean coast for a second maximum. Further details on how WAfriM is defined and used throughout the Report are provided in Annex V.

Global surface temperature See *Global mean surface temperature (GMST)* and *Global mean surface air temperature (GSAT)*. See also *Global warming*.

Global warming Global warming refers to the increase in *global surface temperature* relative to a baseline *reference period*, averaging over a period sufficient to remove interannual variations (e.g., 20 or 30 years). A common choice for the baseline is 1850–1900 (the earliest period of reliable observations with sufficient geographic coverage), with more modern baselines used depending upon the application. See also *Climate change* and *Climate variability*.

Global warming potential (GWP) An index measuring the *radiative forcing* following an emission of a unit mass of a given substance, accumulated over a chosen time horizon, relative to that of the reference substance, *carbon dioxide (CO_2)*. The GWP thus represents the combined effect of the differing times these substances remain in the *atmosphere* and their effectiveness in causing radiative forcing. See also *Lifetime* and *Greenhouse gas emission metric*.

Gravitational, rotational and deformational (GRD) effects See *Sea level change (sea level rise/sea level fall)*.

Gravity Recovery and Climate Experiment (GRACE) A pair of satellites that measured the Earth's gravity field anomalies from 2002 to 2017. These fields have been used, among other things, to study mass changes of the polar *ice sheets* and *glaciers*.

Greenhouse effect The infrared radiative effect of all infrared-absorbing constituents in the *atmosphere*. *Greenhouse gases (GHGs)*, clouds, and some *aerosols* absorb *terrestrial radiation* emitted by the Earth's surface and elsewhere in the atmosphere. These substances emit *infrared radiation* in all directions, but, everything else being equal, the net amount emitted to space is normally less than would have been emitted in the absence of these absorbers because of the decline of temperature with altitude in the *troposphere* and the consequent weakening of emission. An increase in the concentration of GHGs increases the magnitude of this effect; the difference is sometimes called the enhanced greenhouse effect. The change in a GHG concentration because of *anthropogenic emissions* contributes to an instantaneous radiative forcing. Earth's surface temperature and *troposphere* warm in response to this *forcing*, gradually restoring the radiative balance at the top of the atmosphere.

Greenhouse gas emission metric A simplified relationship used to quantify the effect of emitting a unit mass of a given *greenhouse gas* on a specified key measure of *climate change*. A relative GHG emission metric expresses the effect from one gas relative to the effect of emitting a unit mass of a reference GHG on the same measure of climate change. There are multiple emission metrics, and the most appropriate metric depends on the application. GHG emission metrics may differ with respect to (i) the key measure of climate change they consider, (ii) whether they consider climate outcomes for a specified point in time or integrated over a specified time horizon, (iii) the time horizon over which the metric is applied, (iv) whether they apply to a single emission pulse, emissions sustained over a period of time, or a combination of both, and (v) whether they consider the climate effect from an emission compared to the absence of that emission or compared to a reference emissions level or climate state.

Notes: Most relative GHG emission metrics (such as the *global warming potential (GWP)*, global temperature change potential (GTP), global damage potential, and GWP*) use *carbon dioxide (CO_2)* as the reference gas. Emissions of non-CO_2 gases, when expressed using such metrics, are often referred to as 'carbon dioxide equivalent' emissions. A metric that establishes equivalence regarding one key measure of the *climate system* response to emissions does not imply equivalence regarding other key measures. The choice of a metric, including its time horizon, should reflect the policy objectives for which the metric is applied.

Greenhouse gas neutrality Condition in which metric-weighted anthropogenic *greenhouse gas (GHG)* emissions associated with a subject are balanced by metric-weighted anthropogenic GHG removals. The subject can be an entity such as a country, an organization, a district or a commodity, or an activity such as a service or an event. GHG neutrality is often assessed over the life cycle, including indirect ('scope 3') emissions, but can also be limited to the emissions and removals, over a specified period, for which the subject has direct control, as determined by the relevant scheme. The quantification of GHG emissions and removals depends on the GHG emission metric chosen to compare emissions and removals of different gases, as well as the time horizon chosen for that metric.

Note 1: GHG neutrality and net zero GHG emissions are overlapping concepts. The concepts can be applied at global or sub-global scales (e.g., regional, national and sub-national). At a global scale, the terms greenhouse gas neutrality and net zero greenhouse gas emissions are equivalent. At sub-global scales, net zero greenhouse gas emissions is generally applied to emissions and removals under direct control or territorial responsibility of the reporting entity, while greenhouse gas neutrality generally includes emissions and removals within and

or may not be realized. See also *Climate projection*, *Pathways* and *Scenario*.

Proxy A proxy *climate indicator* is any biophysical property of materials formed during the past that is interpreted to represent some combination of climate-related variations back in time. Climate-related data derived in this way are referred to as proxy data, and time series of proxy data are proxy records. Examples of proxy types include pollen assemblages, *tree ring* widths, speleothem and coral geochemistry, and various data derived from marine sediments and *glacier* ice. Proxy data can be calibrated to provide quantitative *climate information*.

Proxy records See *Proxy*.

Quasi-Biennial Oscillation (QBO) A near-periodic oscillation of the equatorial zonal wind between easterlies and westerlies in the tropical *stratosphere* with a mean period of around 28 months. The alternating wind maxima descend from the base of the mesosphere down to the *tropopause* and are driven by wave energy that propagates up from the *troposphere*.

Quaternary The Quaternary Period is the last of three periods that make up the *Cenozoic Era* (66 Ma to present), extending from 2.58 Ma to the present, and includes the *Pleistocene* and *Holocene* Epochs.

Radiative forcing The change in the net, downward minus upward, radiative *flux* (expressed in W m^{-2}) due to a change in an external driver of *climate change*, such as a change in the concentration of *carbon dioxide* (CO_2), the concentration of volcanic *aerosols* or the output of the Sun. The stratospherically adjusted radiative forcing is computed with all tropospheric properties held fixed at their unperturbed values, and after allowing for stratospheric temperatures, if perturbed, to readjust to radiative-dynamical equilibrium. Radiative forcing is called instantaneous if no change in stratospheric temperature is accounted for. The radiative forcing once both stratospheric and tropospheric adjustments are accounted for is termed the effective radiative forcing.

Radiative response (of the climate system) The net top-of-atmosphere radiative flux that opposes a change in *radiative forcing* as a result of *climate feedbacks*. Typical units: W m^{-2}. See also *Earth's energy budget and Climate feedback parameter*.

Rapid dynamical change (of glaciers or ice sheets) Changes in *glacier* or *ice sheet* mass controlled by changes in flow speed and *discharge* rather than by *accumulation* or *ablation*. This can result in a rate of mass change larger than that due to any imbalance between accumulation and ablation. Rapid dynamical change may be initiated by a climatic trigger, such as incursion of warm *ocean* water beneath an *ice shelf*, or thinning of a grounded tide-water terminus, which may lead to reactions within the glacier system that may result in rapid ice loss.

Reanalysis Reanalyses are created by processing past meteorological or oceanographic data using fixed state-of-the-art weather forecasting or *ocean* circulation models with *data assimilation* techniques. They are used to provide estimates of variables such as historical atmospheric temperature and wind or oceanographic temperature and currents, and other quantities. Using fixed data assimilation avoids effects from the changing analysis system that occur in operational analyses. Although continuity is improved, global reanalyses still suffer from changing coverage and biases in the observing systems.

Reasons for concern (RFCs) Elements of a classification framework, first developed in the IPCC Third Assessment Report, which aims to facilitate judgements about what level of *climate change* may be dangerous (in the language of Article 2 of the UNFCCC; UNFCCC, 1992) by aggregating *risks* from various sectors, considering *hazards*, *exposures*, *vulnerabilities*, capacities to adapt, and the resulting *impacts*.

Reconstruction (of climate variable) Approach to reconstructing the past temporal and spatial characteristics of a *climate* variable from predictors. The predictors can be instrumental data if the reconstruction is used to infill missing data or *proxy* data if it is used to develop *paleoclimate* reconstructions. Various techniques have been developed for this purpose: linear multivariate regression-based methods and non-linear Bayesian and analogue methods.

Reference period A time period of interest, or a period over which some relevant statistics are calculated. A reference period can be used as a baseline period or as a comparison to a baseline period.

Baseline period
A time period against which differences are calculated (e.g., expressed as *anomalies* relative to a baseline).

Reference scenario See *Scenario*.

Reforestation Conversion to *forest* of land that has previously contained forests but that has been converted to some other use. [Note: For a discussion of the term forest and related terms such as *afforestation*, reforestation and *deforestation*, see the 2006 IPCC Guidelines for National Greenhouse Gas Inventories and their 2019 Refinement, and information provided by the United Nations Framework Convention on Climate Change (IPCC, 2006, 2019; UNFCCC, 2021a, b).] See also *Afforestation*, *Deforestation*, *Anthropogenic removals* and *Carbon dioxide removal (CDR)*.

Region *Land* and/or *ocean* area characterized by specific geographical and/or climatological features. The *climate* of a region emerges from a multi-scale combination of its own features, remote influences from other regions, and global climate conditions.

Regional climate model (RCM) A *climate model* at higher *resolution* over a limited area. Such models are used in *downscaling* global *climate* results over specific regional domains.

Regional sea level change See *Sea level change (sea level rise/sea level fall)*.

Relative humidity The ratio of actual water vapour pressure to that at saturation with respect to liquid water or ice at the same temperature. See also *Specific humidity*.

Relative sea level (RSL) change See *Sea level change (sea level rise/sea level fall)*.

Remaining carbon budget See *Carbon budget*.

Representative Concentration Pathways (RCPs) See *Pathways*.

of a team of experts) (see Moss and Schneider, 2000; IPCC, 2004; Mastrandrea et al., 2010). See also *Confidence* and *Likelihood*.

Deep uncertainty
A situation of deep uncertainty exists when experts or stakeholders do not know or cannot agree on: (1) appropriate conceptual models that describe relationships among key driving forces in a system; (2) the probability distributions used to represent uncertainty about key variables and parameters; and/or (3) how to weigh and value desirable alternative outcomes (Lempert et al., 2003).

Interpolation uncertainty
Uncertainty arising from a statistical or physical model-based interpolation of a field between available estimates to create a more spatio-temporally complete estimate.

Sampling uncertainty
Uncertainty arising from incomplete or uneven availability of measurements in either space or time or both.

Trend estimates uncertainty
Uncertainty arising from data fitting to a time-series with potential non-linear and autoregressive character.

United Nations Framework Convention on Climate Change (UNFCCC) The UNFCCC was adopted in May 1992 and opened for signature at the 1992 Earth Summit in Rio de Janeiro. It entered into force in March 1994 and as of September 2020 had 197 Parties (196 States and the European Union). The Convention's ultimate objective is the 'stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system' (UNFCCC, 1992). The provisions of the Convention are pursued and implemented by two further treaties: the Kyoto Protocol and the Paris Agreement.

Uptake The transfer of substances (such as carbon) or energy (e.g., heat) from one compartment of a system to another; for example, in the Earth system from the *atmosphere* to the *ocean* or to the land. See also *Pool, carbon and nitrogen*, *Reservoir*, *Sequestration*, *Sequestration potential*, *Sink* and *Source*.

Upwelling region A region of an *ocean* where cold, typically nutrient-rich waters well up from the deep ocean.

Urban heat island (UHI) The relative warmth of a city compared with surrounding rural areas, associated with heat trapping due to the close proximity of tall buildings, the heat-absorbing properties of urban building materials, reduced ventilation, and heat generated directly from human activities. See also *Urbanization*.

Urbanization In the WGI report, urbanization is used to mean the process of soil sealing with the change of natural *land cover* to built environment and urban areas, together with its associated *albedo* changes, and increased surface *runoff* and elevated warming. See also *Urban heat island (UHI)*.

Ventilation The exchange of *ocean* properties with the atmospheric surface layer such that property concentrations are brought closer to equilibrium values with the *atmosphere* (AMS, 2021), and the processes that propagate these properties into the ocean interior.

Vertical land motion (VLM) The change in height of the land surface or the sea floor and can have several causes in addition to elastic deformation associated with contemporary changes in *gravity, rotation and viscoelastic solid Earth deformation (GRD)* and viscoelastic deformation associated with *glacial isostatic adjustment (GIA)*. Subsidence (sinking of the land surface or sea floor) can, for instance, occur through compaction of alluvial sediments in deltaic regions, removal of fluids such as gas, oil, and water, or drainage of peatlands. Tectonic deformation of the Earth's crust can occur as a result of earthquakes and volcanic eruptions. See also *Sea level change (sea level rise/sea level fall)*.

Very short-lived halogenated substances (VSLSs) Very short-lived halogenated substances (VSLSs) are considered to include source gases (very short-lived halogenated substances present in the *atmosphere* in the form they were emitted from natural and *anthropogenic* sources), halogenated product gases arising from source gas degradation, and other sources of *tropospheric* inorganic halogens. VSLSs have tropospheric *lifetimes* of around 0.5 years or less.

Volatile organic compounds (VOCs) Important class of organic chemical air pollutants that are volatile at ambient air conditions. Other terms used to represent VOCs are hydrocarbons (HCs), reactive organic gases (ROGs) and non-methane volatile organic compounds (NMVOCs). NMVOCs are major contributors – together with nitrogen oxides (NO_x), and carbon monoxide (CO) – to the formation of photochemical oxidants such as *ozone* (O_3).

Biogenic volatile organic compounds (BVOCs)
Organic gas-phase compounds emitted from terrestrial and aquatic ecosystems that are critical in ecology and plant physiology, from abiotic and biotic stress functions to integrated components of metabolism. BVOCs are important in atmospheric chemistry as *precursors* for *ozone* (O_3) and secondary organic aerosol formation. Other terms used to represent BVOCs are hydrocarbons (HCs), reactive organic gases (ROGs) and non-methane volatile organic compounds (NMVOCs).

Vulnerability The propensity or predisposition to be adversely affected. Vulnerability encompasses a variety of concepts and elements including sensitivity or susceptibility to harm and lack of capacity to cope and adapt. See also *Exposure*, *Hazard* and *Risk*.

Walker circulation Direct thermally driven zonal overturning circulation in the *atmosphere* over the tropical Pacific Ocean, with rising air in the western and sinking air in the eastern Pacific.

Warm spell See *Heatwave*.

Water cycle See *Hydrological cycle*.

Water mass A body of *ocean* water with identifiable properties (temperature, salinity, density, chemical tracers) resulting from its unique formation process. Water masses are often identified through a vertical or horizontal extremum of a property such as salinity. North Pacific Intermediate Water (NPIW) and Antarctic Intermediate Water (AAIW) are examples of water masses.

Water security 'The capacity of a population to safeguard sustainable access to adequate quantities of acceptable quality water for sustaining livelihoods, human well-being, and socio-

Introduction

This Summary for Policymakers (SPM) presents key findings of the Working Group I (WGI) contribution to the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6)[1] on the physical science basis of climate change. The report builds upon the 2013 Working Group I contribution to the IPCC's Fifth Assessment Report (AR5) and the 2018–2019 IPCC Special Reports[2] of the AR6 cycle and incorporates subsequent new evidence from climate science.[3]

This SPM provides a high-level summary of the understanding of the current state of the climate, including how it is changing and the role of human influence, the state of knowledge about possible climate futures, climate information relevant to regions and sectors, and limiting human-induced climate change.

Based on scientific understanding, key findings can be formulated as statements of fact or associated with an assessed level of confidence indicated using the IPCC calibrated language.[4]

The scientific basis for each key finding is found in chapter sections of the main Report and in the integrated synthesis presented in the Technical Summary (hereafter TS), and is indicated in curly brackets. The AR6 WGI Interactive Atlas facilitates exploration of these key synthesis findings, and supporting climate change information, across the WGI reference regions.[5]

A. The Current State of the Climate

Since AR5, improvements in observationally based estimates and information from paleoclimate archives provide a comprehensive view of each component of the climate system and its changes to date. New climate model simulations, new analyses, and methods combining multiple lines of evidence lead to improved understanding of human influence on a wider range of climate variables, including weather and climate extremes. The time periods considered throughout this section depend upon the availability of observational products, paleoclimate archives and peer-reviewed studies.

A.1 It is unequivocal that human influence has warmed the atmosphere, ocean and land. Widespread and rapid changes in the atmosphere, ocean, cryosphere and biosphere have occurred.

{2.2, 2.3, Cross-Chapter Box 2.3, 3.3, 3.4, 3.5, 3.6, 3.8, 5.2, 5.3, 6.4, 7.3, 8.3, 9.2, 9.3, 9.5, 9.6, Cross-Chapter Box 9.1} (Figure SPM.1, Figure SPM.2)

A.1.1 Observed increases in well-mixed greenhouse gas (GHG) concentrations since around 1750 are unequivocally caused by human activities. Since 2011 (measurements reported in AR5), concentrations have continued to increase in the atmosphere, reaching annual averages of 410 parts per million (ppm) for carbon dioxide (CO_2), 1866 parts per billion (ppb) for methane (CH_4), and 332 ppb for nitrous oxide (N_2O) in 2019.[6] Land and ocean have taken up a near-constant proportion (globally about 56% per year) of CO_2 emissions from human activities over the past six decades, with regional differences (*high confidence*).[7]
{2.2, 5.2, 7.3, TS.2.2, Box TS.5}

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1 Decision IPCC/XLVI-2.

2 The three Special Reports are: Global Warming of 1.5°C: An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty (SR1.5); Climate Change and Land: An IPCC Special Report on climate change, desertification, land degradation, sustainable land management, food security, and greenhouse gas fluxes in terrestrial ecosystems (SRCCL); IPCC Special Report on the Ocean and Cryosphere in a Changing Climate (SROCC).

3 The assessment covers scientific literature accepted for publication by 31 January 2021.

4 Each finding is grounded in an evaluation of underlying evidence and agreement. A level of confidence is expressed using five qualifiers: very low, low, medium, high and very high, and typeset in italics, for example, *medium confidence*. The following terms have been used to indicate the assessed likelihood of an outcome or result: virtually certain 99–100% probability; very likely 90–100%; likely 66–100%; about as likely as not 33–66%; unlikely 0–33%; very unlikely 0–10%; and exceptionally unlikely 0–1%. Additional terms (extremely likely 95–100%; more likely than not >50–100%; and extremely unlikely 0–5%) are also used when appropriate. Assessed likelihood is typeset in italics, for example, *very likely*. This is consistent with AR5. In this Report, unless stated otherwise, square brackets [x to y] are used to provide the assessed very likely range, or 90% interval.

5 The Interactive Atlas is available at https://interactive-atlas.ipcc.ch

6 Other GHG concentrations in 2019 were: perfluorocarbons (PFCs) – 109 parts per trillion (ppt) CF_4 equivalent; sulphur hexafluoride (SF_6) – 10 ppt; nitrogen trifluoride (NF_3) – 2 ppt; hydrofluorocarbons (HFCs) – 237 ppt HFC-134a equivalent; other Montreal Protocol gases (mainly chlorofluorocarbons (CFCs) and hydrochlorofluorocarbons (HCFCs)) – 1032 ppt CFC-12 equivalent). Increases from 2011 are 19 ppm for CO_2, 63 ppb for CH_4 and 8 ppb for N_2O.

7 Land and ocean are not substantial sinks for other GHGs.

Figure SPM.3 | Synthesis of assessed observed and attributable regional changes

The IPCC AR6 WGI inhabited regions are displayed as **hexagons** with identical size in their approximate geographical location (see legend for regional acronyms). All assessments are made for each region as a whole and for the 1950s to the present. Assessments made on different time scales or more local spatial scales might differ from what is shown in the figure. The **colours** in each panel represent the four outcomes of the assessment on observed changes. Striped hexagons (white and light-grey) are used where there is *low agreement* in the type of change for the region as a whole, and grey hexagons are used when there is limited data and/or literature that prevents an assessment of the region as a whole. Other colours indicate at least *medium confidence* in the observed change. The **confidence level** for the human influence on these observed changes is based on assessing trend detection and attribution and event attribution literature, and it is indicated by the number of dots: three dots for *high confidence*, two dots for *medium confidence* and one dot for *low confidence* (single, filled dot: limited agreement; single, empty dot: limited evidence).

Panel (a) For hot extremes, the evidence is mostly drawn from changes in metrics based on daily maximum temperatures; regional studies using other indices (heatwave duration, frequency and intensity) are used in addition. Red hexagons indicate regions where there is at least *medium confidence* in an observed increase in hot extremes.

Panel (b) For heavy precipitation, the evidence is mostly drawn from changes in indices based on one-day or five-day precipitation amounts using global and regional studies. Green hexagons indicate regions where there is at least *medium confidence* in an observed increase in heavy precipitation.

Panel (c) Agricultural and ecological droughts are assessed based on observed and simulated changes in total column soil moisture, complemented by evidence on changes in surface soil moisture, water balance (precipitation minus evapotranspiration) and indices driven by precipitation and atmospheric evaporative demand. Yellow hexagons indicate regions where there is at least *medium confidence* in an observed increase in this type of drought, and green hexagons indicate regions where there is at least *medium confidence* in an observed decrease in agricultural and ecological drought.

For all regions, Table TS.5 shows a broader range of observed changes besides the ones shown in this figure. Note that Southern South America (SSA) is the only region that does not display observed changes in the metrics shown in this figure, but is affected by observed increases in mean temperature, decreases in frost and increases in marine heatwaves.

{11.9, Atlas 1.3.3, Figure Atlas.2, Table TS.5; Box TS.10, Figure 1}

A.4 Improved knowledge of climate processes, paleoclimate evidence and the response of the climate system to increasing radiative forcing gives a best estimate of equilibrium climate sensitivity of 3°C, with a narrower range compared to AR5. {2.2, 7.3, 7.4, 7.5, Box 7.2, 9.4, 9.5, 9.6, Cross-Chapter Box 9.1}

A.4.1 Human-caused radiative forcing of 2.72 [1.96 to 3.48] W m^{-2} in 2019 relative to 1750 has warmed the climate system. This warming is mainly due to increased GHG concentrations, partly reduced by cooling due to increased aerosol concentrations. The radiative forcing has increased by 0.43 W m^{-2} (19%) relative to AR5, of which 0.34 W m^{-2} is due to the increase in GHG concentrations since 2011. The remainder is due to improved scientific understanding and changes in the assessment of aerosol forcing, which include decreases in concentration and improvement in its calculation (*high confidence*). {2.2, 7.3, TS.2.2, TS.3.1}

A.4.2 Human-caused net positive radiative forcing causes an accumulation of additional energy (heating) in the climate system, partly reduced by increased energy loss to space in response to surface warming. The observed average rate of heating of the climate system increased from 0.50 [0.32 to 0.69] W m^{-2} for the period 1971–2006[19] to 0.79 [0.52 to 1.06] W m^{-2} for the period 2006–2018[20] (*high confidence*). Ocean warming accounted for 91% of the heating in the climate system, with land warming, ice loss and atmospheric warming accounting for about 5%, 3% and 1%, respectively (*high confidence*). {7.2, Box 7.2, TS.3.1}

A.4.3 Heating of the climate system has caused global mean sea level rise through ice loss on land and thermal expansion from ocean warming. Thermal expansion explained 50% of sea level rise during 1971–2018, while ice loss from glaciers contributed 22%, ice sheets 20% and changes in land-water storage 8%. The rate of ice-sheet loss increased by a factor of four between 1992–1999 and 2010–2019. Together, ice-sheet and glacier mass loss were the dominant contributors to global mean sea level rise during 2006–2018 (*high confidence*). {9.4, 9.5, 9.6, Cross-Chapter Box 9.1}

A.4.4 The equilibrium climate sensitivity is an important quantity used to estimate how the climate responds to radiative forcing. Based on multiple lines of evidence,[21] the *very likely* range of equilibrium climate sensitivity is between 2°C (*high confidence*) and 5°C (*medium confidence*). The AR6 assessed best estimate is 3°C with a *likely* range of 2.5°C to 4°C (*high confidence*), compared to 1.5°C to 4.5°C in AR5, which did not provide a best estimate. {7.4, 7.5, TS.3.2}

19 Cumulative energy increase of 282 [177 to 387] ZJ over 1971–2006 (1 ZJ = 10^{21} joules).

20 Cumulative energy increase of 152 [100 to 205] ZJ over 2006–2018.

21 Understanding of climate processes, the instrumental record, paleoclimates and model-based emergent constraints (Glossary).

Climatic driver (Climate driver)
A changing aspect of the *climate system* that influences a component of a *human* or *natural* system.

Climatic impact-drivers (CIDs)
Climatic impact-drivers (CIDs) are physical climate system conditions (e.g., means, events, extremes) that affect an element of society or ecosystems. Depending on system tolerance, CIDs and their changes can be detrimental, beneficial, neutral, or a mixture of each across interacting system elements and regions.

CMIP3, CMIP5 and CMIP6
See *Coupled Model Intercomparison Project* (CMIP).

Co-benefit
A positive effect that a *policy* or measure aimed at one objective has on another objective, thereby increasing the total benefit to society or the environment. *Co-benefits* are also referred to as ancillary benefits. See also *Trade-off* and *Adverse side-effect*.

Coast
The *land* near to the sea. The term 'coastal' can refer to that *land* (e.g., as in 'coastal communities'), or to that part of the marine environment that is strongly influenced by *land*-based processes. Thus, coastal seas are generally shallow and near-shore. The landward and seaward limits of the coastal zone are not consistently defined, neither scientifically nor legally. Thus, coastal waters can either be considered as equivalent to territorial waters (extending 12 nautical miles/22.2 km from mean low water), or to the full exclusive economic zone, or to *shelf seas*, with less than 200 m water depth.

Coastal erosion
Coastal erosion, sometimes referred to as shoreline retreat, occurs when a net loss of sediment or bedrock from the shoreline results in landward movement of the high-tide mark.

Communicable disease
Illness due to a specific infectious agent or its toxic products that arises through transmission of that agent or its products from an infected person, animal or reservoir to a susceptible host, either directly or indirectly through an intermediate plant or animal host, vector or the inanimate environment. *Communicable disease* pathogens include bacteria, viruses, fungi, parasites and prions.

Community-based adaptation
See *Adaptation*.

Compound risks
See *Risk*.

Compound weather/climate events
The terms 'compound events', 'compound extremes' and 'compound extreme events' are used interchangeably in the literature and this report and refer to the combination of multiple *drivers* and/or *hazards* that contributes to societal and/or environmental *risk* (Zscheischler et al., 2018).

Concentrations scenario
See *Scenarios*.

Confidence
The robustness of a finding based on the type, amount, quality and consistency of *evidence* (e.g., mechanistic understanding, theory, data, *models*, expert judgment) and on the degree of *agreement* across multiple lines of *evidence*. In this report, *confidence* is expressed qualitatively (Mastrandrea et al., 2010).

Conservation agriculture
A farming system that promotes minimum soil disturbance (e.g., by using no-till practices), maintenance of a permanent soil cover and diversification of plant species. It aims to prevent *land degradation* and regenerate degraded *lands* by enhancing *biodiversity* and natural biological processes above and below the ground surface that contribute to increased water and nutrient use efficiency and improved and sustained crop production (FAO, 2016).

Coping
The use of available skills, resources and opportunities to address, manage and overcome adverse conditions, with the aim of achieving basic functioning of people, *institutions*, organisations and systems in the short to medium term (UNISDR, 2009; IPCC, 2012a).

Coping capacity
The ability of people, *institutions*, organisations and systems, using available skills, values, beliefs, resources and opportunities, to address, manage and overcome adverse conditions in the short to medium term (UNISDR, 2009; IPCC, 2012a). See also *Resilience*.

Coral bleaching
Loss of coral pigmentation through the loss of intracellular symbiotic algae (known as zooxanthellae) and/or loss of their pigments.

Coral reef
An underwater *ecosystem* characterised by structure-building stony corals. Warm-water *coral reefs* occur in shallow seas, mostly in the tropics, with the corals (animals) containing algae (plants) that depend on light and relatively stable temperature conditions. Cold-water *coral reefs* occur throughout the world, mostly at water depths of 50–500 m. In both kinds of reef, living corals frequently grow on older, dead material, predominantly made of calcium carbonate ($CaCO_3$). Both warm- and cold-water *coral reefs* support high *biodiversity* of fish and other groups, and are considered to be especially vulnerable to *climate change*.

Cost–benefit analysis
Monetary assessment of all negative and positive *impacts* associated with a given action. *Cost–benefit analysis* enables comparison of different interventions, investments or strategies and reveals how a given investment or *policy* effort pays off for a particular person, company or country. *Cost–benefit analyses* representing society's point of view are important for *climate change* decision-making, but there are difficulties in aggregating costs and benefits across different actors and across timescales. See also *Discounting*.

Integrated Assessment Scenario Ensemble A set of modelled scenarios from an intercomparison of *integrated assessment models (IAMs)* based on a systematic variation of harmonised scenario designs.

Inter-generational equity See *Equity*.

Internet of Things (IoT) The network of computing devices embedded in everyday objects such as cars, phones and computers, connected via the internet, enabling them to send and receive data.

Irreversibility A perturbed state of a dynamical system is defined as irreversible on a given time scale if the recovery from this state due to natural processes takes substantially longer than the time scale of interest. See also *Tipping point*.

Just transitions A set of principles, processes and practices that aim to ensure that no people, workers, places, sectors, countries or regions are left behind in the transition from a high-carbon to a low-carbon economy. It stresses the need for targeted and proactive measures from governments, agencies, and authorities to ensure that any negative social, environmental or economic impacts of economy-wide transitions are minimised, whilst benefits are maximised for those disproportionally affected. Key principles of just transitions include: respect and dignity for vulnerable groups; fairness in energy access and use, social dialogue and democratic consultation with relevant stakeholders; the creation of decent jobs; social protection; and rights at work. Just transitions could include fairness in energy, land use and climate planning and decision-making processes; economic diversification based on low-carbon investments; realistic training/retraining programmes that lead to decent work; gender-specific policies that promote equitable outcomes; the fostering of international cooperation and coordinated multilateral actions; and the eradication of poverty. Lastly, just transitions may embody the redressing of past harms and perceived injustices (ILO 2015; UNFCCC 2016).

Justice Justice is concerned with ensuring that people get what is due to them, setting out the moral or legal principles of *fairness* and *equity* in the way people are treated, often based on the ethics and values of society.

Climate justice
Justice that links development and human rights to achieve a human-centred approach to addressing *climate change*, safeguarding the rights of the most vulnerable people and sharing the burdens and benefits of climate change and its impacts equitably and fairly (MRFCJ 2018).

Kaya identity In this identity, global emissions are equal to the population size, multiplied by per capita output (gross world product), multiplied by the energy intensity of production, multiplied by the carbon intensity of energy.

Land The terrestrial portion of the biosphere that comprises the natural resources (soil, near-surface air, vegetation and other biota, and water), the ecological processes, topography, and *human settlements* and infrastructure that operate within that system (FAO 2007; UNCCD 1994).

Land cover The biophysical coverage of *land* (e.g., bare soil, rocks, forests, buildings and roads or lakes). Land cover is often categorised in broad land-cover classes (e.g., deciduous forest, coniferous forest, mixed forest, grassland, bare ground).

[Note: In some literature, land cover and land use are used interchangeably, but the two represent distinct classification systems. For example, the land cover class woodland can be under various land uses such as livestock grazing, recreation, conservation, or wood harvest.]

Land cover change Change from one *land cover* class to another, due to change in *land use* or change in natural conditions (Pongratz et al. 2018).

Land degradation A negative trend in land condition, caused by direct or indirect human-induced processes including *anthropogenic* climate change, expressed as long-term reduction or loss of at least one of the following: biological productivity, ecological integrity or value to humans.

[Note: This definition applies to *forest* and non-forest land. Changes in land condition resulting solely from natural processes (such as volcanic eruptions) are not considered to be land degradation. Reduction of biological productivity or ecological integrity or value to humans can constitute degradation, but any one of these changes need not necessarily be considered degradation.]

See also *Desertification*.

Land degradation neutrality A state whereby the amount and quality of land resources necessary to support *ecosystem* functions and services and enhance *food security* remain stable or increase within specified temporal and spatial scales and ecosystems (UNCCD 2020).

Land management Sum of land-use practices (e.g., sowing, fertilising, weeding, harvesting, thinning, clear-cutting) that take place within broader *land-use* categories (Pongratz et al. 2018).

Land management change
A change in land management that occurs within a *land-use* category.

Land potential The inherent, long-term potential of the *land* to sustainably generate *ecosystem services*, which reflects the capacity and *resilience* of the land-based natural capital, in the face of ongoing environmental change (UNEP 2016).

Land rehabilitation Direct or indirect actions undertaken with the aim of reinstating a level of *ecosystem* functionality, where the goal is provision of goods and services rather than ecological restoration (McDonald et al. 2016).

Land restoration The process of assisting the recovery of *land* from a degraded state (IPBES 2018; McDonald et al. 2016).

Land use The total of arrangements, activities and inputs applied to a parcel of *land*. The term land use is also used in the sense of the social and economic purposes for which land is managed (e.g., grazing, timber extraction, conservation and city dwelling). In national *greenhouse gas (GHG)* inventories, land use is classified according to the IPCC land-use categories of forest land, cropland, grassland, wetlands, settlements, other lands (see the 2006 IPCC Guidelines for National GHG Inventories and their 2019 Refinement for details (IPCC 2006, 2019)).

Land use, land-use change and forestry (LULUCF) In the context of national greenhouse gas (GHG) inventories under

IPCC FACTSHEET
What is the IPCC?

The Intergovernmental Panel on Climate Change (IPCC) is the international body for assessing the science related to climate change. The IPCC was set up in 1988 by the World Meteorological Organization (WMO) and United Nations Environment Programme (UNEP) to provide policymakers with regular assessments of the scientific basis of climate change, its impacts and future risks, and options for adaptation and mitigation.

IPCC assessments provide a scientific basis for governments at all levels to develop climate related policies, and they underlie negotiations at the UN Climate Conference – the United Nations Framework Convention on Climate Change (UNFCCC). The assessments are policy-relevant but not policy-prescriptive: they may present projections of future climate change based on different scenarios and the risks that climate change poses and discuss the implications of response options, but they do not tell policymakers what actions to take.

The IPCC embodies a unique opportunity to provide rigorous and balanced scientific information to decision-makers because of its scientific and intergovernmental nature. Participation in the IPCC is open to all member countries of the WMO and United Nations. It currently has 195 members. The Panel, made up of representatives of the member states, meets in Plenary Sessions to take major decisions. The IPCC Bureau, elected by member governments, provides guidance to the Panel on the scientific and technical aspects of the Panel's work and advises the Panel on related management and strategic issues[1].

IPCC assessments are written by hundreds of leading scientists who volunteer their time and expertise as Coordinating Lead Authors and Lead Authors of the reports. They enlist hundreds of other experts as Contributing Authors to provide complementary expertise in specific areas. The authors may work with Chapter Scientists who cross-check between findings presented in different parts of the report, carry out additional fact-checking, and work on reference management among other things. Chapter Scientists are usually early career scientists.

IPCC reports undergo multiple rounds of drafting and review to ensure they are comprehensive and objective and produced in an open and transparent way. Thousands of other experts contribute to the reports by acting as reviewers, ensuring the reports reflect the full range of views in the scientific community. Teams of Review Editors provide a thorough monitoring mechanism for making sure that review comments are addressed. (For more details see *IPCC Factsheet – How does the IPCC select its authors?* and *IPCC Factsheet – How does the IPCC review process work?*).

1 https://www.ipcc.ch/site/assets/uploads/2018/02/TOR_Bureau.pdf

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The IPCC works by assessing published literature (see ***IPCC Factsheet – What literature does the IPCC assess?***). It does not conduct its own scientific research. For all findings, author teams use defined language to characterize their degree of certainty in assessment conclusions[2]. IPCC assessments point to areas of well-established knowledge and of evolving understanding, as well as where multiple perspectives exist in the literature.

The authors producing the reports are currently grouped in three working groups – Working Group I: the Physical Science Basis; Working Group II: Impacts, Adaptation and Vulnerability; and Working Group III: Mitigation of Climate Change – and the Task Force on National Greenhouse Gas Inventories (TFI). As part of the IPCC, a Task Group on Data Support for Climate Change Assessments (TG-Data) provides guidance to the Data Distribution Centre (DDC) on curation, traceability, stability, availability and transparency of data and scenarios related to the reports of the IPCC. TG-Data replaces the Task Group on Data and Scenario Support for Impact and Climate Analysis (TGICA) whose mandate was to facilitates the distribution and application of climate change-related data and scenarios.

IPCC Assessment Reports cover the full scientific, technical and socio-economic assessment of climate change, generally in four parts – one for each of the Working Groups plus a Synthesis Report. Special Reports are assessments of a specific issue. Methodology Reports provide practical guidelines for the preparation of greenhouse gas inventories under the UNFCCC.

2 See *Guidance Note for Lead Authors of the IPCC Fifth Assessment Report on Consistent Treatment of Uncertainties*: https://www.ipcc.ch/site/assets/uploads/2017/08/AR5_Uncertainty_Guidance_Note.pdf

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from historical or *projected* levels, usually allocated by some criteria, as well as sharing the cost burden across countries.

Business as usual (BAU) See *Baseline scenario*.

Carbon budget This term refers to three concepts in the literature: (1) an assessment of *carbon cycle* sources and *sinks* on a global level, through the synthesis of *evidence* for *fossil fuel* and cement emissions, *land-use change* emissions, ocean and land CO_2 sinks, and the resulting atmospheric CO_2 growth rate. This is referred to as the global carbon budget; (2) the estimated cumulative amount of global carbon dioxide emissions that that is estimated to limit global surface temperature to a given level above a *reference period*, taking into account global surface temperature contributions of other *GHGs* and climate forcers; (3) the distribution of the carbon budget defined under (2) to the regional, national, or sub-national level based on considerations of *equity*, costs or efficiency. See also *Remaining carbon budget*.

Carbon cycle The term used to describe the flow of carbon (in various forms, e.g., as *carbon dioxide (CO_2)*, carbon in *biomass*, and carbon dissolved in the ocean as carbonate and bicarbonate) through the *atmosphere*, hydrosphere, terrestrial and marine biosphere and lithosphere. In this report, the reference unit for the global carbon cycle is $GtCO_2$ or GtC (Gigatonne of carbon = 1 GtC = 10^{15} grams of carbon. This corresponds to 3.667 $GtCO_2$).

Carbon dioxide (CO_2) A naturally occurring gas, CO_2 is also a by-product of burning *fossil fuels* (such as oil, gas and coal), of burning *biomass*, of *land-use changes (LUC)* and of industrial processes (e.g., cement production). It is the principal *anthropogenic* greenhouse gas *(GHG)* that affects the Earth's radiative balance. It is the reference gas against which other GHGs are measured and therefore has a global warming potential (GWP) of 1. See also *Greenhouse gas (GHG)*.

Carbon dioxide capture and storage (CCS) A process in which a relatively pure stream of *carbon dioxide (CO_2)* from industrial and energy-related sources is separated (captured), conditioned, compressed and transported to a storage location for long-term isolation from the *atmosphere*. Sometimes referred to as Carbon capture and storage. See also *Carbon dioxide capture and utilisation (CCU)*, *Bioenergy with carbon dioxide capture and storage (BECCS)* and *Uptake*.

Carbon dioxide capture and utilisation (CCU) A process in which CO_2 is captured and then used to produce a new product. If the CO_2 is stored in a product for a *climate*-relevant time horizon, this is referred to as carbon dioxide capture, utilisation and storage (CCUS). Only then, and only combined with CO_2 recently removed from the *atmosphere*, can CCUS lead to *carbon dioxide removal*. CCU is sometimes referred to as carbon dioxide capture and use. See also *Carbon dioxide capture and storage (CCS)*.

Carbon dioxide capture, utilisation and storage (CCUS) See *Carbon dioxide capture and utilisation (CCU)*.

Carbon dioxide removal (CDR) *Anthropogenic* activities removing *CO_2* from the *atmosphere* and durably storing it in geological, terrestrial, or ocean reservoirs, or in products. It includes existing and potential anthropogenic enhancement of biological or geochemical sinks and direct air capture and storage, but excludes natural CO_2 *uptake* not directly caused by human activities. See also *Mitigation (of climate change)*, *Greenhouse gas removal (GGR)*, *Negative emissions*, *Direct air carbon dioxide capture and storage (DACCS)* and *Sink*.

Carbon intensity The amount of emissions of *carbon dioxide (CO_2)* released per unit of another variable such as *gross domestic product (GDP)*, output energy use or transport.

Carbon neutrality See *Net zero CO_2 emissions*.

Carbon price The price for avoided or released *carbon dioxide (CO_2)* or *CO_2-equivalent emissions*. This may refer to the rate of a carbon tax, or the price of emission permits. In many models that are used to assess the economic costs of *mitigation*, carbon prices are used as a proxy to represent the level of effort in mitigation *policies*.

Carbon sequestration The process of storing carbon in a carbon pool. See also *Blue carbon*, *Carbon dioxide capture and storage (CCS)*, *Uptake* and *Sink*.

Carbon sink See *Sink*.

Clean Development Mechanism (CDM) A mechanism defined under Article 12 of the *Kyoto Protocol* through which investors (governments or companies) from developed (Annex B) countries may finance *greenhouse gas (GHG)* emission reduction or removal projects in developing countries (Non-Annex B), and receive Certified Emission Reduction Units (CERs) for doing so. The CERs can be credited towards the commitments of the respective developed countries. The CDM is intended to facilitate the two objectives of promoting *sustainable development (SD)* in developing countries and of helping *industrialised countries* to reach their emissions commitments in a cost-effective way.

Climate Climate in a narrow sense is usually defined as the average weather, or more rigorously, as the statistical description in terms of the mean and variability of relevant quantities over a period of time ranging from months to thousands or millions of years. The classical period for averaging these variables is 30 years, as defined by the World Meteorological Organization. The relevant quantities are most often surface variables such as temperature, precipitation and wind. Climate in a wider sense is the state, including a statistical description, of the *climate system*.

Climate change Climate change refers to a change in the state of the *climate* that can be identified (e.g., by using statistical tests) by changes in the mean and/or the variability of its properties and that persists for an extended period, typically decades or longer. Climate change may be due to natural internal processes or external *forcings* such as modulations of the solar cycles, volcanic eruptions and persistent *anthropogenic* changes in the composition of the *atmosphere* or in *land use*. Note that the *Framework Convention on Climate Change (UNFCCC)*, in its Article 1, defines climate change as: 'a change of climate which is attributed directly or indirectly to human activity that alters the composition of the global atmosphere and which is in addition to natural climate variability observed over comparable time periods.' The UNFCCC thus makes a distinction between climate change attributable to human activities altering the atmospheric composition and climate variability attributable to natural causes. See also *Climate variability*, *Global warming*, *Ocean acidification (OA)* and *Detection and attribution*.

Climate change commitment Climate change commitment is defined as the unavoidable future *climate change* resulting from inertia in the geophysical and socio-economic systems. Different types of climate change commitment are discussed in the literature (see subterms). Climate change commitment is usually quantified in terms of the further change in temperature, but it includes other future changes, for example in the *hydrological cycle*, in *extreme weather events*, in extreme climate events, and in sea level.

Constant composition commitment
The constant composition commitment is the remaining *climate change* that would result if atmospheric composition, and hence *radiative forcing*, were held fixed at a given value. It results from the thermal inertia of the ocean and slow processes in the cryosphere and land surface.

Ecosystem An ecosystem is a functional unit consisting of living organisms, their non-living environment and the interactions within and between them. The components included in a given ecosystem and its spatial boundaries depend on the purpose for which the ecosystem is defined: in some cases they are relatively sharp, while in others they are diffuse. Ecosystem boundaries can change over time. Ecosystems are nested within other ecosystems and their scale can range from very small to the entire biosphere. In the current era, most ecosystems either contain people as key organisms, or are influenced by the effects of human activities in their environment. See also *Ecosystem services*.

Ecosystem services Ecological processes or functions having monetary or non-monetary value to individuals or society at large. These are frequently classified as (1) supporting services such as productivity or *biodiversity* maintenance, (2) provisioning services such as food or fibre, (3) regulating services such as climate regulation or *carbon sequestration*, and (4) cultural services such as tourism or spiritual and aesthetic appreciation.

Effective climate sensitivity See *Climate sensitivity*.

Effective radiative forcing See *Radiative forcing*.

El Niño-Southern Oscillation (ENSO) The term El Niño was initially used to describe a warm-water current that periodically flows along the coast of Ecuador and Peru, disrupting the local fishery. It has since become identified with warming of the tropical Pacific Ocean east of the dateline. This oceanic event is associated with a fluctuation of a global-scale tropical and subtropical surface pressure pattern called the Southern Oscillation. This coupled atmosphere–ocean phenomenon, with preferred time scales of two to about seven years, is known as the El Niño-Southern Oscillation (ENSO). It is often measured by the surface pressure anomaly difference between Tahiti and Darwin and/or the *sea surface temperatures* in the central and eastern equatorial Pacific. During an ENSO event, the prevailing trade winds weaken, reducing upwelling and altering ocean currents such that the sea surface temperatures warm, further weakening the trade winds. This phenomenon has a great impact on the wind, sea surface temperature and precipitation patterns in the tropical Pacific. It has climatic effects throughout the Pacific region and in many other parts of the world, through global teleconnections. The cold phase of ENSO is called La Niña.

Electric vehicle (EV) A vehicle whose propulsion is powered fully or mostly by electricity.

Battery electric vehicle (BEV)
A vehicle whose propulsion is entirely electric without any internal combustion engine.

Plug-in hybrid electric vehicle (PHEV)
A vehicle whose propulsion is mostly electric with batteries re-charged from an electric source but extra power and distance are provided by a hybrid internal combustion engine.

Emission pathways See *Pathways*.

Emission scenario A plausible representation of the future development of emissions of substances that are radiatively active (e.g., *greenhouse gases (GHGs)*, *aerosols*) based on a coherent and internally consistent set of assumptions about driving forces (such as demographic and socio-economic development, technological change, energy and *land use*) and their key relationships. Concentration *scenarios*, derived from emission scenarios, are often used as input to a *climate model* to compute *climate projections*. See also *Baseline scenario*, *Mitigation scenario*, *Socio-economic scenario*, *Scenario*, *Representative Concentration Pathways (RCPs)* (under *Pathways*), *Shared Socio-economic Pathways (SSPs)* (under *Pathways*) and *Transformation pathways* (under *Pathways*).

Emission trajectories A *projected* development in time of the emission of a *greenhouse gas (GHG)* or group of GHGs, *aerosols*, and GHG *precursors*. See also *Emission pathways* (under *Pathways*).

Emissions trading A market-based instrument aiming at meeting a *mitigation* objective in an efficient way. A cap on *GHG* emissions is divided in tradeable emission permits that are allocated by a combination of auctioning and handing out free allowances to entities within the jurisdiction of the trading scheme. Entities need to surrender emission permits equal to the amount of their emissions (e.g., tonnes of CO_2). An entity may sell excess permits to entities that can avoid the same amount of emissions in a cheaper way. Trading schemes may occur at the intra-company, domestic, or international level (e.g., the flexibility mechanisms under the *Kyoto Protocol* and the EU-ETS) and may apply to carbon dioxide (CO_2), other greenhouse gases (GHGs), or other substances.

Enabling conditions Conditions that affect the *feasibility* of *adaptation* and *mitigation* options, and can accelerate and scale-up systemic transitions that would limit temperature increase to 1.5°C and enhance capacities of systems and societies to adapt to the associated *climate change*, while achieving *sustainable development*, eradicating *poverty* and reducing *inequalities*. Enabling conditions include finance, technological innovation, strengthening *policy* instruments, *institutional capacity*, *multilevel governance*, and changes in *human behaviour* and lifestyles. They also include inclusive processes, attention to power asymmetries and unequal opportunities for development and reconsideration of values. See also *Feasibility*.

Energy efficiency The ratio of output or useful energy or energy services or other useful physical outputs obtained from a system, conversion process, transmission or storage activity to the input of energy (measured as kWh kWh^{-1}, tonnes kWh^{-1} or any other physical measure of useful output like tonne-km transported). Energy efficiency is often described by energy intensity. In economics, energy intensity describes the ratio of economic output to energy input. Most commonly energy efficiency is measured as input energy over a physical or economic unit, i.e., kWh USD^{-1} (energy intensity), kWh $tonne^{-1}$. For buildings, it is often measured as kWh m^{-2}, and for vehicles as km $liter^{-1}$ or liter km^{-1}. Very often in policy 'energy efficiency' is intended as the measures to reduce energy demand through technological options such as insulating buildings, more efficient appliances, efficient lighting, efficient vehicles, etc.

Energy security The goal of a given country, or the global community as a whole, to maintain an adequate, stable and predictable energy supply. Measures encompass safeguarding the sufficiency of energy resources to meet national energy demand at competitive and stable prices and the *resilience* of the energy supply; enabling development and deployment of technologies; building sufficient infrastructure to generate, store and transmit energy supplies; and ensuring enforceable contracts of delivery.

Enhanced weathering Enhancing the removal of *carbon dioxide (CO_2)* from the *atmosphere* through dissolution of silicate and carbonate rocks by grinding these minerals to small particles and actively applying them to soils, coasts or oceans.

(Model) Ensemble A group of parallel model simulations characterising historical *climate* conditions, *climate predictions*, or climate projections. Variation of the results across the ensemble members may give an estimate of modelling-based *uncertainty*. Ensembles made with the same model but different initial conditions only characterize the uncertainty associated with internal *climate variability*, whereas multimodel ensembles including simulations by several models also include the impact of model differences. Perturbed parameter ensembles, in which model parameters are varied in a systematic

Inter-generational justice
Justice in the distribution of economic and non-economic costs and benefits across generations.

Procedural justice
Justice in the way outcomes are brought about including who participates and is heard in the processes of decision-making.

Social justice
Just or fair relations within society that seek to address the distribution of wealth, access to resources, opportunity, and support according to principles of justice and fairness.

See also *Equity*, *Ethics*, *Fairness*, and *Human rights*.

Kyoto Protocol The Kyoto Protocol to the *United Nations Framework Convention on Climate Change (UNFCCC)* is an international treaty adopted in December 1997 in Kyoto, Japan, at the Third Session of the *Conference of the Parties* (COP3) to the UNFCCC. It contains legally binding commitments, in addition to those included in the UNFCCC. Countries included in Annex B of the Protocol (mostly OECD countries and countries with economies in transition) agreed to reduce their anthropogenic *greenhouse gas (GHG)* emissions *(carbon dioxide (CO_2)*, *methane (CH_4)*, *nitrous oxide (N_2O)*, hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), and sulphur hexafluoride (SF_6)) by at least 5% below 1990 levels in the first commitment period (2008–2012). The Kyoto Protocol entered into force on 16 February 2005 and as of May 2018 had 192 Parties (191 States and the European Union). A second commitment period was agreed in December 2012 at COP18, known as the Doha Amendment to the Kyoto Protocol, in which a new set of Parties committed to reduce GHG emissions by at least 18% below 1990 levels in the period from 2013 to 2020. However, as of May 2018, the Doha Amendment had not received sufficient ratifications to enter into force. See also *United Nations Framework Convention on Climate Change (UNFCCC)* and *Paris Agreement*.

Land surface air temperature The near-surface air temperature over land, typically measured at 1.25–2 m above the ground using standard meteorological equipment.

Land use Land use refers to the total of arrangements, activities and inputs undertaken in a certain land cover type (a set of human actions). The term land use is also used in the sense of the social and economic purposes for which land is managed (e.g., grazing, timber extraction, conservation and city dwelling). In national *greenhouse gas* inventories, land use is classified according to the IPCC land use categories of forest land, cropland, grassland, wetland, settlements, other. See also *Land-use change (LUC)*.

Land-use change (LUC) Land-use change involves a change from one *land use* category to another.

Indirect land-use change (iLUC)
Refers to market-mediated or policy-driven shifts in *land use* that cannot be directly attributed to land-use management decisions of individuals or groups. For example, if agricultural land is diverted to fuel production, *forest* clearance may occur elsewhere to replace the former agricultural production.

Land use, land-use change and forestry (LULUCF)
In the context of national *greenhouse gas (GHG)* inventories under the *UNFCCC*, LULUCF is a GHG inventory sector that covers *anthropogenic emissions* and removals of GHG from carbon pools in managed lands, excluding non-CO_2 agricultural emissions. Following the 2006 IPCC Guidelines for National GHG Inventories, 'anthropogenic' land-related GHG fluxes are defined as all those occurring on 'managed land', i.e., 'where human interventions and practices have been applied to perform production, ecological or social functions'. Since managed land may include CO_2 removals not considered as 'anthropogenic' in some of the scientific literature assessed in this report (e.g., removals associated with CO_2 fertilization and N deposition), the land-related net GHG emission estimates included in this report are not necessarily directly comparable with LULUCF estimates in National GHG Inventories.

See also *Afforestation*, *Deforestation*, *Reforestation*, and the IPCC Special Report on Land Use, Land-Use Change, and Forestry (IPCC, 2000).

Land use, land-use change and forestry (LULUCF) See *Land-use change (LUC)*.

Life cycle assessment (LCA) Compilation and evaluation of the inputs, outputs and the potential environmental impacts of a product or service throughout its life cycle. This definition builds from ISO (2018).

Likelihood The chance of a specific outcome occurring, where this might be estimated probabilistically. Likelihood is expressed in this report using a standard terminology (Mastrandrea et al., 2010). See Section 1.6 for the list of likelihood qualifiers used. See also *Agreement*, *Evidence*, *Confidence* and *Uncertainty*.

Livelihood The resources used and the activities undertaken in order to live. Livelihoods are usually determined by the entitlements and assets to which people have access. Such assets can be categorised as human, social, natural, physical or financial.

Local knowledge Local knowledge refers to the understandings and skills developed by individuals and populations, specific to the places where they live. Local knowledge informs decision-making about fundamental aspects of life, from day-to-day activities to longer-term actions. This knowledge is a key element of the social and cultural systems which influence observations of, and responses to *climate change*; it also informs *governance* decisions. This definition builds on UNESCO (2018).

Lock-in A situation in which the future development of a system, including infrastructure, technologies, investments, *institutions*, and behavioural norms, is determined or constrained ('locked in') by historic developments.

Long-lived climate forcers (LLCF) Long-lived climate forcers refer to a set of well-mixed *greenhouse gases* with long atmospheric lifetimes. This set of compounds includes *carbon dioxide (CO_2)* and *nitrous oxide (N_2O)*, together with some fluorinated gases. They have a warming effect on *climate*. These compounds accumulate in the *atmosphere* at decadal to centennial time scales, and their effect on climate hence persists for decades to centuries after their emission. On time scales of decades to a century, already emitted emissions of long-lived climate forcers can only be abated by *greenhouse gas removal (GGR)*. See also *Short-lived climate forcers (SLCF)*.

Loss and Damage, and losses and damages Research has taken Loss and Damage (capitalized letters) to refer to political debate under the *UNFCCC* following the establishment of the Warsaw Mechanism on Loss and Damage in 2013, which is to 'address loss and damage associated with impacts of climate change, including extreme events and slow onset events, in developing countries that are particularly vulnerable to the adverse effects of climate change.' Lowercase letters (losses and damages) have been taken to refer broadly to harm from (observed) *impacts* and (projected) *risks* (see Mechler et al., in press).

Maladaptive actions (Maladaptation) Actions that may lead to increased *risk* of adverse climate-related outcomes, including via increased *GHG* emissions, increased *vulnerability* to *climate change*, or diminished welfare, now or in the future. Maladaptation is usually an unintended consequence.

Market exchange rate (MER) The rate at which a currency of one country can be exchanged with the currency of another country. In most economies such rates evolve daily while in others there are official conversion rates that are adjusted periodically. See also *Purchasing power parity (PPP)*.

Market failure When private decisions are based on market prices that do not reflect the real scarcity of goods and services but rather reflect market distortions, they do not generate an efficient allocation of resources but cause welfare losses. A market distortion is any event in which a market reaches a market clearing price that is substantially different from the price that a market would achieve while operating under conditions of perfect competition and state enforcement of legal contracts and the ownership of private property. Examples of factors causing market prices to deviate from real economic scarcity are environmental externalities, public goods, monopoly power, information asymmetry, transaction costs and non-rational behaviour.

Measurement, Reporting and Verification (MRV)

Measurement
'Processes of data collection over time, providing basic datasets, including associated accuracy and precision, for the range of relevant variables. Possible data sources are field measurements, field observations, detection through remote sensing and interviews.' (UN-REDD, 2009).

Reporting
'The process of formal reporting of assessment results to the UNFCCC, according to predetermined formats and according to established standards, especially the IPCC [Intergovernmental Panel on Climate Change] Guidelines and GPG [Good Practice Guidance].' (UN-REDD, 2009)

Verification
'The process of formal verification of reports, for example the established approach to verify national communications and national inventory reports to the UNFCCC.' (UN-REDD, 2009)

Megadrought See *Drought*.

Methane (CH_4) One of the six *greenhouse gases (GHGs)* to be mitigated under the *Kyoto Protocol* and is the major component of natural gas and associated with all hydrocarbon fuels. Significant emissions occur as a result of animal husbandry and agriculture, and their management represents a major *mitigation* option.

Migrant See *Migration*.

Migration The International Organization for Migration (IOM) defines migration as 'The movement of a person or a group of persons, either across an international border, or within a State. It is a population movement, encompassing any kind of movement of people, whatever its length, composition and causes; it includes migration of refugees, displaced persons, economic migrants, and persons moving for other purposes, including family reunification.' (IOM, 2018).

Migrant
The International Organization for Migration (IOM) defines a migrant as 'any person who is moving or has moved across an international border or within a State away from his/her habitual place of residence, regardless of (1) the person's legal status; (2) whether the movement is voluntary or involuntary; (3) what the causes for the movement are; or (4) what the length of the stay is.' (IOM, 2018).

See also *(Internal) Displacement*.

Millennium Development Goals (MDGs) A set of eight time-bound and measurable goals for combating *poverty*, hunger, disease, illiteracy, discrimination against women and environmental degradation. These goals were agreed at the UN Millennium Summit in 2000 together with an action plan to reach the goals by 2015.

Mitigation (of climate change) A human intervention to reduce emissions or enhance the *sinks* of *greenhouse gases*.

Mitigation behaviour See *Human behaviour*.

Mitigation measures In climate *policy*, mitigation measures are technologies, processes or practices that contribute to *mitigation*, for example, renewable energy (RE) technologies, waste minimization processes and public transport commuting practices. See also *Mitigation option*, and *Policies (for climate change mitigation and adaptation)*.

Mitigation option A technology or practice that reduces *GHG* emissions or enhances *sinks*.

Mitigation pathways See *Pathways*.

Mitigation scenario A plausible description of the future that describes how the (studied) system responds to the implementation of *mitigation* policies and measures. See also *Emission scenario*, *Pathways*, *Socio-economic scenario* and *Stabilization (of GHG or CO_2-equivalent concentration)*.

Monitoring and evaluation (M&E) Monitoring and evaluation refers to mechanisms put in place at national to local scales to respectively monitor and evaluate efforts to reduce *greenhouse gas* emissions and/or adapt to the *impacts* of *climate change* with the aim of systematically identifying, characterizing and assessing progress over time.

Motivation (of an individual) An individual's reason or reasons for acting in a particular way; individuals may consider various consequences of actions, including financial, social, affective and environmental consequences. Motivation can come from outside (extrinsic) or from inside (intrinsic) the individual.

Multilevel governance See *Governance*.

Narratives Qualitative descriptions of plausible future world evolutions, describing the characteristics, general logic and developments underlying a particular quantitative set of *scenarios*. Narratives are also referred to in the literature as 'storylines'. See also *Scenario*, *Scenario storyline* and *Pathways*.

Nationally Determined Contributions (NDCs) A term used under the *United Nations Framework Convention on Climate Change (UNFCCC)* whereby a country that has joined the *Paris Agreement* outlines its plans for reducing its emissions. Some countries' NDCs also address how they will adapt to climate change impacts, and what support they need from, or will provide to, other countries to adopt low-carbon pathways and to build climate resilience. According to Article 4 paragraph 2 of the Paris Agreement, each Party shall prepare, communicate and maintain successive NDCs that it intends to achieve. In the lead up to 21st *Conference of the Parties* in Paris in 2015, countries submitted Intended Nationally Determined Contributions (INDCs). As countries join the Paris Agreement, unless they decide otherwise, this INDC becomes their first Nationally Determined Contribution (NDC). See also *United Nations Framework Convention on Climate Change (UNFCCC)* and *Paris Agreement*.

Negative emissions Removal of *greenhouse gases (GHGs)* from the atmosphere by deliberate human activities, i.e., in addition to the removal that would occur via natural *carbon cycle* processes. See also *Net negative emissions*, *Net zero emissions*, *Carbon dioxide removal (CDR)* and *Greenhouse gas removal (GGR)*.

Net negative emissions A situation of net negative emissions is achieved when, as result of human activities, more *greenhouse gases* are removed from the *atmosphere* than are emitted into it. Where multiple greenhouse gases are involved, the quantification of *negative emissions* depends on the climate metric chosen to compare emissions of different gases (such as global warming potential, global temperature change potential, and others, as well as the chosen time horizon). See also *Negative emissions*, *Net zero emissions* and *Net zero CO_2 emissions*.

Net zero CO_2 emissions Net zero *carbon dioxide (CO_2)* emissions are achieved when *anthropogenic* CO_2 emissions are balanced globally by anthropogenic CO_2 removals over a specified period. Net zero CO_2 emissions are also referred to as carbon neutrality. See also *Net zero emissions* and *Net negative emissions*.

Net zero emissions Net zero emissions are achieved when *anthropogenic emissions* of *greenhouse gases* to the *atmosphere* are balanced by *anthropogenic removals* over a specified period. Where multiple greenhouse gases are involved, the quantification of net zero emissions depends on the climate metric chosen to compare emissions of different gases (such as global warming potential, global temperature change potential, and others, as well as the chosen time horizon). See also *Net zero CO_2 emissions*, *Negative emissions* and *Net negative emissions*.

Nitrous oxide (N_2O) One of the six *greenhouse gases (GHGs)* to be mitigated under the *Kyoto Protocol*. The main *anthropogenic* source of N_2O is agriculture (soil and animal manure management), but important contributions also come from sewage treatment, *fossil fuel* combustion, and chemical industrial processes. N_2O is also produced naturally from a wide variety of biological sources in soil and water, particularly microbial action in wet tropical *forests*.

Non-CO_2 emissions and radiative forcing Non-CO_2 emissions included in this report are all *anthropogenic emissions* other than CO_2 that result in *radiative forcing*. These include *short-lived climate forcers*, such as *methane (CH_4)*, some fluorinated gases, *ozone (O_3)* precursors, *aerosols* or aerosol *precursors*, such as *black carbon* and sulphur dioxide, respectively, as well as long-lived *greenhouse gases*, such as *nitrous oxide (N_2O)* or other fluorinated gases. The radiative forcing associated with non-CO_2 emissions and changes in surface *albedo* is referred to as non-CO_2 radiative forcing.

Non-overshoot pathways See *Pathways*.

Ocean acidification (OA) Ocean acidification refers to a reduction in the *pH* of the ocean over an extended period, typically decades or longer, which is caused primarily by uptake of *carbon dioxide (CO_2)* from the *atmosphere*, but can also be caused by other chemical additions or subtractions from the ocean. *Anthropogenic* ocean acidification refers to the component of pH reduction that is caused by human activity (IPCC, 2011, p. 37).

Ocean fertilization Deliberate increase of nutrient supply to the near-surface ocean in order to enhance biological production through which additional *carbon dioxide (CO_2)* from the *atmosphere* is sequestered. This can be achieved by the addition of micro-nutrients or macro-nutrients. Ocean fertilization is regulated by the London Protocol.

Overshoot See *Temperature overshoot*.

Overshoot pathways See *Pathways*.

Ozone (O_3) Ozone, the triatomic form of oxygen (O_3), is a gaseous atmospheric constituent. In the *troposphere*, it is created both naturally and by photochemical reactions involving gases resulting from human activities (smog). Tropospheric ozone acts as a *greenhouse gas*. In the *stratosphere*, it is created by the interaction between solar ultraviolet radiation and molecular oxygen (O_2). Stratospheric ozone plays a dominant role in the stratospheric radiative balance. Its concentration is highest in the ozone layer.

Paris Agreement The Paris Agreement under the *United Nations Framework Convention on Climate Change (UNFCCC)* was adopted on December 2015 in Paris, France, at the 21st session of the *Conference of the Parties (COP)* to the UNFCCC. The agreement, adopted by 196 Parties to the UNFCCC, entered into force on 4 November 2016 and as of May 2018 had 195 Signatories and was ratified by 177 Parties. One of the goals of the Paris Agreement is 'Holding the increase in the global average temperature to well below 2°C above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5°C above pre-industrial levels', recognising that this would significantly reduce the risks and impacts of climate change. Additionally, the Agreement aims to strengthen the ability of countries to deal with the impacts of climate change. The Paris Agreement is intended to become fully effective in 2020. See also *United Nations Framework Convention on Climate Change (UNFCCC)*, *Kyoto Protocol* and *Nationally Determined Contributions (NDCs)*.

Participatory governance See *Governance*.

Pathways The temporal evolution of natural and/or *human systems* towards a future state. Pathway concepts range from sets of quantitative and qualitative *scenarios* or *narratives* of potential futures to solution-oriented decision-making processes to achieve desirable societal goals. Pathway approaches typically focus on biophysical, techno-economic, and/or socio-behavioural trajectories and involve various dynamics, goals and actors across different scales.

1.5°C pathway
A pathway of emissions of *greenhouse gases* and other climate forcers that provides an approximately one-in-two to two-in-three chance, given current knowledge of the climate response, of *global warming* either remaining below 1.5°C or returning to 1.5°C by around 2100 following an *overshoot*. See also *Temperature overshoot*.

Adaptation pathways
A series of *adaptation* choices involving trade-offs between short-term and long-term goals and values. These are processes of deliberation to identify solutions that are meaningful to people in the context of their daily lives and to avoid potential *maladaptation*.

Development pathways
Development pathways are trajectories based on an array of social, economic, cultural, technological, *institutional* and biophysical features that characterise the interactions between human and natural systems and outline visions for the future, at a particular scale.

Emission pathways
Modelled trajectories of global *anthropogenic emissions* over the 21st century are termed emission pathways.

Mitigation pathways
A mitigation pathway is a temporal evolution of a set of *mitigation scenario* features, such as *greenhouse gas* emissions and socio-economic development.

Overshoot pathways
Pathways that exceed the stabilization level (concentration, *forcing*, or temperature) before the end of a time horizon of interest (e.g., before 2100) and then decline towards that level by that time. Once the target level is exceeded, removal by sinks of *greenhouse gases* is required. See also *Temperature overshoot*.